

Online Appendix

The Cost of Charitable Care: Resource Externalities and the Value of Insurance

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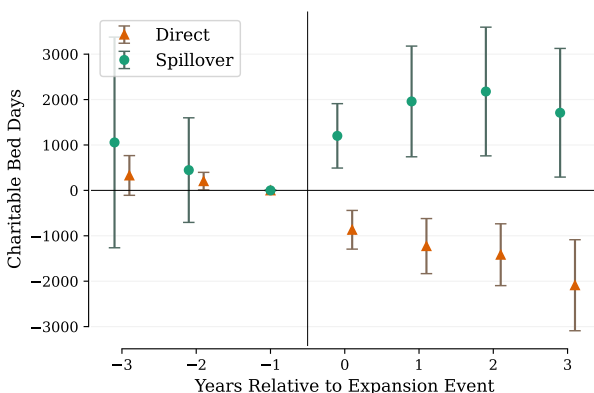
This Online Appendix accompanies the article of the same title.

A Additional Figures and Tables

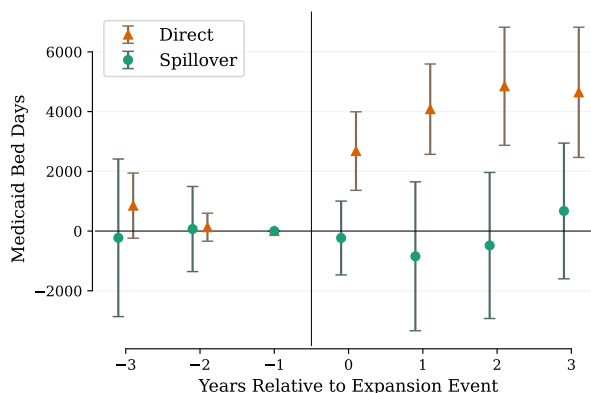
Table A.1: Medicaid Expansion Effects on Charitable Care

	Panel A: Direct Treatment				Panel B: Spillover Treatment			
	Charitable		State Charitable		Charitable		State Charitable	
	Admissions	DRG	Admissions	Spending	Admissions	DRG	Admissions	Spending
Expansion Effect (α)	-348.113*** (72.171)	-446.040*** (82.103)	-4681.280*** (853.197)	-108.198*** (13.583)	256.221*** (91.020)	287.658** (142.274)	-35.697 (809.488)	33.693 (21.040)
Mean Dep.	678.005	809.760	8767.576	285.124	1120.333	1267.883	19225.818	547.119
N	2081	2081	959	771	3065	3065	763	613
R ²	0.948	0.951	0.991	0.961	0.978	0.967	0.996	0.944

Notes: Admission and DRG-based measures include only admissions from patients residing in the same state as the associated hospital. “Charitable Admissions” corresponds to the count of charitable care admissions. “Charitable DRG” is the sum of DRG weights for charitable care patients. “State Charitable Admissions” corresponds to the total number of charitable care admissions in the state. “State Charitable Spending” corresponds to the total reported expenditure on charitable care by non-profit systems in each state, as reported in their tax returns. The first two columns of each panel are estimates from Equation (2) with two-way cluster-robust standard errors at the event and hospital levels. We estimate the state-level outcomes using Equation (3) with heteroskedasticity-robust standard errors. For state-level regressions, we keep only one observation per state-year. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$



(a) Effects on Charity Care Admissions



(b) Effects on Medicaid Admissions

Figure A.1: Medicaid Expansion Effects on Charity Care and Medicaid Admissions

Notes: The two figures show the estimated effects of Medicaid expansion on hospitals in expansion states (triangles) and on hospitals in non-expansion states linked to expansion states through system membership (circles). These estimates match the event-study variant of the stacked event-study panel regression (2) and support the evidence in the main text Section 4.2.

Table A.2: Spillover Exposure Effects

	Panel A: Charitable Care at Spillover-Treated			Panel B: Admissions at Local-Untreated (bed days)			
	Admissions (1)	Bed Days (2)	Bed Day Share (3)	Charitable (4)	Gov. (5)	Private (6)	Total (7)
Expansion Effect (α)	40.262 (66.117)	-231.764 (277.356)	-0.002 (0.005)				
$\hookrightarrow \times$ Exposure	11.214** (5.473)	77.091*** (27.392)	0.001*** (0.000)				
Local Exposure (α^e)				-1009.057** (424.829)	1173.584 (1604.996)	1681.869* (909.450)	2807.092 (1963.974)
Mean Exposure	18.240	18.240	18.240	4.539	4.539	4.539	4.539
Std. Exposure	10.228	10.228	10.228	9.335	9.335	9.335	9.335
Mean Dep.	1120.333	3938.259	0.055	8442.128	63565.617	23601.596	98385.344
N	3065	3065	3065	1156	1156	1156	1156
R^2	0.978	0.962	0.910	0.944	0.988	0.974	0.992

Notes: This table complements the main text Table 2. Panel A presents how exposure, measured as tens of millions of dollars spent by systems in Medicaid expansion states in 2013, is associated with charitable care at spillover-treated hospitals. The estimates follow regression Equation (2), with standard errors clustered at the event and hospital levels. Panel B presents the effect of the instrument—Local Exposure—as defined in the main text, on admissions. These correspond to estimates of Equation (4) with HRR-clustered standard errors. Column 4 is thus the first stage of our IV regression in Equation (5). “Mean Dep.” in Panel A is the unweighted treated-hospital mean in the year before the event. “Mean Dep.” in Panel B is the unweighted mean among exposed hospitals in the year before their first local exposure; hospitals without an observed outcome in that baseline year are omitted from this mean. Exposure statistics are unweighted: Panel A uses treated hospital-event observations at $t = -1$ with positive exposure; Panel B summarizes first positive local exposure among exposed hospitals observed at $t = -1$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

Table A.3: Charitable Care Effects at Local Untreated Hospitals (OLS)

Panel A: Effects on Insured Care (OLS) (bed-days)								
	Overall		Emergency Care			Scheduled Care		
	Gov. (1)	Private (2)	Total (3)	Gov. (4)	Private (5)	Total (6)	Gov. (7)	Private (8)
Charitable Care Admissions (β)	1.434 (1.198)	0.081 (0.684)	1.813 (1.507)	1.676 (1.132)	0.137 (0.492)	-0.298 (0.941)	-0.242 (0.621)	-0.056 (0.446)
Mean Dep.	63565.617	23601.596	46258.277	36798.787	9459.489	40908.936	26766.830	14142.106
N	1156	1156	1156	1156	1156	1156	1156	1156
R ²	0.987	0.970	0.967	0.968	0.924	0.975	0.967	0.972
Panel B: Effects on Cost and Revenue (OLS) (thousands)								
	Profit	Revenue			Cost			
	Total (1)	Total (2)	Medicare (3)	Private (proxy) (4)	Total (5)	Labor (6)	Uncompensated Care (7)	Charity Care (8)
Charitable Care Admissions (β)	-3.159 (2.178)	2.917 (3.161)	3.427* (2.036)	0.714 (6.875)	6.076*** (2.258)	6.430* (3.316)	24.125** (11.644)	11.136* (6.134)
Mean Dep. (millions)	42.806	263.981	78.850	150.250	221.176	204.553	54.509	36.461
N	977	977	537	537	977	977	977	977
R ²	0.892	0.987	0.989	0.961	0.992	0.988	0.663	0.505

Notes: OLS estimates from Equation (5) matching main text Table 2. See the main text table for details. Standard errors in parentheses are heteroskedasticity-robust. Panel B, column 3 reports Medicare inpatient reimbursement. "Mean Dep." is the unweighted mean for exposed hospitals in the year before first local exposure, within each column's estimation sample; missing baseline outcomes are omitted from this mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Medicaid Expansion Effects on Length of Stay and Care Delivery

Panel A: Average Length of Stay (days per admission)								
	Charitable		Gov.		Private		Total	
	Spillover (1)	Local (2)	Spillover (3)	Local (4)	Spillover (5)	Local (6)	Spillover (7)	Local (8)
Expansion Effect (α)	-0.074 (0.219)	-0.067 (0.111)	-0.145 (0.135)	0.003 (0.091)	-0.150 (0.126)	0.052 (0.065)	-0.109 (0.126)	0.018 (0.082)
Mean Dep.	3.540	3.774	4.569	4.891	3.522	3.918	4.273	4.509
N	3031	1133	3065	1133	3065	1133	3065	1133
R^2	0.688	0.728	0.716	0.844	0.632	0.810	0.714	0.845

Panel B: Emergency and Scheduled Care (bed-days)								
	Emergency Charitable		Scheduled Charitable		Emergency Insured		Scheduled Insured	
	Spillover (1)	Local (2)	Spillover (3)	Local (4)	Spillover (5)	Local (6)	Spillover (7)	Local (8)
Expansion Effect (α)	845.229* (486.538)	-533.411 (456.736)	407.643* (229.791)	-475.646* (248.549)	-450.050 (2579.399)	1445.506 (2454.844)	-678.589 (1999.858)	1409.948 (1751.036)
Mean Dep.	3163.407	5640.064	774.852	2802.064	39703.965	46258.277	26428.814	40908.938
N	3065	1156	3065	1156	3065	1156	3065	1156
R^2	0.952	0.928	0.913	0.889	0.993	0.972	0.987	0.977

Notes: Companion to Panel B of Table 1. "Local" stands for local untreated. All measures include only patients residing in the hospital's state. Panel A reports hospital-year bed-days divided by admissions of the same type. Panel B reports total bed-days by emergency status; scheduled care is the complement of emergency care. "Charitable" includes free-care and self-pay patients, "Gov." is Medicare plus Medicaid, "Insured" is government plus private, and "Total" includes all payer types. Averages with nonpositive or missing admissions are undefined; Local columns use a common sample within each panel with all four outcomes observed. Spillover estimates use Equation (2), with two-way clustered standard errors at the event and hospital levels. Local estimates use Equation (4), with hospital and state-year fixed effects and standard errors clustered by hospital referral region. "Mean Dep." is the unweighted mean of the hospital-year outcome among treated units in the year before expansion treatment for Spillover columns, and among exposed hospitals in the year before their first local exposure for Local columns. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table A.5: System Targets and Reaction to Expansion Effects

	Panel A: Charitable Care Admissions at Spillover-Treated Hospitals		Panel B: Expansion Effect on System Charity Spending	
	<u>Exposure Metric:</u>		<u>Treated Definition:</u>	
	Share of Charitable Expenditure in Expansion States	Change in Medicaid Admissions in Expansion States	Spillover	Direct-only
Exposure Tercile 1	99.227 (84.486)	73.400 (72.399)		
Exposure Tercile 2	330.872** (134.904)	320.427** (132.761)		
Exposure Tercile 3	150.091 (133.504)	254.215* (130.412)		
Expansion Effect (α)			-1.581 (9.854)	-16.155*** (2.347)
N	3065	3065	920	10381
R ²	0.978	0.978	0.966	0.898
Mean Dep.	1120.333	1120.333	119.510	20.471
Mean Exposure	0.370	3608.114		

Notes: This table shows evidence on how systems respond to Medicaid expansions. Panel A shows how charitable care admissions at spillover-treated hospitals change in response to spillover treatment, allowing for heterogeneity across terciles of two different exposure metrics. The share of charitable care expenditure in expansion states was computed in 2013. The change in Medicaid admissions in expansion states corresponds to the difference between the averages in 2012–2013 and 2014–2015. Both exposure metrics are defined at the system level. Panel B presents the effect of Medicaid expansion on spillover-treated and direct-only treated hospitals, where the outcome is the system’s total spending on charitable care, as reported to the IRS. The direct-only group excludes hospitals whose systems have members in non-expansion states in the treatment year. All regressions follow the design in Equation (2), with standard errors clustered two-way at the system and event levels. Control units are weighted to sum to one in each event, following Wing et al. (2024). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

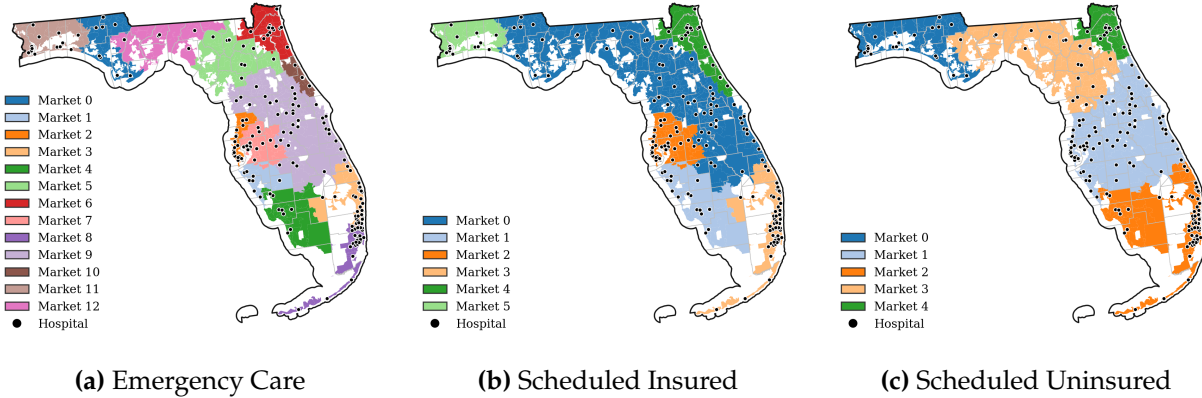
Table A.6: Grant Response to Charitable Care Spending

	Panel A: System State				Panel B: System			
	Gov. Grants		All Grants		Gov. Grants		All Grants	
	Contemp.	Lag.	Contemp.	Lag.	Contemp.	Lag.	Contemp.	Lag.
Charitable Care Spending (α)	0.144** (0.072)	0.027 (0.039)	0.293*** (0.099)	0.112* (0.064)	0.270*** (0.074)	0.086* (0.045)	0.824*** (0.249)	0.245 (0.152)
N	11922	10217	11922	10217	11386	11386	11386	11386
R^2	0.932	0.943	0.971	0.971	0.881	0.878	0.813	0.807
Mean Dep.	5.512	5.512	12.866	12.866	5.886	5.886	13.738	13.738
Mean Spending	10.971	11.489	10.971	11.489	11.472	10.339	11.472	10.339

Notes: This table shows the relationship between grants given to hospitals and their charitable care spending. Panel A reports estimates of α from the regression:

$$y_{smt} = \alpha c_{smt} + \gamma_{sm} + \delta_t + \varepsilon_{smt},$$

where s indexes a system, c_{smt} is the system's spending in state m in year t , γ_{sm} are system-state fixed effects, δ_t are year fixed effects, and errors are clustered at the system level. Panel B is the associated estimate when aggregating grants and spending at the system level and adjusting the regression to have system and year fixed effects. Standard errors in Panel B are heteroskedasticity-robust. "Contemp." means contemporaneous charitable care spending is used in c_{smt} , while "Lag." means spending from the previous year is used. Government grants exclude payments for services rendered, such as DSH payments. All Grants include direct donations and exclude any payments for services rendered. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

**Figure A.2:** Market Definitions by Service Line

Notes: These figures show how we group HRRs by market in our structural analysis. Our algorithm clusters contiguous HRRs such that at least 90% of demand originating within the market is retained within the market. White regions in the interior of Florida correspond to ZIP codes where we do not observe sufficient consumers in our data to meet masking thresholds.

Table A.7: Demand Model Estimates of Disutility of Travel Parameters

	Emergency Care												Scheduled Insured						Scheduled Uninsured					
Market	0	1	2	3	4	5	6	7	8	9	10	11	12	0	1	2	3	4	5	0	1	2	3	4
Age: 6–17	–0.219 (0.129)	2.455 (0.352)	–2.652 (0.349)	1.894 (0.116)	1.060 (0.115)	0.540 (0.186)	–0.336 (0.094)	0.854 (0.192)	–0.235 (0.231)	0.662 (0.101)	–0.819 (0.160)	–0.226 (0.040)	0.174 (0.074)	0.372 (0.039)	0.209 (0.067)	–0.030 (0.061)	–0.016 (0.059)	–0.599 (0.093)	–0.260 (0.020)	–0.782 (0.350)	0.089 (0.358)	–4.253 (0.244)	0.841 (0.677)	3.852 (0.939)
Age: 18–24	–1.616 (0.135)	1.582 (0.299)	–14.309 (0.330)	–3.356 (0.107)	0.507 (0.100)	–0.966 (0.147)	–3.098 (0.077)	–3.121 (0.169)	1.272 (0.176)	–3.219 (0.089)	–3.240 (0.161)	–3.116 (0.038)	–1.062 (0.073)	–3.448 (0.038)	–2.371 (0.067)	–7.025 (0.058)	–1.082 (0.050)	–3.717 (0.087)	–2.308 (0.023)	–1.055 (0.275)	–0.520 (0.311)	–3.074 (0.179)	–1.261 (0.603)	4.598 (0.673)
Age: 25–34	–2.431 (0.118)	–0.527 (0.293)	–14.802 (0.316)	–4.385 (0.100)	–0.519 (0.097)	–1.450 (0.140)	–5.346 (0.074)	–3.475 (0.157)	3.466 (0.163)	–4.346 (0.084)	–3.724 (0.139)	–4.599 (0.035)	–1.407 (0.065)	–3.494 (0.036)	–2.614 (0.063)	–7.420 (0.055)	–3.161 (0.049)	–4.832 (0.081)	–2.842 (0.022)	–1.076 (0.259)	–0.934 (0.308)	–3.522 (0.174)	–0.681 (0.589)	5.065 (0.660)
Age: 35–44	–2.572 (0.109)	–2.226 (0.291)	–16.938 (0.314)	–5.735 (0.100)	–1.450 (0.095)	–2.053 (0.140)	–6.345 (0.074)	–4.097 (0.154)	2.445 (0.161)	–5.703 (0.083)	–4.679 (0.141)	–5.279 (0.035)	–1.931 (0.063)	–2.008 (0.036)	–1.476 (0.062)	–4.161 (0.054)	–0.010 (0.049)	–3.422 (0.081)	–1.358 (0.019)	–1.162 (0.262)	0.322 (0.308)	–3.271 (0.176)	–0.566 (0.592)	4.311 (0.662)
Age: 45–54	–3.075 (0.103)	–4.716 (0.287)	–19.396 (0.311)	–7.642 (0.096)	–2.668 (0.093)	–2.983 (0.136)	–7.862 (0.071)	–6.334 (0.151)	1.154 (0.159)	–7.169 (0.081)	–6.364 (0.134)	–5.544 (0.032)	–2.681 (0.062)	–2.048 (0.034)	–1.790 (0.060)	–3.171 (0.051)	1.943 (0.048)	–3.544 (0.077)	–1.405 (0.018)	–0.801 (0.258)	–0.121 (0.307)	–3.211 (0.174)	–1.065 (0.591)	4.082 (0.657)
Age: 55–64	–3.526 (0.099)	–6.572 (0.285)	–21.969 (0.311)	–9.240 (0.096)	–3.637 (0.092)	–3.424 (0.135)	–10.298 (0.072)	–7.974 (0.151)	0.865 (0.158)	–8.557 (0.081)	–8.096 (0.133)	–5.797 (0.030)	–3.043 (0.062)	–2.577 (0.033)	–2.788 (0.058)	–5.228 (0.051)	2.036 (0.048)	–4.086 (0.074)	–1.633 (0.017)	–1.381 (0.263)	–0.617 (0.308)	–3.503 (0.175)	–1.413 (0.591)	2.795 (0.675)
Age: 65–74	–4.260 (0.100)	–9.028 (0.284)	–25.092 (0.311)	–11.900 (0.096)	–5.386 (0.092)	–4.043 (0.136)	–12.745 (0.073)	–10.123 (0.152)	–2.112 (0.158)	–10.036 (0.081)	–10.537 (0.135)	–6.968 (0.031)	–3.488 (0.064)	–3.421 (0.033)	–4.204 (0.058)	–7.388 (0.050)	0.968 (0.048)	–5.265 (0.074)	–2.163 (0.018)	–4.344 (0.326)	0.682 (0.321)	–2.087 (0.204)	0.845 (0.651)	–1.211 (0.718)
Age: 75+	–5.692 (0.097)	–14.216 (0.282)	–30.158 (0.311)	–19.240 (0.093)	–7.557 (0.091)	–5.179 (0.135)	–15.558 (0.073)	–13.112 (0.150)	–5.905 (0.157)	–12.618 (0.080)	–14.948 (0.134)	–10.354 (0.032)	–4.584 (0.063)	–5.133 (0.034)	–6.771 (0.059)	–9.757 (0.051)	–2.156 (0.048)	–7.111 (0.077)	–4.944 (0.023)	–3.735 (0.385)	0.951 (0.324)	–2.524 (0.206)	0.147 (0.651)	–23.579 (1.455)
Male	–2.964 (0.096)	–8.675 (0.281)	–10.858 (0.310)	–11.902 (0.088)	–6.868 (0.091)	–3.142 (0.131)	–4.099 (0.069)	–12.081 (0.144)	–22.906 (0.156)	–7.126 (0.078)	–6.596 (0.132)	–1.196 (0.027)	–0.903 (0.056)	–3.531 (0.032)	–6.271 (0.059)	–4.815 (0.049)	–5.489 (0.047)	–5.105 (0.073)	–1.017 (0.018)	–2.458 (0.257)	–6.684 (0.306)	–4.483 (0.173)	–3.414 (0.588)	–10.717 (0.667)
Female	–3.490 (0.097)	–9.771 (0.281)	–12.319 (0.311)	–13.450 (0.090)	–7.647 (0.091)	–4.532 (0.132)	–6.480 (0.070)	–13.372 (0.145)	–24.327 (0.155)	–8.187 (0.079)	–7.147 (0.132)	–2.534 (0.029)	–1.233 (0.057)	–4.751 (0.032)	–7.498 (0.059)	–7.367 (0.049)	–8.088 (0.047)	–6.439 (0.073)	–1.704 (0.017)	–2.693 (0.261)	–7.574 (0.306)	–6.980 (0.170)	–4.130 (0.589)	–11.742 (0.649)
Low income	–0.920 (0.038)	–4.786 (0.053)	0.588 (0.051)	–2.548 (0.035)	–3.582 (0.024)	–2.021 (0.036)	–2.719 (0.024)	–3.293 (0.047)	–7.247 (0.043)	–0.934 (0.024)	1.593 (0.049)	–1.483 (0.013)	–2.062 (0.028)	–1.092 (0.012)	–1.703 (0.017)	–0.429 (0.016)	–1.830 (0.009)	0.299 (0.027)	–0.381 (0.009)	–0.578 (0.058)	–1.619 (0.039)	–2.935 (0.039)	–0.765 (0.095)	–1.671 (0.127)
N	99082	248260	555772	549865	335779	224124	487632	489475	560721	538205	158142	250850	109635	492068	433851	521488	533472	352353	158523	5918	36954	72212	10892	11401
Clusters	76143	164710	356222	346137	238748	178574	336692	344649	348140	368549	104370	183542	90308	357615	275023	358210	346066	231152	112947	5782	35305	63045	10462	10939

Notes: Standard errors clustered at the choice event level in parentheses.

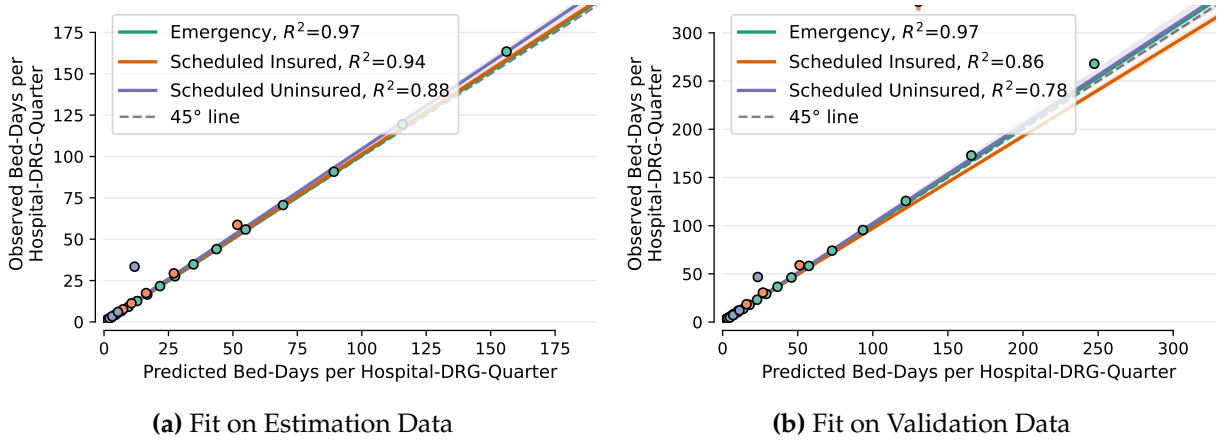


Figure A.3: Demand Estimation Fit

Notes: These figures show the fit of the demand model on the data used for estimation and on a random 20% sample of market-line-quarters, which we exclude from estimation. To predict on the training data, we use the estimated patient preference parameters and recover admission rates by one iteration of the admission fixed point. The admission fixed point matches the aggregate length of stay at the hospital-line; hence, we plot the fit at a much finer level: the hospital-line-diagnosis group level.

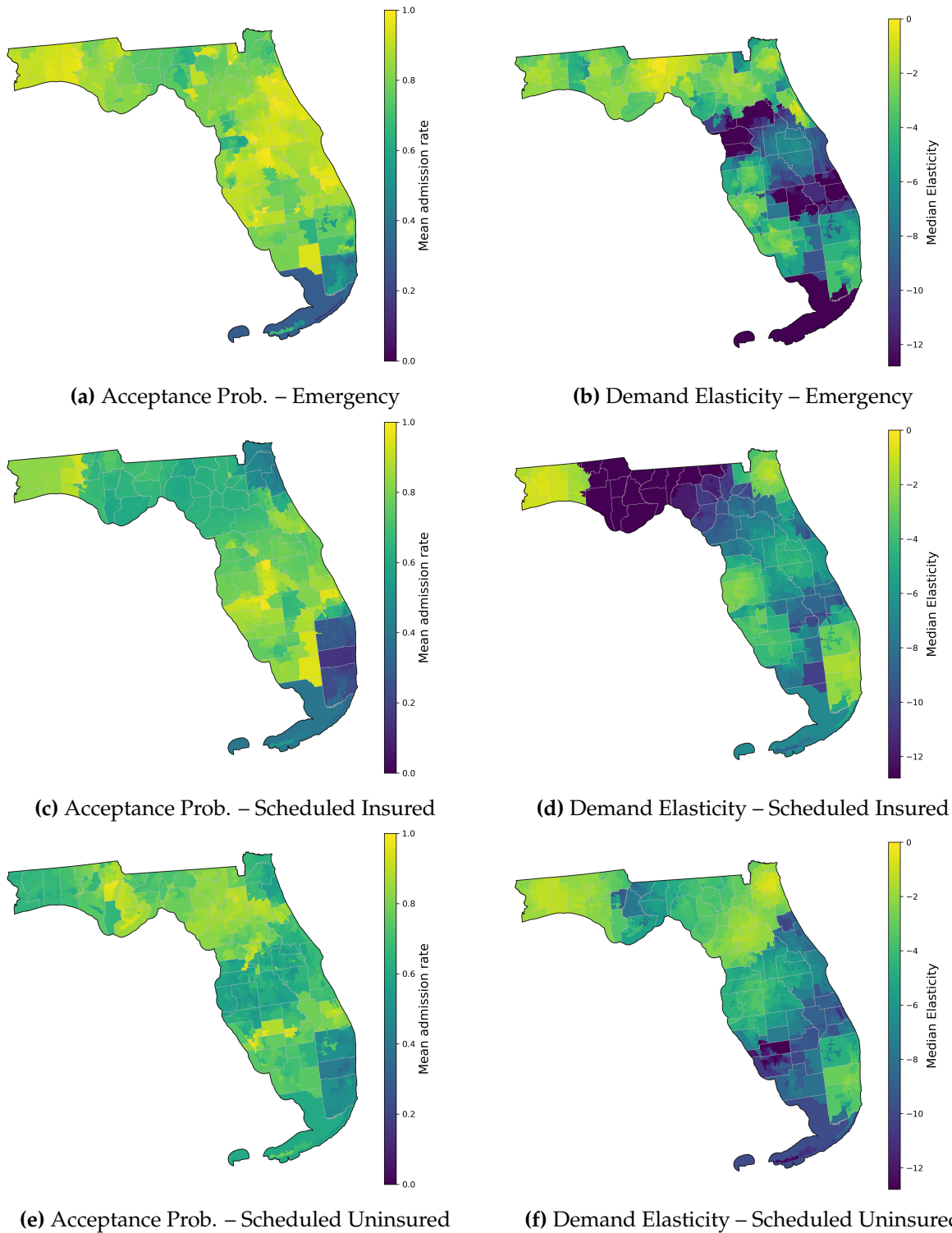


Figure A.4: Preferred Admission Probabilities and Distance Elasticities

Notes: The figures on the left show the frequency-weighted mean estimated probability of admission to the most preferred option within each ZCTA, including the outside option. The figures on the right show the median estimated distance elasticity within each ZCTA.

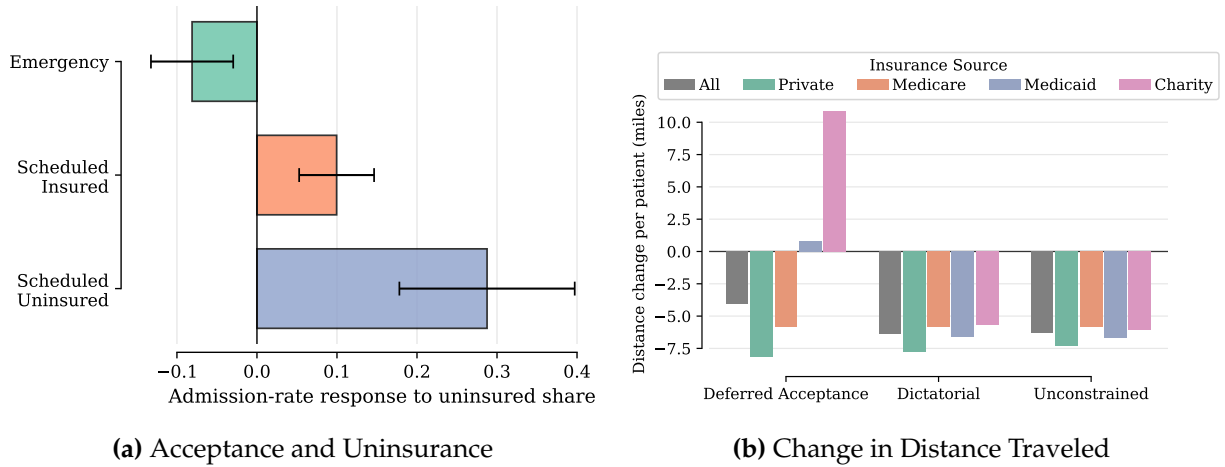


Figure A.5: Acceptance Rate and Allocative Efficiency Holding Supply Fixed

Notes: Figure A.5a shows how the probability of patients being admitted to their most preferred hospital changes with the local share of uninsured. The plot presents the estimate and 95% confidence interval for the coefficient from a regression of the ZIP Code Tabulation Area (ZCTA) average admission rate to the preferred hospital on the local uninsurance rate, conditional on ZCTA-by-service-line fixed effects and a year fixed effect to account for state-level admission trends. Standard errors are clustered at the ZCTA level. Figure A.5b displays the estimated allocative efficiency of the 2019 allocations relative to three alternative benchmarks: patient-proposed deferred acceptance matching, dictatorial surplus-maximizing assignment, and unconstrained surplus-maximizing assignment. The dictatorial problem is a Generalized Assignment Problem, which is solved via the algorithm of Achterberg (2009).

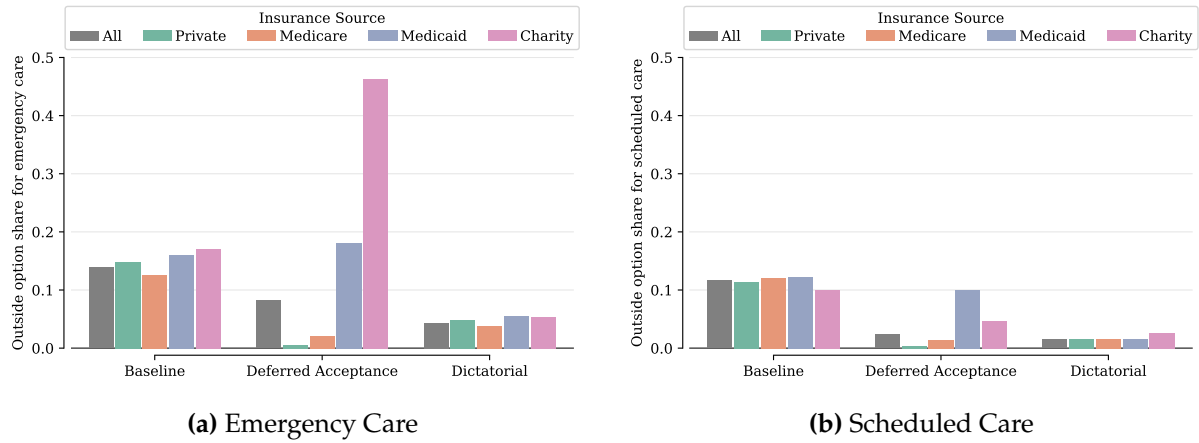
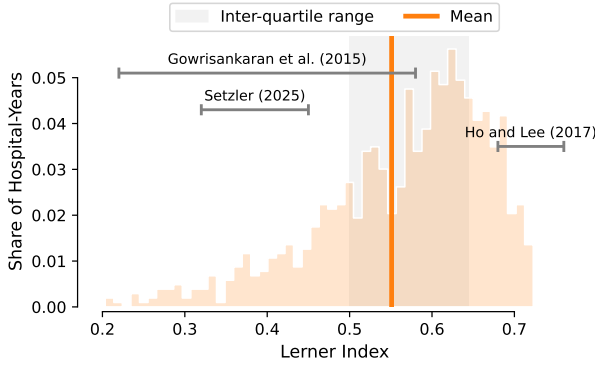
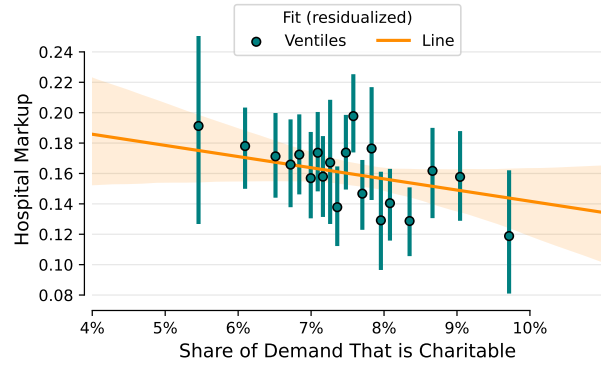


Figure A.6: Outside-of-Market Shares Under Alternative Allocative Regimes

Notes: These figures show the share of demand by insurance type and emergency status that is allocated to the out-of-market option under alternative allocative regimes, as discussed in Section 7.1. Note that we exclude here patients whose preference is to be assigned to the outside option, as all direct allocation regimes would assign them out of the market.



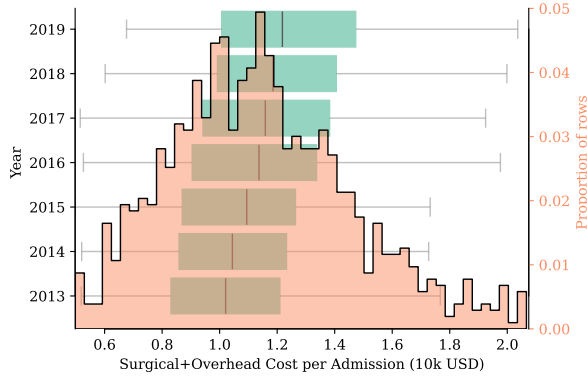
(a) Markup on Private Insurance Prices



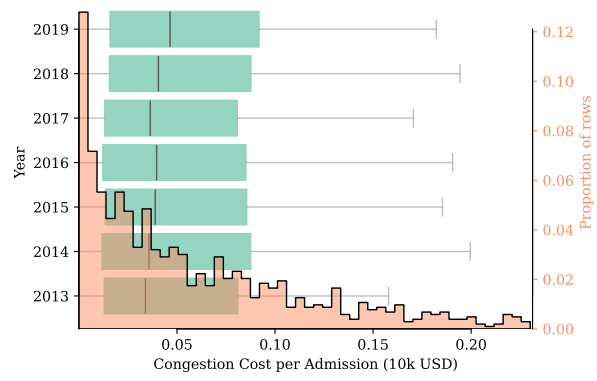
(b) Markups and Local Charity Burden

Figure A.7: Comparison of Estimated Markups and Correlation with Charity Burden

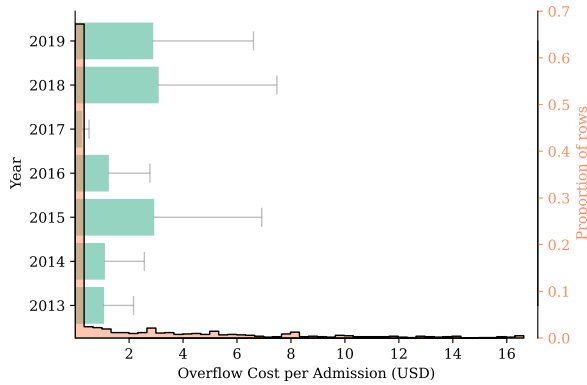
Notes: These figures display markup statistics implied by the structural cost estimates. Figure A.7a shows the distribution of estimated Lerner indices, measured as private-payer profit divided by private-payer revenue. The shaded gray area represents the interquartile range of our estimates, and the solid orange line represents the mean. Hospital-years receive equal weight; only the histogram bars omit the bottom and top 2.5%. Each reference estimate uses a different sample of hospitals, insurers, and years: Gowrisankaran et al. (2015) use a sample of 11 hospitals from Northern Virginia between 2003 and 2006, with data from four large private insurers. Ho and Lee (2017) study the California market in 2004 with data from large private insurers. We note that the article does not report hospital markups directly, so we infer the range from reported estimates. Setzler (2025) studies a large collection of hospitals from 1996 to 2022, infers private prices from HCRIS reports, and reports markups by hospital size categories. Figure A.7b shows estimates from regressing a hospital's markup over total cost (profit divided by total cost) on its predicted charitable-admission share, conditional on hospital fixed effects. The green dots represent binned averages after removing hospital fixed effects. The orange line shows the fit over the residualized data. The estimated slope is -0.735 with a standard error of 0.274 . This figure complements the main text Figure 6.



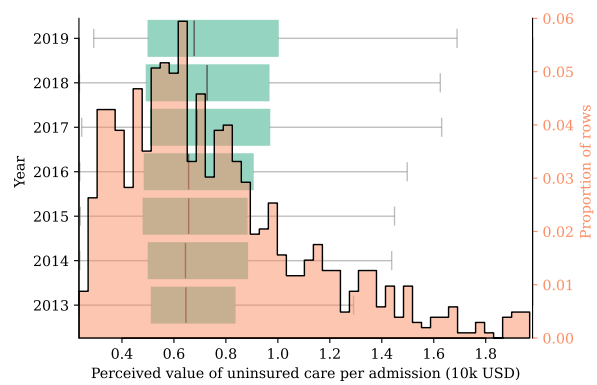
(a) Surgical Cost



(b) Congestion Cost



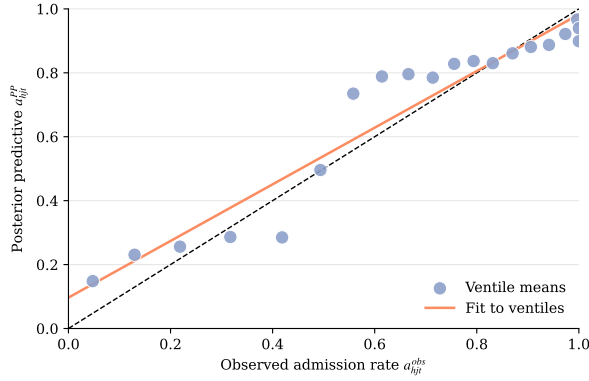
(c) Overflow Cost



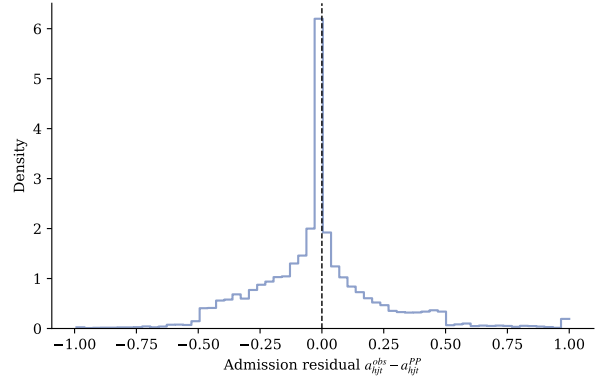
(d) Perceived Value of Uninsured Care

Figure A.8: Distribution of Estimated Cost Components

Notes: These figures show the distribution of hospitals' cost estimates from the structural estimates. In each figure, the histogram shows the overall distribution of estimates, while the horizontal boxplots show the yearly distribution. Figure (a) shows the estimates associated with components c_{hjt}^S, c_{hjt}^F . Figure (b) shows the cost associated with c_{hjt}^V . Figure (c) shows the costs associated with component c_{ht}^o . Figure (d) shows the perceived revenue from uninsured care, associated with component τ_{hjt} .



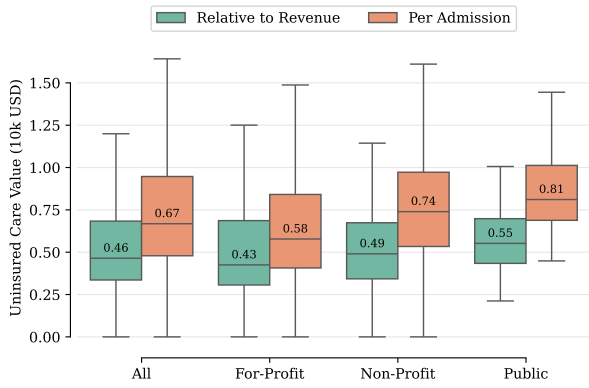
(a) Fit



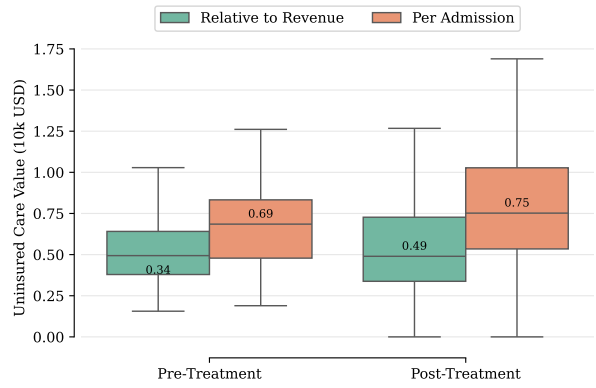
(b) Residual

Figure A.9: Cost Estimates Fit — Implied and Observed Admission Rates

Notes: These figures show how well our structural model fits the data. Using the estimated cost and τ_{hjt} components, we compute each hospital's optimal admission rate and contrast it with the observed rate.



(a) By Hospital Type



(b) By Spillover Treatment

Figure A.10: Estimate of Hospitals' Preference for Uninsured Care

Notes: These figures show estimates associated with τ_{hjt} , the hospital's preference for improving uninsured surplus.

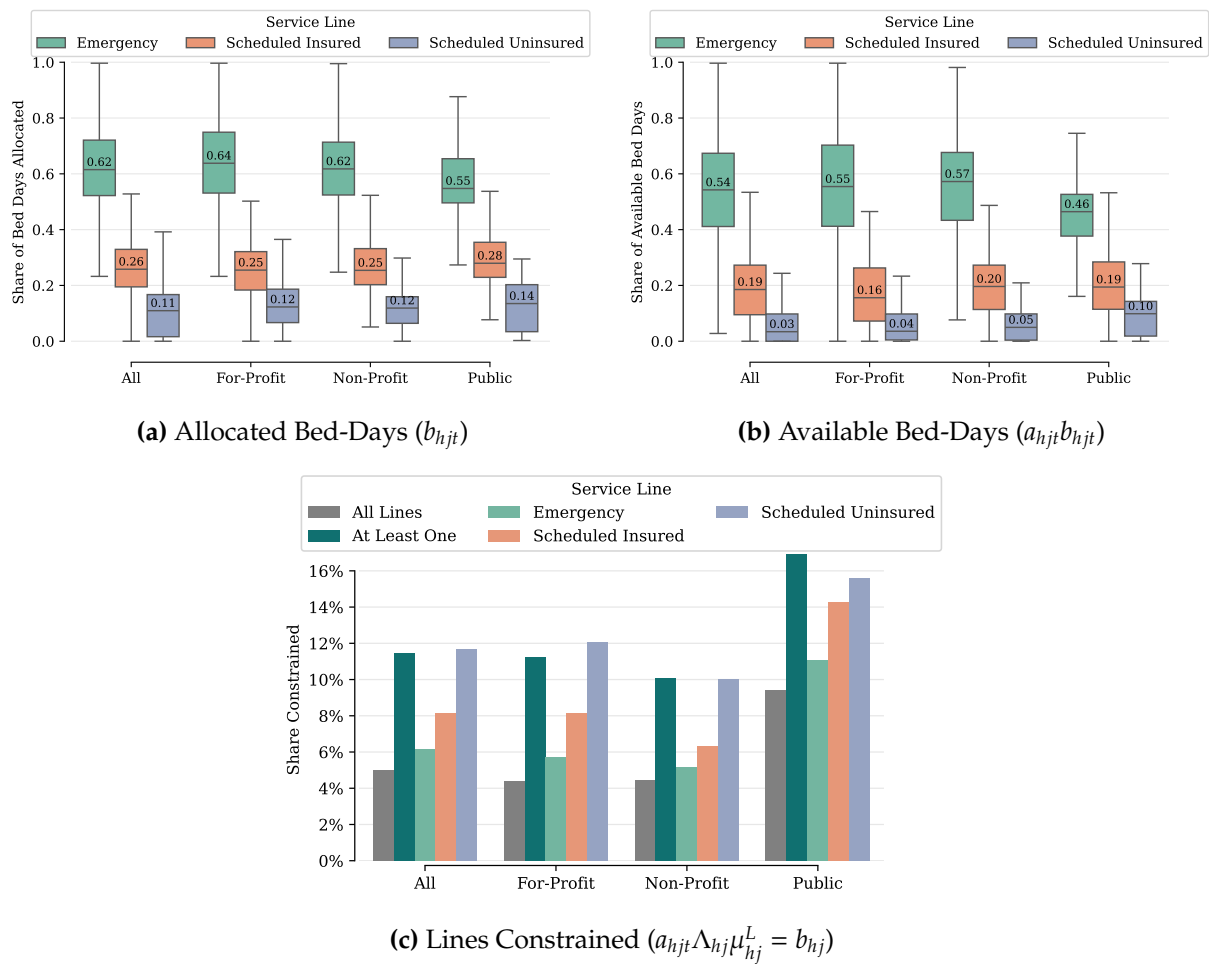


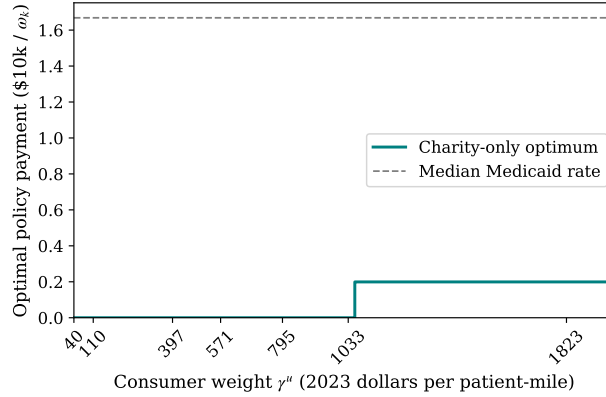
Figure A.11: Estimated Resource Allocation

Notes: These figures show the estimated resource allocation to hospitals by service line.

Table A.8: Counterfactual Per-Capita Surplus and Allocative Efficiency

	Per-capita surplus change (\$10k/patient)				Share of Dictatorial allocative gains (%)				
	Uninsured	Private	Medicare	Medicaid	Uninsured	Private	Medicare	Medicaid	Overall
Panel A: Pay-per-Complexity									
Medicaid-equivalent	0.1235	0.0486	0.0467	0.0617	10.559	2.305	5.235	3.729	4.284
Medicare-equivalent	0.1789	0.0654	0.0644	0.0820	17.663	3.355	7.743	5.315	6.444
Optimum at $\gamma^\pi = 0$	0.1538	0.0583	0.0561	0.0748	14.043	2.846	6.209	4.777	5.301
Optimum at $\gamma^\pi = 0.93$	0.2059	0.0710	0.0706	0.0906	22.512	4.153	9.054	6.545	7.823
Panel B: Floor-and-Trade									
Unconstrained	0.1236	0.0355	0.0459	0.0421	15.664	2.485	8.616	3.564	5.993
Homogeneous	0.0320	0.0011	0.0050	0.0030	4.219	-0.040	0.990	0.126	0.707

Notes: Panel A reports the change in per-capita consumer surplus in \$10,000 per patient. Panel A values patients on the modeled logit surplus and Panel B on the deterministic value of the assigned option; both exclude testing-data sentinel choices from their numerator and their denominator. Panel B is the percentage of the baseline-to-Dictatorial allocation gain captured by each policy, by payor, followed by the pooled share over all payors.

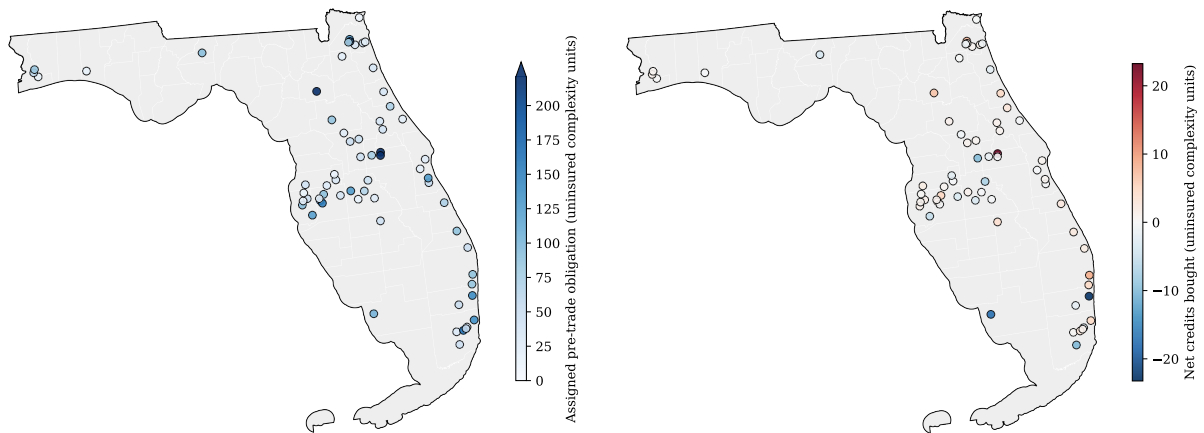
**Figure A.12: Optimal Pay-Per-Complexity Charity-Surplus-Only Policy**

Notes: This figure shows the optimal policy when $\gamma^\pi = 0$ and the regulator only values uninsured patient surplus.

Table A.9: Cumulative True Medicaid Welfare Effects

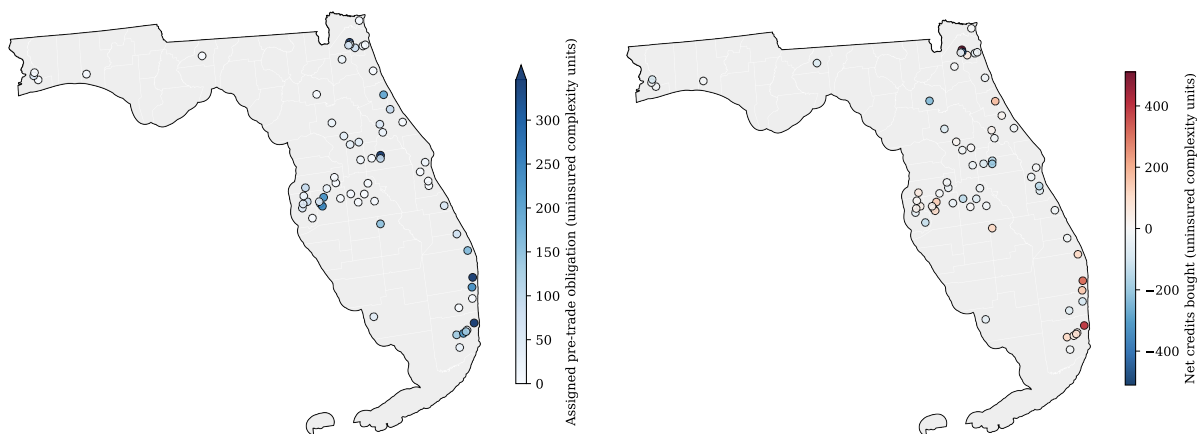
	Regulatory	Δ Welfare (\$100M)				Δ Available Capacity (1k bed-days)				
	Step	Patients		Hospitals	Gov.	Emergency	Scheduled		Idle	
		All	Charity				Insured	Uninsured		(p.p.)
Cumulative Medicaid Expansion										
Medicaid Rates	1	2.943	0.591	1.313	1.949	153.521	−5.286	144.756	−8.605	
↪ + Medicaid Risk Mix	2	5.428	3.224	1.433	1.974	146.767	−4.642	122.872	−7.783	
↪ + Medicaid Networks	3	5.402	3.198	1.429	1.977	148.656	−2.120	118.083	−7.772	
↪ + Change of Service Line	4	5.354	3.206	1.227	1.990	313.908	184.760	−211.698	−8.428	

Notes: Each row is a cumulative endpoint of the Medicaid transition; all changes are relative to the original status quo baseline, not to the preceding row. Welfare is measured in hundreds of millions of dollars. Patient surplus and fiscal cost cover solver choices and fully fixed choices. The Patients columns report the gains for all patients and charitable-care patients, respectively, with “Gov.” representing the change in fiscal cost. Available capacity is the sum across hospitals of the product of each line’s allocated bed-days and its acceptance rate, measured in thousands. The idle capacity ratio is defined as $(\sum_h B_h - \sum_{h,j} a_{hj} b_{hj}) / \sum_h B_h$, and the change is presented in percentage points. The second column reports the transition step. All values are quarterly.



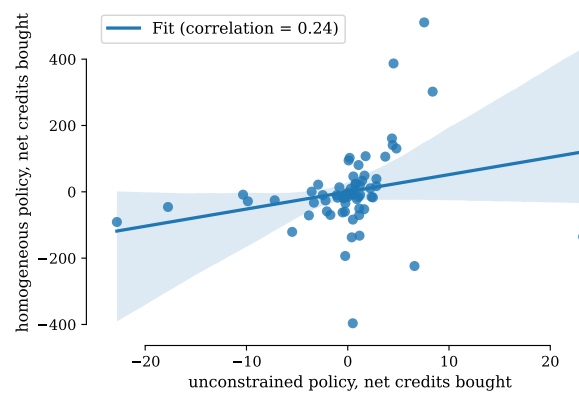
(a) Unconstrained Credit Obligations

(b) Unconstrained Credit Trade



(c) Homogeneous Credit Obligations

(d) Homogeneous Credit Trade



(e) Homogeneous and Unconstrained Correlation

Figure A.13: Floor-and-Trade Credit Trade

Notes: These figures are for 2019Q4. Figure A.13a shows the assigned obligations under the optimal unconstrained floor-and-trade policy, while Figure A.13b shows the trading position in equilibrium of participating hospitals. Figures A.13c and A.13d show the same for the homogeneous policy. Finally, Figure A.13e shows the correlation between trading positions in the two policies.

Table A.10: Tax-at-Risk Sensitivity: Floor-and-Trade Welfare Effects

Tax at risk	Trading	Δ Welfare (\$100M)				Δ Available Capacity (1k bed-days)			
	Price	Patients		Hospitals	Gov.	Emergency	Scheduled		Idle
	z^*	All	Charity				Insured	Uninsured	(p.p.)
Unconstrained									
5%	11.317	2.501	0.543	-0.169	0.000	118.675	-18.918	110.240	-6.167
10%	11.686	2.516	0.545	-0.172	0.000	119.247	-19.013	110.235	-6.181
15%*	11.686	2.516	0.545	-0.172	0.000	119.247	-19.013	110.235	-6.181
20%*	11.686	2.516	0.545	-0.172	0.000	119.247	-19.013	110.235	-6.181
Homogeneous									
5%	0.126	0.203	0.117	-0.002	0.000	6.453	-2.483	22.959	-0.791
10%	0.251	0.310	0.141	-0.002	0.000	13.831	-5.085	25.268	-0.999
15%	0.370	0.432	0.178	-0.003	0.000	21.885	-7.769	29.220	-1.273
20%	0.474	0.613	0.258	-0.018	0.000	30.059	-8.931	76.451	-2.866

Notes: This table shows the effect of increasing the tax-at-risk for non-profit hospitals in the floor-and-trade policies. Welfare and capacity changes are relative to the status quo. All values are quarterly. An asterisk marks an outcome that coincides with the 10% solution at the precision reported. We note that the last two rows in the unconstrained solve are mechanically identical to the 10% solution because the slope of the planner's objective in trading-price space is flat to numerical precision at the 10% solution. However, increasing the trading price above this level yields minimal gains, and the effect appears only in the fifth decimal place of the welfare objectives.

B Descriptive Analysis Appendix

B.1 Spillover Effects on Quality and Distance Traveled

In this subsection, we explore how spillovers from Medicaid expansion to non-expansion states affected patients' quality and distance. Intuitively, if the spillover effects documented in the main text stem from a supply expansion of emergency care at spillover-treated hospitals, then demand should reallocate in favor of patients' preferences. Because patient preferences are not directly identified without further assumptions, a natural proxy is whether admissions occurred closer to home or at better hospitals.

To measure changes in quality, we examine how the average quality of care changes for patients in treated regions. In particular, we use mortality and readmission metrics estimated and published by CMS and examine how the county-level weighted average quality changes with treatment. We weight each quality metric by the number of admissions in the population of interest, to capture the average quality experienced by patients with different insurance coverage. Formally, we compute $\bar{q}_{ict} = \sum_{h \in \mathcal{H}_c} q_{ht} s_{ihct}$, where q_{ht} is the quality outcome of hospital h in period t , $\mathcal{H}_c$ is the set of hospitals found in county c , and s_{ihct} is the share of admissions of patients with insurance source i from county c that are admitted to hospital h at time t . We measure quality at this aggregate level rather than at the hospital level, because quality metrics vary little over time and our interest is in capturing how the reallocation of patients benefited or disadvantaged certain population groups. To account for local changes in overall quality, we explore the changes in quality perceived by those seeking charitable care and those on government insurance, relative to the privately insured. Thus, we define $\Delta \bar{q}_{ict} = \bar{q}_{ict} - \bar{q}_{\text{private},ct}$ and conduct the regressions

$$\Delta \bar{q}_{ict} = \alpha T_{ct} + \gamma_c + \delta_{s(c)t} + \varepsilon_{ict}$$

where T_{ct} indicates that there is at least one hospital in the county that experiences spillover treatment in a period prior to or equal to t , γ_c is a county fixed effect, and $\delta_{s(c)t}$ is a state-by-year fixed effect. Table A.11 shows the estimates.

The estimates indicate that, relative to privately insured individuals, charitable care patients are admitted to hospitals with lower mortality rates across five of six dimensions. While the magnitude of the effects is modest, the direction supports the idea that charitable care patients move toward better hospitals following Medicaid expansion spillovers. This is consistent with hospitals gaining a larger share of the charitable care population by expanding bed-day reservations for emergency-department admissions, and with those patients being more likely to seek charity care.

We now turn our attention to the effect of spillover on distance traveled. To assess whether expanding charitable care benefited charity patients by offering care closer to home, we use the average distance among admitted charitable care patients as the dependent variable in the

Table A.11: Medicaid Spillover Effect on the Quality of Hospitals Admitting Charitable Care Patients

Panel A: Mortality Measures						
	AMI (1)	CABG (2)	COPD (3)	Heart Failure (4)	Pneumonia (5)	Stroke (6)
Spillover Effect (α)	-0.196** (0.096)	0.017 (0.080)	-0.085 (0.064)	-0.089* (0.052)	-0.102** (0.045)	-0.042 (0.057)
<i>N</i>	853	853	853	853	853	853
<i>R</i> ²	0.641	0.465	0.668	0.269	0.642	0.872
Mean Dep.	11.745	1.499	7.930	11.485	11.762	14.066
Panel B: Readmission Measures						
	AMI (1)	CABG (2)	COPD (3)	Heart Failure (4)	Pneumonia (5)	Stroke (6)
Spillover Effect (α)	-0.062 (0.159)	0.030 (0.279)	0.163* (0.096)	-0.053 (0.201)	-0.069 (0.063)	0.111 (0.114)
<i>N</i>	853	853	853	853	853	853
<i>R</i> ²	0.631	0.494	0.678	0.262	0.462	0.658
Mean Dep.	12.857	6.810	19.566	21.409	16.752	11.792

Notes: This table presents the estimated effect of local exposure to spillover effects from Medicaid expansions in non-expansion states on the quality of hospitals accessed by patients seeking charitable care. Quality outcomes are measured by CMS on a rate basis, per 100 patients. The outcome is the county-level average quality for the population relative to the average quality for privately insured patients. All regressions include county fixed effects and state-year fixed effects. Standard errors are clustered at the HRR level. “Mean Dep.” shows the average quality outcome experienced by charitable care patients, $\bar{q}_{ict}$. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

regression of Equation (2). Table A.12 shows that, consistent with the hypothesis above, charitable care patients are the only patient type experiencing a decrease in average distance at spillover hospitals.

Overall, we observe that both distance and quality improved for charitable care patients. This sorting of demand thus appears consistent with elements that might plausibly correlate with their hospital preferences. We note that the interaction between quality and distance might attenuate the effect documented here if higher-quality hospitals are located farther from where the uninsured live. This could help explain the imprecision in the estimated distance effect, as it mixes patients sorting toward quality and away from distance, and vice versa.

Finally, we examine the medical conditions associated with the increase in charitable care at spillover-treated hospitals. We reestimate the stacked difference-in-differences specification in Equation (2) separately for each major diagnostic category (MDC), using total bed-days from charitable care admissions in that category as the outcome. The hospital sample, matched control groups, regression weights, and fixed effects are unchanged. As in the main analysis, we include

Table A.12: Medicaid Spillover Effect on Admitted Distance at Spillover-Treated Hospitals

	Source of Insurance			
	Private	Gov.	Charitable	All
Expansion Effect (α)	0.117 (0.226)	0.332 (0.231)	-0.544* (0.319)	0.116 (0.173)
N	3065	3065	3031	3065
R^2	0.992	0.997	0.971	0.998
Mean Dep.	13.051	11.709	11.659	12.088

Notes: This table presents the estimated spillover effects of Medicaid expansion on spillover-treated hospitals in non-expansion states. The estimates show the effect on the distance patients travel between their ZIP code of residence and the hospital admitting them, as estimated from Equation (2). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

only patients residing in the hospital's state and cluster standard errors at the hospital and event levels.

Table A.13 shows statistically significant increases in charitable care across several diagnostic categories, including HIV, neoplasms, blood and immune disorders, respiratory conditions, digestive conditions, and hepatobiliary and pancreatic conditions. The estimates for circulatory conditions and pregnancy are not statistically distinguishable from zero, but their imprecision does not rule out increases in these categories. Together with the quality and distance patterns above, these results are consistent with expanded access to charitable care at hospitals better aligned with patients' preferences.

Table A.13: Medicaid Expansion Effects on Charitable Care by MDC

	Nervous	Eye	ENT	Respiratory	Circulatory
Expansion Effect (α)	52.959 (45.835)	3.547 (2.259)	18.137*** (6.978)	152.857*** (42.371)	154.113 (99.162)
Mean Dep.	261.259	9.593	61.222	405.370	533.148
N	3065	3065	3065	3065	3065
R ²	0.930	0.871	0.949	0.941	0.939
	Digestive	Hep./Pancreas	MSK	Skin/Breast	Endocrine
Expansion Effect (α)	147.007*** (53.025)	88.712*** (31.990)	22.634 (37.997)	38.772 (26.075)	41.352 (31.156)
Mean Dep.	550.333	286.037	227.370	245.704	171.519
N	3065	3065	3065	3065	3065
R ²	0.968	0.959	0.964	0.969	0.936
	Kidney/Urinary	Male Repro.	Female Repro.	Pregnancy	Newborns
Expansion Effect (α)	66.676** (30.672)	0.339 (4.151)	36.169** (16.879)	9.083 (26.156)	91.677 (88.304)
Mean Dep.	173.741	22.185	61.852	71.481	145.815
N	3065	3065	3065	3065	3065
R ²	0.934	0.783	0.898	0.938	0.726
	Blood/Immune	Neoplasms	Infectious	Mental Health	Substance Use
Expansion Effect (α)	30.130** (12.152)	29.676** (13.631)	110.223 (84.794)	23.492 (77.038)	26.505 (23.768)
Mean Dep.	57.741	37.852	249.889	101.889	93.111
N	3065	3065	3065	3065	3065
R ²	0.888	0.878	0.887	0.858	0.853
	Injuries	Burns	Health Factors	Trauma	HIV
Expansion Effect (α)	15.686 (15.105)	1.833 (2.764)	65.434 (44.866)	-33.692 (27.464)	38.224** (19.172)
Mean Dep.	101.000	1.926	26.630	12.889	28.704
N	3065	3065	3065	3065	3065
R ²	0.953	0.964	0.665	0.878	0.815

Notes: This table presents the estimated effect of Medicaid expansion on spillover-treated hospitals. Each cell represents an estimate from an independent regression, where the dependent variable is the total length of stay for charitable care admissions associated with the given MDC. ENT refers to Ears, Nose, and Throat. MSK refers to the musculoskeletal system. Hep refers to the hepatobiliary system. All regressions include admissions only from patients residing in the same state as the associated hospital. Standard errors are clustered at the hospital and event levels (two-way). * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.2 Spillover Effects on Access and Crowdouts

This subsection supplements the evidence on demand reallocation in Section 4. We examine how spillovers affected total care and its composition at hospitals in non-expansion states, distinguishing spillover-treated hospitals from local untreated hospitals. We also report alternative outcome measures and the alternative local-hospital sample referenced in the main text.

B.2.1 Spillover-Treated Hospitals. For spillover-treated hospitals, we use the stacked regression design of Equation (2), retaining the main analysis's sample, matched controls, weights, and fixed effects. We include only admissions by patients residing in the hospital's state and cluster standard errors at the hospital and event levels.

Panel A of Table A.14 reports effects on total, emergency, and non-emergency care, measured in bed-days.¹ The estimated change in total care is small relative to baseline volume and statistically insignificant. The point estimates indicate an increase in emergency care and a decrease in non-emergency care, but both are imprecise.

Panel B examines insurance composition using shares of bed-days. The overall privately insured share falls by 2.6 percentage points, while the estimated change in the Medicare and Medicaid share is statistically insignificant. Within emergency care, the charitable care share rises by 1.3 percentage points, and the private share falls by 2.7 percentage points, with no statistically significant change in the government-insured share. Within non-emergency care, the charitable care share does not change significantly, while the private share falls and the government-insured share rises.

Together, these patterns suggest substitution toward charitable care patients within spillover-treated hospitals. Given that Medicaid expansions are unlikely to reduce insured demand in non-expansion states and insured patients are more profitable to serve, this substitution is consistent with hospitals facing resource constraints.² The volume estimates are also consistent with resources moving from non-emergency to emergency care, although they do not establish that reallocation precisely. The relative protection of Medicare and Medicaid admissions is consistent with non-profit hospitals seeking to maintain good standing in their reported community benefits, which include these services.

B.2.2 Local Untreated Hospitals. Panel C of Table A.14 reports the corresponding changes at local untreated hospitals more than 5 miles and no more than 15 miles from a spillover-treated

¹HCUP classifies the emergency status of an admission based on the patient's need for immediate care, and this classification covers more than visits to the emergency department. Conceptually, we consider all patients in need of emergency care in hospitals with an emergency department (ED) to be protected by EMTALA, since even those not routed through the ED could re-enter the hospital via the ED to obtain regulatory protection.

²Otherwise, a hospital expanding charitable care would retain its profitable privately insured patients.

hospital. Following Equation (4), these estimates include hospital and state-year fixed effects, with standard errors clustered at the hospital referral region level. Charitable care declines by approximately 1,009 bed-days, while privately insured care increases by approximately 1,682 bed-days; the latter estimate is less precise. Estimated increases in Medicare and Medicaid care, and in total care, are statistically insignificant. These changes mirror the patient reallocation described in the main text. They are reduced-form responses to spillover exposure; Section 4.4 separately estimates the effect of an additional charitable care admission using the spillover instrument.

Table A.14: Spillover Effects on Admissions and Insurance Composition

Panel A: Spillover-Treated Hospital Admissions (bed days)									
	Total Admitted			Emergency Admissions			Non-Emergency Admissions		
	(1)			(2)			(3)		
Expansion Effect (α)	486.549 (3654.607)			698.406 (2882.950)			-211.857 (2158.025)		
N 3065	Mean Dep.	72637.6	R ² 0.996	Mean Dep.	44389.4	R ² 0.992	Mean Dep.	28248.2	R ² 0.987
Panel B: Spillover-Treated Hospitals' Insurance Composition (share of bed days)									
	Total Admitted		Emergency Admissions			Non-Emergency Admissions			
	Gov.	Private	Charitable	Private	Gov.	Charitable	Private	Gov.	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	
Expansion Effect (α)	0.007 (0.014)	-0.026*** (0.009)	0.013** (0.006)	-0.027** (0.012)	0.008 (0.016)	0.003 (0.004)	-0.032*** (0.008)	0.023* (0.013)	
Mean Dep.	0.737	0.181	0.076	0.139	0.761	0.030	0.248	0.691	
N	3065	3065	3060	3060	3060	3065	3065	3065	
R ²	0.876	0.914	0.880	0.835	0.793	0.825	0.926	0.907	
Panel C: Local Untreated Hospital Admissions (bed days)									
	Charitable		Gov.		Private		Total		
	(1)		(2)		(3)		(4)		
Expansion Effect (α)	-1009.057** (424.829)		1173.584 (1604.996)		1681.869* (909.450)		2807.092 (1963.974)		
N 1156	Mean Dep.	8442.1	R ² 0.944	Mean Dep.	63565.6	R ² 0.988	Mean Dep.	23601.6	R ² 0.974
							Mean Dep.	98385.3	R ² 0.992

Notes: This table presents the effects of Medicaid expansion in non-expansion states. Panels A and B show effects on spillover-treated hospitals, as estimated from Equation (2) with standard errors clustered at the event and hospital levels (two-way). Panel C shows the effect on local untreated hospitals within 15 miles of a spillover-treated hospital but more than 5 miles away. These are estimates of Equation (4), with standard errors clustered at the HRR level. Table A.15 shows a variant with Panel A in logs, Panel B measured in bed-days, and Panel C without the 5-mile exclusion and restricted to system hospitals. "Gov." represents admissions from Medicare and Medicaid enrollees. All demand metrics incorporate only demand from patients residing in the same state as the affected hospitals. "Mean Dep." in Panels A and B is the treated-unit mean in the year before expansion treatment. "Mean Dep." in Panel C is the unweighted mean among exposed hospitals in the year before their first local exposure; hospitals without an observed outcome in that baseline year are omitted from this mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.2.3 Alternative Outcomes and Local Hospital Sample. Table A.15 presents the alternative specifications referenced in the main text. For spillover-treated hospitals, Panel A measures care in log bed-days and Panel B measures care by insurance source in bed-days rather than

shares. The estimated changes in total, emergency, and non-emergency volume remain statistically insignificant. In levels, charitable care increases in both emergency and non-emergency care, with a larger point estimate for emergency care, while the estimated declines in privately insured care are imprecise.

For local untreated hospitals, Panel C drops the 5-mile exclusion and restricts the sample to system hospitals within 15 miles of a spillover-treated hospital. These two sample changes are applied jointly. Charitable care falls by approximately 1,116 bed-days, and privately insured care rises by approximately 2,393 bed-days, preserving the direction of the baseline reallocation patterns. The estimates for government-insured care and total care remain imprecise.

Table A.15: Spillover Effects: Alternative Outcomes and Local Hospital Sample

Panel A: Spillover-Treated Hospital Admissions (log bed days)												
		Total Admitted			Emergency Admissions				Non-Emergency Admissions			
		(1)			(2)				(3)			
Expansion Effect (α)		0.016 (0.046)			0.198 (0.137)				−0.092 (0.071)			
N 3065	Mean Dep.	10.643	R ²	0.956	Mean Dep.	9.873	R ²	0.948	Mean Dep.	9.610	R ²	0.939
Panel B: Spillover-Treated Hospitals’ Insurance Composition (bed days)												
		Total Admitted		Emergency Admissions			Non-Emergency Admissions					
		Gov. (1)	Private (2)	Charitable (3)	Private (4)	Gov. (5)	Charitable (6)	Private (7)	Gov. (8)			
Expansion Effect (α)		345.021 (2316.662)	−1473.660 (1453.212)	845.229* (486.538)	−861.456 (725.756)	411.406 (2015.353)	407.643* (229.791)	−612.204 (884.010)	−66.385 (1224.427)			
Mean Dep.		50885.332	15247.444	3163.407	6831.593	32872.371	774.852	8415.852	18012.963			
N		3065	3065	3065	3065	3065	3065	3065	3065			
R ²		0.996	0.989	0.952	0.986	0.992	0.913	0.986	0.984			
Panel C: Local Untreated Hospital Admissions (bed days)												
		Charitable		Gov.		Private		Total				
		(1)		(2)		(3)		(4)				
Expansion Effect (α)		−1116.019* (592.085)		1146.643 (2200.531)		2392.507** (1090.029)		2911.283 (2791.728)				
N 724	Mean Dep.	8189.415	R ²	0.948	Mean Dep.	60137.535	R ²	0.987	Mean Dep.	26920.732	R ²	0.977
									Mean Dep.	98281.438	R ²	0.992

Notes: This table presents the estimated effect of Medicaid expansion on hospitals in non-expansion states. It complements Table A.14. Panel A shows the total change in admissions, measured in log bed-days. Panel B shows admissions (measured in bed-days) by insurance source, with “Gov.” denoting the sum of Medicare and Medicaid. Panels A and B are estimates of Equation (2) with standard errors clustered at the event and hospital levels (two-way). Panel C shows the effect on local untreated hospitals, defined as all system hospitals located within 15 miles of a spillover-treated hospital. The estimates use Equation (4), with standard errors clustered at the HRR level. All admissions are from patients residing within the same state as the associated hospital. “Mean Dep.” in Panel C is the unweighted mean among exposed hospitals in the year before their first local exposure; hospitals without an observed outcome in that baseline year are omitted from this mean. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$

B.3 Private Revenue Proxy

Hospital cost reports do not separate inpatient revenue by payer, except for Medicare. Column 4 of panel B in the main text Table 2 therefore uses a residual proxy in the spirit of Dafny (2009): we value all inpatient care at the hospital's own realized payment rate and subtract what can be attributed to public payers. All inputs are hospital fiscal-year items from CMS Form 2552-10. Let C_{hf} be inpatient charges, the sum of general routine, intensive-care, and ancillary charges (Worksheet G-2, column 1, lines 1, 16, and 18), and let T_{hf} and N_{hf} be total patient charges and net patient revenue (Worksheet G-3, column 1, lines 1 and 3). We impute inpatient net revenue as $\rho_{hf}C_{hf}$, where $\rho_{hf} = N_{hf}/T_{hf}$ is the hospital's realization rate, and subtract Medicare primary-payer payments and amounts payable (Worksheet E, Part A, column 1, lines 60 and 61) together with Medicaid net revenue D_{hf} (Worksheet S-10, line 2), scaled by the inpatient share of charges because the S-10 item covers all settings:

$$\widehat{R}_{hf}^{\text{priv}} = \underbrace{\rho_{hf}C_{hf}}_{\text{inpatient net revenue}} - \underbrace{\left(M_{hf}^{\text{pay}} + M_{hf}^{\text{payable}} + \frac{C_{hf}}{T_{hf}}D_{hf}\right)}_{\text{attributed to Medicare and Medicaid}}. \quad (\text{A.1})$$

We align fiscal and calendar years by weighting each report by the fraction of its period falling within the year, keeping only hospital-years whose reports cover the year without overlap and with no missing components. We deflate by the GDP deflator, require ρ_{hf} and the inpatient charge share to lie in $[0, 1]$ and the residual to be non-negative, and cap the result at its 97.5th percentile. These screens bind: the two payer columns are estimated on 537 hospital-years against 977 for the other financial outcomes, and the surviving sample tilts toward larger hospitals filing complete reports. The Medicare column is not a residual; it is reported inpatient reimbursement (Worksheet E, Part A, column 1, line 71).

Equation (A.1) assigns to private insurers everything the two public terms miss. Medicare Advantage is generally not booked on the traditional Medicare lines of Worksheet E and so falls into the residual, and its share grew over our period; TRICARE, the VA, and workers' compensation are included as well; and, most relevant here, so are direct payments from uninsured patients and recoveries on previously uncompensated accounts, which leaves the proxy not cleanly orthogonal to the regressor of Table 2. The Medicaid subtraction is itself noisy, since S-10 reporting is uneven across states and mixes fee-for-service with managed care. The imputation adds two further approximations: ρ_{hf} is facility-wide, so applying it to inpatient charges assumes that inpatient and outpatient care are discounted alike and that discounts do not vary by payer, when in fact public discounts are deepest; and scaling Medicaid by the inpatient charge share assumes its inpatient-outpatient split matches the all-payer split. Neither assumption is testable in these data, and neither is innocuous: across full-year filings in our period, ρ_{hf} has a median of 0.29, with a tenth-to-ninetieth percentile range of 0.16 to 0.46, and the residual is roughly half of imputed

inpatient net revenue at the median, so modest errors in ρ_{hf} move the proxy substantially.

The proxy is a revenue flow rather than a price: it moves with admitted volume, case mix, and charge-setting as well as negotiated rates, and scale dominates its variation, with the correlation between its logarithm and log inpatient charges at 0.91. Dranove et al. (2026) evaluate residual proxies of this family against transaction prices and find moderate cross-sectional but very limited time-series correlation, and because our specification absorbs hospital fixed effects, it leans on precisely that time-series variation. We therefore read column 4 as evidence on where the revenue loss falls, through its sign and rough magnitude, rather than as a precise decomposition or as a statement about commercial prices. The proxy is confined to these reduced-form results. It plays no role in the structural analysis, which reconstructs Florida private prices from the Health Price Finder files and Medicare and Medicaid payments from each year's payment rules, as described in Section 3.2.

B.4 Spillover Instrument

This subsection describes the construction of the spillover instrument used in the main text and the arguments behind its validity. We build our main spillover instrument in three steps. First, we measure the intensity of each spillover event by the charitable care spending the system had in expansion states in 2013. Second, we distribute the intensity of each spillover-treated hospital among untreated hospitals within 15 miles. We exclude those found within five miles to avoid noise from local transfer agreements and local shocks. In distributing the spillover, we give each local untreated hospital a share inversely proportional to its distance to the spillover-treated hospital. Third, we accumulate the exposure share at each local untreated hospital to form its unique spillover-exposure amount. In estimation, we further interact the instrument with the local untreated hospital's ownership status to account for how public hospitals' response to spillover might differ from that of privately owned non-profits.

Our estimates concern the effects of spillover-induced charitable care reallocation on local untreated hospitals, allowing insured patients and hospital resources to adjust. They do not isolate the effect of changing a hospital's charitable admissions while holding its neighbors' operations fixed. Formally, let U_{ht}^0 and y_{ht}^0 denote charitable admissions and an outcome at hospital h in year t in the absence of the spillover-induced changes in neighboring systems' charitable care supply, and let U_{ht} , y_{ht} denote the outcomes with spillovers. Both scenarios allow the local market to reach its corresponding equilibrium, holding other contemporaneous market primitives fixed. Define the spillover changes, $\Delta U_{ht} = U_{ht} - U_{ht}^0$ and $\Delta y_{ht} = y_{ht} - y_{ht}^0$. Using the local-exposure indicator T_{ht} defined in the main text, the average equilibrium effect on exposed local untreated hospitals, normalized by their average change in charitable admissions, is

$$\rho_T^{\text{eq}} = \frac{\mathbb{E}[\Delta y_{ht} \mid T_{ht} = 1]}{\mathbb{E}[\Delta U_{ht} \mid T_{ht} = 1]}. \quad (\text{A.2})$$

Expectations are over hospital-years in the estimation population. Since exposure reduces charitable admissions, the ratio expresses the outcome change per additional admission when comparing the same equilibria in reverse.

Given the evidence presented in the main text, we assume no anticipation and that the exposure captures spillovers, so that $\Delta U_{ht} = \Delta y_{ht} = 0$ when $T_{ht} = 0$. This requires that comparison hospitals do not experience omitted geographic or system spillovers. Equilibrium outcomes must be well defined, either by uniqueness or a stable selection rule. Exclusion requires expansion exposure to affect local untreated hospitals through changes in neighboring systems' charitable care supply and the resulting market adjustment, rather than through independent changes in local coverage, reimbursement, technology, or financial transfers. Insured patients redirected by a neighbor's charitable care expansion are therefore part of the target effect. Exclusion does not require this response to operate through the focal hospital's own admission count.

The exclusion restriction requires no exposure-related counterfactual trends in charitable admissions and the outcome. Baseline charitable spending and geographic proximity limit contemporaneous reverse causality but do not establish these restrictions; as in exposure-based designs, differential exposure must be unrelated to omitted counterfactual changes (Goldsmith-Pinkham et al., 2020). Restricting the sample to non-expansion states and hospitals outside treated systems limits direct coverage and own-system channels, while the acquisition exclusions limit merger-related confounding. Relevance, in turn, requires a nonzero residual first stage, assessed using Appendix Table A.2, and the normalization requires $E[\Delta U_{ht} | T_{ht} = 1] \neq 0$. To interpret the denominator as admissions displaced, we additionally assume $\Delta U_{ht} \leq 0$, with strict inequality for some exposed hospitals. The opposing charitable care movements in Table 1, together with the absence of a statistically detectable state-level increase, support the reallocation mechanism.

To connect the desired β_T^{eq} to the 2SLS estimate, denote v_{ht} as the population first-stage fitted value of residualized U_{ht} using excluded instruments $\tilde{Z}_{ht}$. Under the preceding orthogonality restrictions, finite moments, and relevance, the population 2SLS coefficient satisfies

$$\beta^{\text{2SLS}} = \frac{E[v_{ht} \Delta y_{ht} | T_{ht} = 1]}{E[v_{ht} \Delta U_{ht} | T_{ht} = 1]}. \quad (\text{A.3})$$

Consequently, it equals β_T^{eq} if and only if

$$\text{Cov}(v_{ht}, \Delta y_{ht} - \beta_T^{\text{eq}} \Delta U_{ht} | T_{ht} = 1) = 0. \quad (\text{A.4})$$

A common per-admission equilibrium response, $\Delta y_{ht} = \beta_T^{\text{eq}} \Delta U_{ht}$, is sufficient; the covariance restriction permits heterogeneity but requires the additional IV weighting not to select systematically on deviations from the treated-average response. This is an assumption about total equilibrium responses, not homogeneous clinical costs. Without it, the coefficient is an IV-weighted ratio of equilibrium effects, potentially with signed weights, rather than the normalized average effect on

all exposed hospitals.³

C Model Appendix

This appendix presents developments associated with the model described in Section 5.

C.1 Admission Rate Approximation

Our main model assumes that hospitals set the admission rate rather than control a queuing process. In this appendix, we present two models of hospital behavior in which the hospital line controls a queue rather than the admission rate directly. The first model is simpler and yields the same payoffs as our main model, while the second is more realistic and yields payoffs arbitrarily close to those of our main model. We present the model below for a single hospital, service line, period, and market, and omit the associated subindices. We begin by restating our main model in a simplified way that captures the core economics of the problem and minimizes the notational burden of our empirically oriented model.

Main Model: In this model, all patients arrive at the hospital at the beginning of the period, and the number of admission requests follows a Poisson distribution with rate λ . The line has capacity b , and each admitted patient yields a profit of v and occupies a bed for the full period. The hospital decides in advance on a fraction of patients to admit, a , and pays a rental cost r for any additional capacity it requires to serve its admitted population. As in our main model, the implied admission count is Poisson with rate $a\lambda$. Without loss of generality, we can consider the hospital choosing the admission rate $x = a\lambda$, subject to $x \in [0, \lambda]$. The expected profit of the hospital is

$$\pi^{main}(x) = vx - r\mathbb{E}[(N(x) - b)_+]$$

where $N(x)$ is the number of admitted patients, and $(y)_+$ denotes the positive part of y .

Posted Wait-Time Model: In this simple queuing model, the hospital manages a queue by posting a wait time $w \geq 0$, indicating how long patients must wait for service. Patients arrive continuously according to a Poisson process at rate λ and, upon requesting admission, draw a patience time T_i , which indicates the maximum time they are willing to wait for service. Therefore, a patient waits in the queue only if $T_i \geq w$. The hospital has b beds, and each admitted patient stays for one unit of time. Any patient who must be admitted when no beds are available (due to wait-time commitments) must be served in a rented bed. As before, the hospital earns a profit of v per admission and rents beds at a rate of r . Denote $a(w) = \mathbb{P}(T_i \geq w)$ and assume that T_i is i.i.d. in the population with positive density on the positive reals. Then the number of patients in

³A convex average of hospital-level responses $\Delta y_{ht}/\Delta U_{ht}$ additionally requires $\Delta y_{ht} = 0$ whenever $\Delta U_{ht} = 0$ and non-negative weights proportional to $v_{ht}\Delta U_{ht}$. These conditions are not required for the normalized target in Equation (A.2), but are needed for this more restrictive average-response interpretation.

the committed queue is Poisson with rate $x(w) \equiv a(w)\lambda$. In stationarity, the number of patients requiring beds at any instant is the number of committed services that start in the previous unit of time, which is a Poisson random variable of rate $x(w)$. Therefore, the hospital experiences the same occupancy demand per period as in our main model. Since the patience distribution is supported on the positive real line, the hospital can indirectly control the rate $a(w) \in [0, 1]$ by adjusting the posted wait time. Hence, this model yields the same payoff as our main model.

Despite its simplifications, this model more closely reflects real hospital behavior. In practice, patients arrive continuously over time and are given wait times rather than being admitted at random. While wait times are typically not uniform, the homogeneity of w can be treated as a plug-in approximation of the mean wait time, which patients use to decide whether to commit to the queue and which hospitals use to simplify their analysis. The main caveat, however, is that at the beginning of the period, beds sit idle because every patient is made to wait even when capacity is available. The next model corrects this issue.

FCFS with Wait Time Commitment Model: This model has the same structure as the *Posted Wait-Time Model* except that hospitals commit to having patients wait at most $w \geq 0$ units of time before service, but treat patients as soon as capacity is available on a first-come, first-served (FCFS) basis. As before, patients join the queue only if their patience exceeds the posted wait-time guarantee (i.e., they are pessimistic about wait times) and occupy a bed for a full unit of time. Under this model, profits net of rental costs are the same as in the posted wait-time model in stationarity: patients commit to the queue at rate $x(w)$, and thus, in expectation, the hospital is paid $vx(w)$. The main difference is the rental cost, since the hospital uses idle capacity more effectively with an FCFS queue.

The difference in rental cost between the main model, which serves patients immediately, and the FCFS model, which queues patients, depends on how likely a patient is to occupy a resource at any given time. To quantify this, fix a time t and consider the indicator that marks whether she is occupying a bed under the FCFS model, versus the indicator that marks her occupancy under the main model. First, if the patient arrived between $t - w$ and t , the main model might mark her as using a resource, while the FCFS model might have her waiting for service. Alternatively, the patient might have arrived between $t - 1 - w$ and $t - 1$, which implies that the immediate service model already served and discharged the patient, while the FCFS model is still serving her. Each window has length w , and patients arrive at rate $x(w)$ per period, so the expected number of patients subject to these discrepancies is $2x(w)w$. Now, since for any y, z, b we have $|(y - b)_+ - (z - b)_+| \leq |y - z|$, the discrepancy between the models' overflow in a period of length one is bounded above by $2x(w)w$. Therefore, denoting $\pi^{FCFS}(w)$ as the expected profit of the firm under the FCFS model, we have that

$$|\pi^{FCFS}(w) - \pi^{main}(x(w))| \leq 2rx(w)w.$$

Now suppose that $\mathbb{P}(T_i \geq w) = \frac{1}{1+\eta w}$ (i.e., patience is distributed Pareto type II). Then $x(w) = \lambda/(1 + \eta w)$, so $w x(w) = (\lambda - x(w))/\eta \leq \lambda/\eta$. Hence,

$$|\pi^{FCFS}(w) - \pi^{main}(x(w))| \leq \frac{2r\lambda}{\eta}.$$

Therefore, the profit discrepancies between the two models decay to zero as consumer patience becomes small ($\eta \rightarrow \infty$).

Discussion: These two models show clearly how our admission-rate control model approximates a queue-control model. In particular, admission-rate control yields payoffs similar to those from controlling patients' wait time by managing queues and rented capacity. However, our admission-rate control model implies a significant simplification relative to a queue management model once we incorporate competition. In estimation, a queue model requires us to distinguish between patients who are admitted immediately into available capacity and patients who must wait. Without wait-time data, we cannot resolve this without stronger assumptions. Moreover, under competition, it is natural to assume that patients' patience is not exogenous but dictated by the wait times offered at rival hospitals. This links demand across firms in a complex way, making analysis intractable without further simplifications or extensive simulation. Hence, our approach offers a midpoint between realism and tractability. As our estimates show, the model yields accurate cost estimates for hospitals, suggesting that our approximations have a modest impact on characterizing optimal resource allocation.

C.2 Poisson Rate Approximation

In this section, we discuss briefly why we consider the assumption of a Poisson illness process to be mild in our setting. In what follows, we take the independence assumption as given and discuss only the difference between our model and the standard model, which treats each illness draw as a Bernoulli random variable.

First, we note that the population in the empirical application of hospital demand is often very large. For example, in our empirical analysis, we consider Florida, which, in 2019, had a population of over 21.2 million. Second, relative to the population, hospitalizations are rare, with about 2.6 million hospitalization events at general hospitals. By the Poisson limit theorem (Feller, 1968), if n is the size of the population and p_n is the probability of illness, then as $p_n \rightarrow 0$ and $np_n \rightarrow \lambda$, the distribution of the number of illnesses at any period converges to a Poisson distribution of rate λ .

To illustrate the precision of this assumption, we simulate admission decisions in a market calibrated to Madison, WI. We simulate a large sample of potential patients, each choosing hospitals based on distance and some hospital-specific quality. Following our demand model, we assume patients differ in their distaste for distance or out-of-market consumption, with individual characteristics drawn independently from log-normal distributions. Patients choose among the four

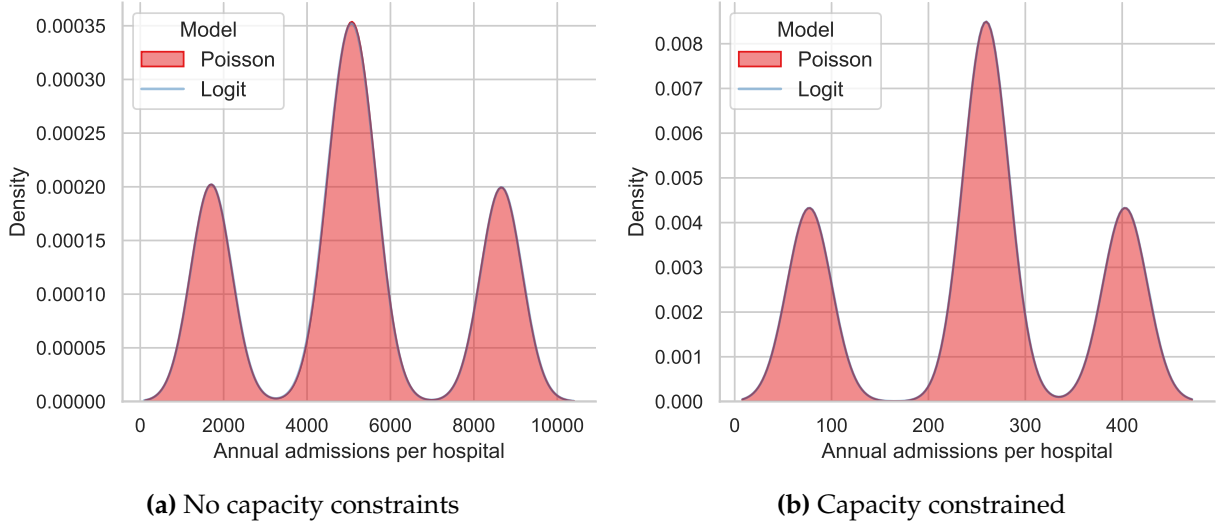


Figure A.14: Admission rate comparison between multinomial logit model and Poisson

Notes: These figures compare patient arrival rates at hospitals in an environment calibrated to approximate the city of Madison, WI. The distribution is based on 800 Monte Carlo replicates. The simulated city has a population of 283,000 and four large hospitals with capacities of 672, 440, 426, and 129. We assume a 60% capacity factor and a 9% probability of needing inpatient care. In each simulation, we randomly locate consumers and hospitals across a 9×9 grid, representing square miles in the city. The logit and Poisson are nearly overlapping, making the logit distribution appear as the contour of the Poisson distribution.

general inpatient hospitals located in the city and an out-of-market option. Figure A.14 shows the arrival distribution implied by the true model with and without a capacity constraint. As shown, the distributions are almost identical, as expected from Poisson's law.

C.3 Derivations

Here, we derive Equation (10). Given that the derivation is for a single market period, we omit all t and m indices. We begin by deriving the uninsured surplus, then derive the expectations of the other terms.

Let $\mathcal{K}_{hj}^u$ denote the subset of consumer types that are uninsured and require charity care. The

expected surplus of the uninsured population is:

$$\begin{aligned}
\mathbb{E}[S_{hj}^u(a_{hj})] &= \sum_{k \in \mathcal{K}_{hj}^u} \lambda_k \left[\mathbb{E}_{i,a} [\ln(1 + \sum_{h' \in \mathcal{H}} \tilde{a}_{ih'} e^{v_{ih'}}) | (m_i, d_i) = (m_k, d_k)] + \gamma^{euler} \right] \\
&= \sum_{k \in \mathcal{K}_{hj}^u} \lambda_k \left[\mathbb{E}_{i,a_{-h}} [a_{hj} \ln(1 + e^{v_{ih}} + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{ih'} e^{v_{ih'}}) \right. \\
&\quad \left. + (1 - a_{hj}) \ln(1 + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{ih'} e^{v_{ih'}}) | (m_i, d_i) = (m_k, d_k)] + \gamma^{euler} \right] \\
&= \sum_{k \in \mathcal{K}_{hj}^u} \lambda_k \left[\mathbb{E}_{i,a_{-h}} [a_{hj} \ln(\frac{1 + e^{v_{ih}} + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{ih'} e^{v_{ih'}}}{1 + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{ih'} e^{v_{ih'}}}) + \ln(1 + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{ih'} e^{v_{ih'}}) | (m_i, d_i) = (m_k, d_k)] + \gamma^{euler} \right] \\
&=: a_{hj} \Delta S_{hj}^u + S_{j,-h}^u.
\end{aligned}$$

Since $S_{j,-h}^u$ does not depend on hospital h 's actions, we omit it from the objective in Equation (10).

We now proceed with the remaining terms. To recall, we seek to simplify

$$\mathbb{E}[\sum_{k \in \mathcal{K}_j} n_{hk} (p_{hk} - c_{hk}) - c_{hj}^V N_{hj}^2 - \exp(c_h^o (N_{hj}^L(a) - b))].$$

First, by the Poisson distribution of n_{hk} , we have that $\mathbb{E}[n_{hk}] = a_{hj} \lambda_k R_{hk}$. Second, recall that N_{hj} is distributed Poisson with rate $a_{hj} \Lambda_{hj}$. Therefore, by the standard expression for the second uncentered moment of a Poisson random variable, we have that $\mathbb{E}[N_{hj}^2] = a \Lambda_{hj} + a^2 \Lambda_{hj}^2$. Finally, the term $\mathbb{E}[\exp(c_h^o (N_{hj}^L(a) - b))]$ can be derived by noting that it is closely related to the moment generating function of the random variable $N_{hj}^L(a)$, which is distributed as a compound Poisson. In particular, we have that $N_{hj}^L = \sum_{i=1}^{N_{hj}} \sum_{k \in \mathcal{K}_j} \frac{\lambda_k R_{hk}}{\Lambda_{hj}} e_i^k$ and thus that the moment-generating function of the length of stay of a randomly chosen admitted patient is $M_{hj}^L(t) = \sum_{k \in \mathcal{K}_j} \frac{\lambda_k R_{hk}}{\Lambda_{hj}} \mathbb{E}[e^{t \ell_i^k}]$. Therefore, $\mathbb{E}[e^{t N_{hj}^L} | N_{hj} = n] = (M_{hj}^L(t))^n$, and thus

$$\mathbb{E}[e^{t N_{hj}^L}] = \sum_{n=0}^{\infty} e^{-a_{hj} \Lambda_{hj}} \frac{(a_{hj} \Lambda_{hj})^n}{n!} (M_{hj}^L(t))^n = \exp(a_{hj} \Lambda_{hj} (M_{hj}^L(t) - 1))$$

where the second equality follows from the series formula for the exponential. Finally, applying it to the term $\mathbb{E}[\exp(c_h^o (N_{hj}^L(a) - b))]$ yields $\exp(a \Lambda_{hj} (M_{hj}^L(c_h^o) - 1) - c_h^o b)$, which is the expression shown in the main text.

C.4 Model Properties

This section proves Proposition 1 from the main text and discusses sufficient conditions for equilibrium uniqueness. We work in a single period and market, with finitely many hospitals, service lines, and consumer types. We maintain the set of assumptions stated in the main text model section and at the beginning of the proposition throughout this analysis.

For compactness, denote the linear and quadratic payoff coefficients in Equation (10) by

$$\begin{aligned}\pi_{hj}^L &:= \sum_{k \in \mathcal{K}_j} \lambda_k R_{hk} (p_{hk} - c_{hk}) - c_{hj}^V \Lambda_{hj} + \tau_{hj} \Delta S_{hj}^u, \\ \pi_{hj}^V &:= c_{hj}^V \Lambda_{hj}^2.\end{aligned}\tag{A.5}$$

For the first three items in the proposition, we prove the result for a single hospital, holding rivals' admission rates fixed; hence, we suppress the h index except in the set $\mathcal{J}_h$. We restore it when proving equilibrium existence and discussing uniqueness. Moreover, many of the properties pertain to a single service line; hence, we suppress the j index unless it is relevant for clarification. The line's problem can then be expressed as

$$\pi(b) = \max_{a \in [0,1]} a\pi^L - a^2\pi^V - \exp(a\chi - c^o b) \quad \text{s.t.} \quad a\Lambda\mu^L \leq b.$$

Here π^L is finite and $\pi^V, c^o, \chi > 0$. In particular, the moment condition and positive length of stay imply $M^L(c^o) - 1 \geq c^o\mu^L > 0$. We use this simplified notation below.

Proof of Proposition 1. We start by deriving the closed-form expressions for $a^o(b)$ and $a^*(b)$, which are useful for the empirical analysis but unused in the rest of the proof. For $a^o(b)$, we relax both the interval restriction on a and the ex-ante capacity constraint, so the maximization is over $a \in \mathbb{R}$. Its first-order condition is

$$\pi^L - 2a\pi^V - \chi \exp(a\chi - c^o b) = 0.$$

Define $y = \pi^L - 2\pi^V a$; then the first-order condition simplifies to

$$\begin{aligned}
y &= \chi \exp\left(\chi \frac{\pi^L - y}{2\pi^V} - c^o b\right) \\
y \exp\left(\frac{\chi y}{2\pi^V}\right) &= \chi \exp\left(\frac{\chi \pi^L}{2\pi^V} - c^o b\right) \\
\frac{\chi y}{2\pi^V} \exp\left(\frac{\chi y}{2\pi^V}\right) &= \frac{\chi^2}{2\pi^V} \exp\left(\frac{\chi \pi^L}{2\pi^V} - c^o b\right) \\
\frac{\chi y}{2\pi^V} &= W_0\left(\frac{\chi^2}{2\pi^V} \exp\left(\frac{\chi \pi^L}{2\pi^V} - c^o b\right)\right) \\
\Rightarrow a^o(b) &= \frac{\pi^L}{2\pi^V} - \frac{1}{\chi} W_0\left(\frac{\chi^2}{2\pi^V} \exp\left(\frac{\chi \pi^L}{2\pi^V} - c^o b\right)\right),
\end{aligned}$$

where W_0 is the principal real branch of the Lambert W function. Its argument is strictly positive. The derivative of the objective with respect to a is strictly decreasing from $+\infty$ to $-\infty$, so this first-order condition has exactly one solution, which is the unconstrained maximizer. Restricting the solution to the feasible interval gives

$$a^*(b) = \min\left\{1, \frac{b}{\Lambda \mu^L}, \max\{0, a^o(b)\}\right\}.$$

This concludes the preliminary part of the proposition.

Denote the objective in the maximization by $\tilde{\pi}(a, b) = a\pi^L - a^2\pi^V - \exp(a\chi - c^o b)$. The Hessian of the objective is

$$\nabla^2 \tilde{\pi} = \begin{pmatrix} -2\pi^V - \chi^2 \exp(a\chi - c^o b) & \chi c^o \exp(a\chi - c^o b) \\ \chi c^o \exp(a\chi - c^o b) & -(c^o)^2 \exp(a\chi - c^o b) \end{pmatrix}.$$

Therefore, for any non-zero vector $\mathbf{x} = (x_1, x_2)$, we have that

$$\mathbf{x}'(\nabla^2 \tilde{\pi})\mathbf{x} = -2\pi^V x_1^2 - \exp(a\chi - c^o b)(\chi x_1 - c^o x_2)^2 < 0.$$

Hence, the objective is strictly concave. Now consider the hospital's problem

$$\begin{aligned}
\max_{\mathbf{a}, \mathbf{b}} \quad & \sum_{j \in \mathcal{J}_h} \tilde{\pi}_j(a_j, b_j) \\
\text{s.t.} \quad & 0 \leq a_j \leq 1, \\
& 0 \leq a_j \Lambda_j \mu_j^L \leq b_j, \\
& \sum_{j \in \mathcal{J}_h} b_j \leq B.
\end{aligned}$$

The objective has a negative-definite Hessian that is block diagonal across lines, and the constraint

set is convex because it is defined by affine inequalities. Since the objective is continuous and the constraint set is nonempty and compact, a solution exists; strict concavity implies uniqueness. This proves the first item of the proposition.

The second item follows by implicitly differentiating the unconstrained first-order condition:

$$\frac{da^o(b)}{db} = \frac{\chi c^o \exp(a^o(b)\chi - c^o b)}{2\pi^V + \chi^2 \exp(a^o(b)\chi - c^o b)} > 0.$$

On any interval where $a^*(b) \in (0, 1)$, the constrained solution is $\min\{a^o(b), b/(\Lambda\mu^L)\}$. Both functions are strictly increasing in b , so their minimum is strictly increasing, including when the binding constraint changes. This proves the second item of the proposition.

We now prove the third item of the proposition with greater generality. First, we prove that $\pi(b)$ is concave. To see this, take $b_1, b_2 \geq 0$, and denote $a_1 = a^*(b_1)$ and $a_2 = a^*(b_2)$. Let $\lambda \in (0, 1)$, and denote $a_\lambda = \lambda a_1 + (1 - \lambda)a_2$ and $b_\lambda = \lambda b_1 + (1 - \lambda)b_2$. Since a_1 is feasible under b_1 and a_2 is feasible under b_2 , we have that $a_\lambda \Lambda\mu^L \leq b_\lambda$. Then

$$\begin{aligned} \pi(b_\lambda) &\geq \tilde{\pi}(a_\lambda, b_\lambda) \\ &= \tilde{\pi}(\lambda a_1 + (1 - \lambda)a_2, \lambda b_1 + (1 - \lambda)b_2) \\ &\geq \lambda \tilde{\pi}(a_1, b_1) + (1 - \lambda)\tilde{\pi}(a_2, b_2) \\ &= \lambda \pi(b_1) + (1 - \lambda)\pi(b_2), \end{aligned}$$

where the first inequality follows from the optimality of $a^*(b_\lambda)$ and the feasibility of a_λ , and the second inequality from the concavity of $\tilde{\pi}$.

The finite concave value function $\pi(b)$ is differentiable almost everywhere on $(0, \infty)$ and has a nonempty superdifferential there, denoted by $\partial\pi(b)$. It is also strictly increasing: increasing b raises the objective at the previously optimal admission rate, which remains feasible. Hence the hospital uses all of its resources at the optimum. Since $B > 0$, the resource-allocation problem admits a strictly feasible allocation with positive resources for every line. The KKT conditions for this concave program therefore apply. Denote $\eta \geq 0$ as the multiplier on the hospital's total resource constraint. Stationarity gives

$$\eta \in \partial\pi_j(b_j^*) \quad \forall j \in \mathcal{J}_h \text{ such that } b_j^* > 0.$$

This implies that for any j, j' with $b_j^*, b_{j'}^* > 0$, we have that $\eta \in \partial\pi_j(b_j^*) \cap \partial\pi_{j'}(b_{j'}^*)$. An immediate implication is that, at any point of differentiability,

$$\left. \frac{\partial\pi_j(b)}{\partial b} \right|_{b=b_j^*} = \left. \frac{\partial\pi_{j'}(b)}{\partial b} \right|_{b=b_{j'}^*}.$$

This proves the third item of the proposition.

We now prove the fourth item. We now restore the hospital index and make the dependence on rivals' admission rates explicit. All demand and payoff objects in Equations (6) and (10) are evaluated at $\mathbf{a}_{-h}$. Denote the expected bed-day load of line j when its admission rate equals one as

$$L_{hj}(\mathbf{a}_{-h}) := \Lambda_{hj}(\mathbf{a}_{-h}) \mu_{hj}^L(\mathbf{a}_{-h}).$$

Let $J_h = |\mathcal{J}_h|$ denote the number of service lines at hospital h . Hospital h 's feasible set, conditional on $\mathbf{a}_{-h}$, is

$$\begin{aligned} \Gamma_h(\mathbf{a}_{-h}) := \left\{ (\mathbf{a}_h, \mathbf{b}_h) : \mathbf{a}_h \in [0, 1]^{J_h}, \quad \mathbf{b}_h \geq \mathbf{0}, \quad \sum_{j \in \mathcal{J}_h} b_{hj} \leq B_h, \right. \\ \left. a_{hj} L_{hj}(\mathbf{a}_{-h}) \leq b_{hj} \quad \text{for every } j \in \mathcal{J}_h \right\}. \end{aligned} \quad (\text{A.6})$$

Let

$$\text{BR}_h(\mathbf{a}_{-h}) = \left(\text{BR}_h^a(\mathbf{a}_{-h}), \text{BR}_h^b(\mathbf{a}_{-h}) \right)$$

denote the solution to hospital h 's joint admission and resource-allocation problem obtained by combining Equations (7) and (10). The first item established above implies that this solution is unique for every $\mathbf{a}_{-h}$. An equilibrium is a profile $(\mathbf{a}^*, \mathbf{b}^*)$ such that $(\mathbf{a}_h^*, \mathbf{b}_h^*) = \text{BR}_h(\mathbf{a}_{-h}^*)$ for every hospital h .⁴

The request probabilities in Equation (6) are continuous in $\mathbf{a}_{-h}$. Indeed, after conditioning on a patient's preferences, the expectation over the finitely many Bernoulli acceptance realizations is multilinear in the rivals' admission rates. The same argument applies to the demand aggregates and uninsured-surplus term in Equation (10); the moment and integrability conditions ensure that every coefficient is finite. These objects are therefore polynomials in rivals' rates. In particular, the hospital objective and each $L_{hj}(\mathbf{a}_{-h})$ are continuous in $\mathbf{a}_{-h}$, and positive demand and length of stay imply $L_{hj}(\mathbf{a}_{-h}) > 0$ at every rival profile.

For every $\mathbf{a}_{-h}$, the set $\Gamma_h(\mathbf{a}_{-h})$ is nonempty because it contains $(\mathbf{0}, \mathbf{0})$, and it is compact and convex. It also varies continuously with $\mathbf{a}_{-h}$. All values lie in the fixed compact set $[0, 1]^{J_h} \times \{\mathbf{b}_h \geq \mathbf{0} : \sum_j b_{hj} \leq B_h\}$. Upper hemicontinuity therefore follows from its closed graph: limits of feasible points satisfy all the constraints by continuity of L_{hj} . To establish lower hemicontinuity, take $\mathbf{a}_{-h}^n \rightarrow \mathbf{a}_{-h}$ and $(\mathbf{a}_h, \mathbf{b}_h) \in \Gamma_h(\mathbf{a}_{-h})$, and define

$$t_n := \min \left\{ 1, \min_{\{j: a_{hj} > 0\}} \frac{b_{hj}}{a_{hj} L_{hj}(\mathbf{a}_{-h}^n)} \right\},$$

⁴Because the expected-load constraints in Equation (A.6) depend on rivals' admission rates, the hospital game is formally a generalized Nash game. We use "equilibrium" for brevity.

where the inner minimum over an empty set is interpreted as $+\infty$. If $a_{hj} > 0$, feasibility implies $b_{hj} \geq a_{hj}L_{hj}(\mathbf{a}_{-h}) > 0$, so $t_n \in (0, 1]$ and $(t_n \mathbf{a}_h, \mathbf{b}_h) \in \Gamma_h(\mathbf{a}_{-h}^n)$. Each ratio in the minimum converges to at least one. Thus $t_n \rightarrow 1$, and these feasible points converge to $(\mathbf{a}_h, \mathbf{b}_h)$.

Berge's maximum theorem and the strict concavity established in the first item therefore imply that BR_h is single-valued and continuous (Aliprantis and Border, 2006, Theorems 17.31 and 17.6). Define

$$\mathcal{A} := \prod_{h \in \mathcal{H}} [0, 1]^{J_h}, \quad \mathcal{T}(\mathbf{a}) := \left(\text{BR}_h^{\mathbf{a}}(\mathbf{a}_{-h}) \right)_{h \in \mathcal{H}}.$$

The set $\mathcal{A}$ is nonempty, compact, and convex, and $\mathcal{T}$ is a continuous self-map of $\mathcal{A}$. By Brouwer's fixed-point theorem, a fixed point $\mathbf{a}^*$ exists. Letting $\mathbf{b}_h^* := \text{BR}_h^{\mathbf{a}^*}(\mathbf{a}_{-h}^*)$ for each h then completes an equilibrium. This proves the fourth item and completes the proof. $\square$

Equilibrium uniqueness. We now derive sufficient conditions for equilibrium uniqueness. We state the conditions in terms of utilities, illness rates, lengths of stay, resources, and payoff parameters. We first bound these objects uniformly in rivals' admission rates, then bound each hospital's response to a change in those rates. Throughout, $\|\cdot\|_2$ denotes the Euclidean norm for vectors and the spectral norm for matrices, $\|\cdot\|_F$ denotes the Frobenius norm, and $x^+ := \max\{x, 0\}$. All inequalities between vectors are componentwise.

Uniform demand and payoff bounds. Write $\mu_k^L := \mathbb{E}[\ell_i^k]$ and $M_k(t) := \mathbb{E}[e^{t\ell_i^k}]$ for the type-specific mean and moment-generating function of length of stay. The line aggregates in the main text satisfy

$$\begin{aligned} \Lambda_{hj}(\mathbf{a}_{-h}) &= \sum_{k \in \mathcal{K}_j} \lambda_k R_{hk}(\mathbf{a}_{-h}), \\ L_{hj}(\mathbf{a}_{-h}) &= \sum_{k \in \mathcal{K}_j} \lambda_k R_{hk}(\mathbf{a}_{-h}) \mu_k^L, \\ \chi_{hj}(\mathbf{a}_{-h}) &= \sum_{k \in \mathcal{K}_j} \lambda_k R_{hk}(\mathbf{a}_{-h}) (M_k(c_h^o) - 1). \end{aligned} \tag{A.7}$$

In particular, both the mean length of stay and its moment-generating function at the line level may change with rivals' rates through the composition of admitted patients.

The request probability can be written as

$$R_{hk}(\mathbf{a}_{-h}) = \mathbb{E}_{i, \mathbf{a}_{-h}} \left[\frac{e^{v_{ih}}}{1 + e^{v_{ih}} + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{h'i} e^{v_{ih'}}} \right], \tag{A.8}$$

where, for a type- k patient, $\tilde{a}_{h'i}$ has mean $a_{h'j(k)}$, using the service-line assignment $j(k)$ in the main text. The expectation is over the patient's preferences and the rivals' acceptance draws. The

all-open and all-closed rival profiles therefore give

$$\underline{R}_{hk} := \mathbb{E}_i \left[\frac{e^{v_{ih}}}{1 + \sum_{h' \in \mathcal{H}} e^{v_{ih'}}} \right] \leq R_{hk}(\mathbf{a}_{-h}) \leq \bar{R}_{hk} := \mathbb{E}_i \left[\frac{e^{v_{ih}}}{1 + e^{v_{ih}}} \right]. \quad (\text{A.9})$$

Replacing R_{hk} in Equation (A.7) by $\underline{R}_{hk}$ or $\bar{R}_{hk}$ defines $\underline{\Delta}_{hj}$, $\bar{\Delta}_{hj}$, $\underline{L}_{hj}$, $\bar{L}_{hj}$, and $\underline{\chi}_{hj}$, $\bar{\chi}_{hj}$. Each pair bounds the corresponding aggregate at every rival profile. All lower bounds are strictly positive: every line has positive potential demand, and $M_k(c_h^o) - 1 \geq c_h^o \mu_k^L > 0$. The maintained moment condition ensures the upper bounds are finite. Also set

$$\underline{\pi}_{hj}^V := c_{hj}^V \underline{\Delta}_{hj}^2, \quad \bar{\pi}_{hj}^V := c_{hj}^V \bar{\Delta}_{hj}^2, \quad \underline{L}_h := \min_{j \in \mathcal{J}_h} \underline{L}_{hj} > 0. \quad (\text{A.10})$$

The expected contribution to uninsured surplus is nonnegative and is largest when all rivals are closed, so

$$0 \leq \Delta S_{hj}^u(\mathbf{a}_{-h}) \leq \bar{\Delta S}_{hj}^u := \sum_{k \in \mathcal{K}_j^u} \lambda_k \mathbb{E}_i[\log(1 + e^{v_{ih}})] < \infty.$$

Sorting the price–cost terms in π_{hj}^L by sign gives

$$\begin{aligned} \bar{\pi}_{hj}^L &:= \sum_{\substack{k \in \mathcal{K}_j \\ p_{hk} > c_{hk}}} \lambda_k \bar{R}_{hk}(p_{hk} - c_{hk}) + \sum_{\substack{k \in \mathcal{K}_j \\ p_{hk} \leq c_{hk}}} \lambda_k \underline{R}_{hk}(p_{hk} - c_{hk}) - c_{hj}^V \underline{\Delta}_{hj} + \tau_{hj} \bar{\Delta S}_{hj}^u, \\ \underline{\pi}_{hj}^L &:= \sum_{\substack{k \in \mathcal{K}_j \\ p_{hk} > c_{hk}}} \lambda_k \underline{R}_{hk}(p_{hk} - c_{hk}) + \sum_{\substack{k \in \mathcal{K}_j \\ p_{hk} \leq c_{hk}}} \lambda_k \bar{R}_{hk}(p_{hk} - c_{hk}) - c_{hj}^V \bar{\Delta}_{hj}. \end{aligned} \quad (\text{A.11})$$

Thus $\underline{\pi}_{hj}^L \leq \pi_{hj}^L(\mathbf{a}_{-h}) \leq \bar{\pi}_{hj}^L$ for every $\mathbf{a}_{-h}$.

A fixed feasible set and an admission ceiling. For a fixed rival profile, let $q_{hj} := a_{hj} L_{hj}(\mathbf{a}_{-h})$ denote the expected bed-day load of line j . Dropping temporarily the upper bound $a_{hj} \leq 1$, the hospital chooses (q_h, b_h) from

$$\mathcal{Y}_h := \left\{ (q_h, b_h) \geq \mathbf{0} : q_{hj} \leq b_{hj} \text{ for every } j \in \mathcal{J}_h, \quad \sum_{j \in \mathcal{J}_h} b_{hj} \leq B_h \right\}. \quad (\text{A.12})$$

This set is nonempty, compact, convex, and independent of rivals' rates. The objective becomes

$$\tilde{U}_h(q_h, b_h; \mathbf{a}_{-h}) := \sum_{j \in \mathcal{J}_h} \left[q_{hj} \frac{\pi_{hj}^L}{L_{hj}} - q_{hj}^2 \frac{\pi_{hj}^V}{L_{hj}^2} - \exp\left(q_{hj} \frac{\chi_{hj}}{L_{hj}} - c_h^o b_{hj}\right) \right], \quad (\text{A.13})$$

where the dependence of the coefficients on $\mathbf{a}_{-h}$ is suppressed. For fixed rivals, the coordinate change is a linear bijection. Hence $\tilde{U}_h$ inherits strict concavity from the hospital's original objective and has a unique maximizer over $\mathcal{Y}_h$. Denote it by $\mathbf{y}_h^*(\mathbf{a}_{-h}) = (q_h^*(\mathbf{a}_{-h}), b_h^*(\mathbf{a}_{-h}))$, and write $\mathbf{y}_h =$

(q_h, b_h) and $L_h = (L_{hj})_{j \in \mathcal{J}_h}$.

Define the uniform admission bound

$$\bar{a}_h := \max_{j \in \mathcal{J}_h} \min \left\{ \frac{B_h}{L_{hj}}, \frac{(\bar{\pi}_{hj}^L)^+}{2c_{hj}^V \Lambda_{hj}^2} \right\}. \quad (\text{A.14})$$

The two terms bound admissions using total resources and quadratic congestion costs, respectively. At the relaxed maximizer, $q_{hj}^* \leq b_{hj}^* \leq B_h$. Moreover, if $q_{hj}^* > 0$, lowering q_{hj} while holding beds fixed is feasible, so

$$0 \leq \frac{\partial \tilde{U}_h}{\partial q_{hj}}(\mathbf{y}_h^*; \mathbf{a}_{-h}) = \frac{1}{L_{hj}} \left[\pi_{hj}^L - 2 \frac{q_{hj}^*}{L_{hj}} \pi_{hj}^V - \chi_{hj} \exp \left(q_{hj}^* \frac{\chi_{hj}}{L_{hj}} - c_h^o b_{hj}^* \right) \right].$$

The exponential term is strictly positive. Together with the resource bound, this yields

$$\frac{q_{hj}^*(\mathbf{a}_{-h})}{L_{hj}(\mathbf{a}_{-h})} \leq \min \left\{ \frac{B_h}{L_{hj}(\mathbf{a}_{-h})}, \frac{(\pi_{hj}^L(\mathbf{a}_{-h}))^+}{2c_{hj}^V \Lambda_{hj}(\mathbf{a}_{-h})^2} \right\} \leq \bar{a}_h. \quad (\text{A.15})$$

The inequality also holds when $q_{hj}^* = 0$.

Assumption A.1 (Admission ceiling). *For every hospital h , $\bar{a}_h \leq 1$.*

Adding $q_{hj} \leq L_{hj}(\mathbf{a}_{-h})$ to $\mathcal{Y}_h$ gives the original feasible set in bed-day coordinates. Under Assumption A.1, Equation (A.15) shows that the relaxed maximizer satisfies these added constraints. It therefore also solves the original problem, and

$$\text{BR}_{hj}^a(\mathbf{a}_{-h}) = \frac{q_{hj}^*(\mathbf{a}_{-h})}{L_{hj}(\mathbf{a}_{-h})}, \quad \text{BR}_h^b(\mathbf{a}_{-h}) = b_h^*(\mathbf{a}_{-h}). \quad (\text{A.16})$$

This is the only role of the admission ceiling. Equality is allowed: the argument requires the relaxed optimizer to be feasible for the original problem, not the upper bound on admissions to be slack.

A uniform bound on own curvature. Write

$$\Xi_{hj} := \exp \left(q_{hj} \frac{\chi_{hj}}{L_{hj}} - c_h^o b_{hj} \right).$$

The negative Hessian of $\tilde{U}_h$ is block diagonal across lines, with line- j block

$$-\nabla_{(q_{hj}, b_{hj})}^2 \tilde{U}_h = \begin{pmatrix} \frac{2\pi_{hj}^V + \chi_{hj}^2 \Xi_{hj}}{L_{hj}^2} & -c_h^o \frac{\chi_{hj}}{L_{hj}} \Xi_{hj} \\ -c_h^o \frac{\chi_{hj}}{L_{hj}} \Xi_{hj} & (c_h^o)^2 \Xi_{hj} \end{pmatrix}. \quad (\text{A.17})$$

Its determinant is $2\pi_{hj}^V (c_h^o)^2 \Xi_{hj} / L_{hj}^2 > 0$. For a positive definite 2×2 matrix, the smallest eigenvalue is at least its determinant divided by its trace. For the block above, this ratio is

$$\frac{\det}{\text{tr}} = \frac{(c_h^o)^2}{\Xi_{hj}^{-1} + \frac{\chi_{hj}^2 + (c_h^o)^2 L_{hj}^2}{2\pi_{hj}^V}}.$$

On $\mathcal{Y}_h$, $\Xi_{hj} \geq e^{-c_h^o B_h}$. The uniform demand bounds therefore imply

$$\alpha_h := \min_{j \in \mathcal{J}_h} \frac{(c_h^o)^2}{e^{c_h^o B_h} + \frac{\bar{\chi}_{hj}^2 + (c_h^o)^2 \bar{L}_{hj}^2}{2c_{hj}^V \Lambda_{hj}^2}} > 0, \quad -\nabla_{\mathbf{y}_h}^2 \tilde{U}_h \geq \alpha_h \mathbf{I}. \quad (\text{A.18})$$

Consequently, $\tilde{U}_h$ is α_h -strongly concave on $\mathcal{Y}_h$, uniformly in $\mathbf{a}_{-h}$. In particular, integrating its Hessian along the segment between any $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}_h$ gives

$$\langle \nabla_{\mathbf{y}_h} \tilde{U}_h(\mathbf{y}; \mathbf{a}_{-h}) - \nabla_{\mathbf{y}_h} \tilde{U}_h(\mathbf{y}'; \mathbf{a}_{-h}), \mathbf{y} - \mathbf{y}' \rangle \leq -\alpha_h \|\mathbf{y} - \mathbf{y}'\|_2^2. \quad (\text{A.19})$$

Cross-hospital interaction. For $h' \neq h$, define bounds on the reduction in request probability and uninsured surplus when rival h' accepts a patient:

$$\begin{aligned} \overline{\partial R}_{hh'k} &:= \mathbb{E}_i \left[\frac{e^{v_{ih}} e^{v_{ih'}}}{(1 + e^{v_{ih}})(1 + e^{v_{ih}} + e^{v_{ih'}})} \right], \\ \overline{\partial S}_{hh'k} &:= \mathbb{E}_i \left[\log \frac{(1 + e^{v_{ih}})(1 + e^{v_{ih'}})}{1 + e^{v_{ih}} + e^{v_{ih'}}} \right]. \end{aligned} \quad (\text{A.20})$$

The second bound applies only to uninsured types. The integrands are zero when either hospital is outside the patient's network. Both bounds are finite on their respective domains; the second integrand lies between zero and $\log(1 + e^{v_{ih}})$.

To verify the bounds, fix a type k served by line j' of rival h' and condition on preferences and all other acceptance draws. Write $\Sigma_{ik} := \sum_{h'' \in \mathcal{H} \setminus \{h, h'\}} \tilde{a}_{h''i} e^{v_{ih''}}$. Expanding over the two values of $\tilde{a}_{h'i}$ in Equation (A.8) shows that R_{hk} is affine in $a_{h'j'}$ and

$$-\frac{\partial R_{hk}}{\partial a_{h'j'}} = \mathbb{E} \left[\frac{e^{v_{ih}} e^{v_{ih'}}}{(1 + e^{v_{ih}} + \Sigma_{ik})(1 + e^{v_{ih}} + e^{v_{ih'}} + \Sigma_{ik})} \right] \leq \overline{\partial R}_{hh'k}.$$

The integrand is nonnegative and decreasing in Σ_{ik} , so the upper bound is attained when the other rivals are closed. The derivative is zero if $k \notin \mathcal{K}_{j'}$.

For the type- k contribution to uninsured surplus, the reduction when h' accepts is

$$g_{ik}(\Sigma) := \log \frac{1 + e^{v_{ih}} + \Sigma}{1 + \Sigma} - \log \frac{1 + e^{v_{ih}} + e^{v_{ih'}} + \Sigma}{1 + e^{v_{ih'}} + \Sigma}.$$

Writing $u = 1 + \Sigma$, $x = e^{v_{ih}}$, and $y = e^{v_{ih'}}$,

$$g'_{ik}(\Sigma) = -\frac{x}{u(u+x)} + \frac{x}{(u+y)(u+x+y)} \leq 0.$$

Since $g_{ik}(\Sigma) \rightarrow 0$ as $\Sigma \rightarrow \infty$, it lies between zero and $g_{ik}(0)$, whose expectation is $\bar{\partial S}_{hh'k}$. Thus the same reasoning bounds the derivative of uninsured surplus. We can take all these derivatives after evaluating the finite sum over acceptance realizations; the maintained integrability condition ensures finite coefficients.

Aggregate these bounds over the types shared by lines j and j' :

$$\begin{aligned} \bar{\partial \Lambda}_{hj,h'j'} &:= \sum_{k \in \mathcal{K}_j \cap \mathcal{K}_{j'}} \lambda_k \bar{\partial R}_{hh'k}, \\ \bar{\partial L}_{hj,h'j'} &:= \sum_{k \in \mathcal{K}_j \cap \mathcal{K}_{j'}} \lambda_k \bar{\partial R}_{hh'k} \mu_k^L, \\ \bar{\partial \chi}_{hj,h'j'} &:= \sum_{k \in \mathcal{K}_j \cap \mathcal{K}_{j'}} \lambda_k \bar{\partial R}_{hh'k} (M_k(c_h^o) - 1), \\ \bar{\partial \Delta S}_{hj,h'j'} &:= \sum_{k \in \mathcal{K}_j^u \cap \mathcal{K}_{j'}} \lambda_k \bar{\partial S}_{hh'k}. \end{aligned} \tag{A.21}$$

The corresponding payoff bounds are

$$\begin{aligned} \bar{\partial \pi^L}_{hj,h'j'} &:= \sum_{k \in \mathcal{K}_j \cap \mathcal{K}_{j'}} \lambda_k \bar{\partial R}_{hh'k} (|p_{hk} - c_{hk}| + c_{hj}^V) + \tau_{hj} \bar{\partial \Delta S}_{hj,h'j'}, \\ \bar{\partial \pi^V}_{hj,h'j'} &:= 2c_{hj}^V \bar{\Lambda}_{hj} \bar{\partial \Lambda}_{hj,h'j'}. \end{aligned} \tag{A.22}$$

For $X \in \{\Lambda, L, \chi\}$, the derivative satisfies $0 \leq -\partial X_{hj}/\partial a_{h'j'} \leq \bar{\partial X}_{hj,h'j'}$. The absolute derivatives of π_{hj}^L and π_{hj}^V are bounded by the respective terms in Equation (A.22). These statements follow by differentiating the finite sums and applying the triangle inequality; for π_{hj}^V use $\partial \pi_{hj}^V = 2c_{hj}^V \bar{\Lambda}_{hj} \partial \Lambda_{hj}$.

We next bound the mixed derivatives of the transformed objective over the entire fixed set. On $\mathcal{Y}_h$, define

$$\begin{aligned} \bar{a}_{hj}^{\mathcal{Y}} &:= \frac{B_h}{\underline{L}_{hj}}, & \bar{\Xi}_{hj} &:= \exp(\bar{a}_{hj}^{\mathcal{Y}} \bar{\chi}_{hj}), \\ \bar{\partial a}_{hj,h'j'}^{\mathcal{Y}} &:= \frac{\bar{a}_{hj}^{\mathcal{Y}} \bar{\partial L}_{hj,h'j'}}{\underline{L}_{hj}}, & \bar{G}_{hj} &:= \max\{|\underline{\pi}_{hj}^L|, |\bar{\pi}_{hj}^L|\} + 2\bar{a}_{hj}^{\mathcal{Y}} \bar{\pi}_{hj}^V + \bar{\chi}_{hj} \bar{\Xi}_{hj}. \end{aligned} \tag{A.23}$$

Here $a_{hj} = q_{hj}/L_{hj} \leq \bar{a}_{hj}^{\mathcal{Y}}$ and $\Xi_{hj} \leq \bar{\Xi}_{hj}$. Holding $\mathbf{y}_h$ fixed,

$$\frac{\partial a_{hj}}{\partial a_{h'j'}} = -\frac{a_{hj}}{L_{hj}} \frac{\partial L_{hj}}{\partial a_{h'j'}}, \quad \left| \frac{\partial a_{hj}}{\partial a_{h'j'}} \right| \leq \bar{\partial a}_{hj,h'j'}, \quad |\pi_{hj}^L - 2a_{hj}\pi_{hj}^V - \chi_{hj}\Xi_{hj}| \leq \bar{G}_{hj}.$$

The bound $\bar{a}_{hj}^{\mathcal{Y}}$ may exceed one. The admission ceiling bounds the optimizer at each rival profile, but not every point of $\mathcal{Y}_h$ evaluated at every other profile. The mixed-derivative bounds must allow for this distinction.

Using $\partial_{a_{h'j'}}$ for a derivative at fixed $\mathbf{y}_h$, differentiation gives

$$\begin{aligned} \frac{\partial^2 \tilde{U}_h}{\partial q_{hj} \partial a_{h'j'}} &= -\frac{\partial_{a_{h'j'}} L_{hj}}{L_{hj}^2} [\pi_{hj}^L - 2a_{hj}\pi_{hj}^V - \chi_{hj}\Xi_{hj}] \\ &\quad + \frac{1}{L_{hj}} \left[\partial_{a_{h'j'}} \pi_{hj}^L - 2(\partial_{a_{h'j'}} a_{hj})\pi_{hj}^V - 2a_{hj}\partial_{a_{h'j'}} \pi_{hj}^V - (\partial_{a_{h'j'}} \chi_{hj})\Xi_{hj} \right. \\ &\quad \left. - \chi_{hj}\Xi_{hj}((\partial_{a_{h'j'}} a_{hj})\chi_{hj} + a_{hj}\partial_{a_{h'j'}} \chi_{hj}) \right], \quad (\text{A.24}) \\ \frac{\partial^2 \tilde{U}_h}{\partial b_{hj} \partial a_{h'j'}} &= c_h^o \Xi_{hj} ((\partial_{a_{h'j'}} a_{hj})\chi_{hj} + a_{hj}\partial_{a_{h'j'}} \chi_{hj}). \end{aligned}$$

Bounding each factor by the preceding inequalities yields the entrywise bounds

$$\begin{aligned} W_{hj,h'j'}^q &:= \frac{\bar{\partial L}_{hj,h'j'}}{L_{hj}^2} \bar{G}_{hj} + \frac{1}{L_{hj}} \left[\bar{\partial \pi}_{hj,h'j'}^L + 2\bar{\partial a}_{hj,h'j'} \bar{\pi}_{hj}^V + 2\bar{a}_{hj}^{\mathcal{Y}} \bar{\partial \pi}_{hj,h'j'}^V \right. \\ &\quad \left. + \bar{\partial \chi}_{hj,h'j'} \bar{\Xi}_{hj} + \bar{\chi}_{hj} \bar{\Xi}_{hj} (\bar{\partial a}_{hj,h'j'} \bar{\chi}_{hj} + \bar{a}_{hj}^{\mathcal{Y}} \bar{\partial \chi}_{hj,h'j'}) \right], \quad (\text{A.25}) \\ W_{hj,h'j'}^b &:= c_h^o \bar{\Xi}_{hj} (\bar{\partial a}_{hj,h'j'} \bar{\chi}_{hj} + \bar{a}_{hj}^{\mathcal{Y}} \bar{\partial \chi}_{hj,h'j'}), \end{aligned}$$

so that the absolute values of the two derivatives in Equation (A.24) are at most $W_{hj,h'j'}^q$ and $W_{hj,h'j'}^b$ respectively. Define

$$\begin{aligned} \beta_{hh'}^q &:= \left(\sum_{j \in \mathcal{J}_h} \sum_{j' \in \mathcal{J}_{h'}} [(W_{hj,h'j'}^q)^2 + (W_{hj,h'j'}^b)^2] \right)^{1/2}, \quad (\text{A.26}) \\ \beta_{hh'}^L &:= \left(\sum_{j \in \mathcal{J}_h} \sum_{j' \in \mathcal{J}_{h'}} (\bar{\partial L}_{hj,h'j'})^2 \right)^{1/2}. \end{aligned}$$

Since the spectral norm is at most the Frobenius norm, these entrywise bounds imply, uniformly on $\mathcal{Y}_h \times \prod_{g \neq h} [0, 1]^{J_g}$,

$$\|\nabla_{\mathbf{y}_h, \mathbf{a}_{h'}}^2 \tilde{U}_h\|_2 \leq \beta_{hh'}^q, \quad \left\| \frac{\partial \mathbf{L}_h}{\partial \mathbf{a}_{h'}} \right\|_2 \leq \beta_{hh'}^L. \quad (\text{A.27})$$

All constants above are explicit functions of the model primitives; their construction requires

neither a best response nor an equilibrium.

Bounding the rate best response. For $h' \neq h$, define

$$\phi_{hh'} := \frac{\beta_{hh'}^q}{\underline{L}_h \alpha_h} + \frac{\bar{a}_h \beta_{hh'}^L}{\underline{L}_h}. \quad (\text{A.28})$$

The first term accounts for changes in the optimal bed-day load, while the second accounts for changes in converting that load to an admission rate.

Lemma A.1. *Under the maintained conditions and Assumption A.1, for any two admission-rate profiles $\mathbf{a}, \mathbf{a}' \in \mathcal{A}$,*

$$\|\text{BR}_h^a(\mathbf{a}_{-h}) - \text{BR}_h^a(\mathbf{a}'_{-h})\|_2 \leq \sum_{h' \neq h} \phi_{hh'} \|\mathbf{a}_{h'} - \mathbf{a}'_{h'}\|_2. \quad (\text{A.29})$$

Proof. Fix h and write $\mathbf{y} = \mathbf{y}_h^*(\mathbf{a}_{-h})$, $\mathbf{y}' = \mathbf{y}_h^*(\mathbf{a}'_{-h})$, and $\Delta = \mathbf{y} - \mathbf{y}'$. Both points belong to the same convex set $\mathcal{Y}_h$. First-order optimality therefore gives

$$\begin{aligned} \langle \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}; \mathbf{a}_{-h}), \mathbf{y}' - \mathbf{y} \rangle &\leq 0, \\ \langle \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}'_{-h}), \mathbf{y} - \mathbf{y}' \rangle &\leq 0. \end{aligned}$$

Adding these inequalities, inserting the gradient at $(\mathbf{y}'; \mathbf{a}_{-h})$, and applying Equation (A.19), we obtain

$$\begin{aligned} \alpha_h \|\Delta\|_2^2 &\leq \langle \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}_{-h}) - \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}'_{-h}), \Delta \rangle \\ &\leq \|\nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}_{-h}) - \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}'_{-h})\|_2 \|\Delta\|_2. \end{aligned}$$

If $\Delta \neq 0$, divide by $\alpha_h \|\Delta\|_2$; if $\Delta = 0$, the resulting bound is immediate. These optimality inequalities hold even when constraints bind or the set of binding constraints changes.

Let $\mathbf{a}_{-h}(t) = \mathbf{a}'_{-h} + t(\mathbf{a}_{-h} - \mathbf{a}'_{-h})$, $t \in [0, 1]$. The segment lies in $\prod_{g \neq h} [0, 1]^{J_g}$. Holding $\mathbf{y}'$ fixed, integration along this segment gives

$$\begin{aligned} &\nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}_{-h}) - \nabla_{\mathbf{y}_h} \widetilde{U}_h(\mathbf{y}'; \mathbf{a}'_{-h}) \\ &= \int_0^1 \sum_{h' \neq h} \nabla_{\mathbf{y}_h, \mathbf{a}_{h'}}^2 \widetilde{U}_h(\mathbf{y}'; \mathbf{a}_{-h}(t)) (\mathbf{a}_{h'} - \mathbf{a}'_{h'}) dt. \end{aligned}$$

The derivatives are continuous because the demand and payoff coefficients are polynomials, L_{hj} is bounded away from zero, and the exponential is smooth. Taking norms and using Equation (A.27) yields

$$\|\mathbf{q}_h^*(\mathbf{a}_{-h}) - \mathbf{q}_h^*(\mathbf{a}'_{-h})\|_2 \leq \|\mathbf{y} - \mathbf{y}'\|_2 \leq \frac{1}{\alpha_h} \sum_{h' \neq h} \beta_{hh'}^q \|\mathbf{a}_{h'} - \mathbf{a}'_{h'}\|_2. \quad (\text{A.30})$$

The same integration argument applied to L_h gives

$$\|L_h(\mathbf{a}_{-h}) - L_h(\mathbf{a}'_{-h})\|_2 \leq \sum_{h' \neq h} \beta_{hh'}^L \|\mathbf{a}_{h'} - \mathbf{a}'_{h'}\|_2. \quad (\text{A.31})$$

Finally, write $L_{hj} = L_{hj}(\mathbf{a}_{-h})$, $L'_{hj} = L_{hj}(\mathbf{a}'_{-h})$, $q_{hj}^* = q_{hj}^*(\mathbf{a}_{-h})$, and $q_{hj}^{*'} = q_{hj}^*(\mathbf{a}'_{-h})$. Equation (A.16) implies

$$\text{BR}_{hj}^a(\mathbf{a}_{-h}) - \text{BR}_{hj}^a(\mathbf{a}'_{-h}) = \frac{q_{hj}^* - q_{hj}^{*'}}{L_{hj}} + \frac{q_{hj}^{*'} L'_{hj} - L_{hj}}{L'_{hj} L_{hj}}.$$

Since $q_{hj}^{*'} / L'_{hj} \leq \bar{a}_h$ and $L_{hj} \geq \underline{L}_h$, the triangle inequality gives

$$\begin{aligned} & \|\text{BR}_h^a(\mathbf{a}_{-h}) - \text{BR}_h^a(\mathbf{a}'_{-h})\|_2 \\ & \leq \frac{1}{\underline{L}_h} \|\mathbf{q}_h^*(\mathbf{a}_{-h}) - \mathbf{q}_h^*(\mathbf{a}'_{-h})\|_2 + \frac{\bar{a}_h}{\underline{L}_h} \|L_h(\mathbf{a}_{-h}) - L_h(\mathbf{a}'_{-h})\|_2. \end{aligned}$$

Substituting Equations (A.30) and (A.31) proves the claimed bound with $\phi_{hh'}$ as defined in Equation (A.28). $\square$

A sufficient condition for uniqueness. Let $\Phi = (\phi_{hh'})_{h,h' \in \mathcal{H}}$, with off-diagonal entries given by Equation (A.28) and $\phi_{hh} = 0$. This is a nonnegative matrix. Its spectral radius, $\rho(\Phi)$, is the largest modulus of its eigenvalues.

Assumption A.2 (Weak interaction). *The interaction matrix satisfies*

$$\rho(\Phi) < 1. \quad (\text{A.32})$$

For a nonnegative matrix, this condition is equivalent to the existence of weights $w_h > 0$ such that

$$\kappa_w := \max_{h \in \mathcal{H}} \frac{1}{w_h} \sum_{h' \neq h} \phi_{hh'} w_{h'} < 1. \quad (\text{A.33})$$

Indeed, if $\rho(\Phi) < 1$, the Neumann series gives $\mathbf{w} = (\mathbf{I} - \Phi)^{-1} \mathbf{1} = \sum_{m \geq 0} \Phi^m \mathbf{1} \geq \mathbf{1}$, and $\Phi \mathbf{w} = \mathbf{w} - \mathbf{1} < \mathbf{w}$. Conversely, if such weights exist, the matrix norm induced by $\max_h |x_h| / w_h$ equals κ_w for $\Phi \geq 0$, so $\rho(\Phi) \leq \kappa_w < 1$. In particular, the simpler row-sum condition

$$\kappa := \max_{h \in \mathcal{H}} \sum_{h' \neq h} \phi_{hh'} < 1$$

is sufficient, taking $\mathbf{w} = \mathbf{1}$.

Another sufficient condition is $\lambda_{\min}(\Psi) > 0$, where $\Psi := \mathbf{I} - \frac{1}{2}(\Phi + \Phi')$. To see this, take a unit

nonnegative Perron eigenvector v of Φ . Then

$$\rho(\Phi) = v' \Phi v = v' \frac{1}{2}(\Phi + \Phi') v \leq \lambda_{\max}\left(\frac{1}{2}(\Phi + \Phi')\right) = 1 - \lambda_{\min}(\Psi).$$

The symmetrized condition is in turn implied by $\frac{1}{2} \sum_{h' \neq h} (\phi_{hh'} + \phi_{h'h}) < 1$ for every h : Gershgorin's theorem places every eigenvalue of the symmetric matrix Ψ in a strictly positive interval centered at its unit diagonal. These are alternative sufficient tests; the spectral-radius condition can hold even when the unweighted or symmetrized tests fail.

Proposition A.1 (Uniqueness). *Under the maintained conditions of Proposition 1 and Assumptions A.1 and A.2, the game has exactly one equilibrium in admission rates and resource allocations.*

Proof. Existence follows from the fourth item of Proposition 1. Under Assumption A.2, choose positive weights satisfying Equation (A.33). Equip $\mathcal{A}$ with the norm

$$\|a\|_w := \max_{h \in \mathcal{H}} \frac{\|a_h\|_2}{w_h}.$$

The set $\mathcal{A}$ is closed in finite-dimensional space and hence complete in this norm. Lemma A.1 implies, for every hospital h ,

$$\begin{aligned} \frac{\|\text{BR}_h^a(a_{-h}) - \text{BR}_h^a(a'_{-h})\|_2}{w_h} &\leq \frac{1}{w_h} \sum_{h' \neq h} \phi_{hh'} w_{h'} \frac{\|a_{h'} - a'_{h'}\|_2}{w_{h'}} \\ &\leq \kappa_w \|a - a'\|_w. \end{aligned}$$

Taking the maximum over hospitals gives $\|\mathcal{T}(a) - \mathcal{T}(a')\|_w \leq \kappa_w \|a - a'\|_w$. Since $\kappa_w < 1$, the rate best-response map is a contraction and has a unique fixed point. The corresponding resource allocations are also unique because each hospital's joint best response is unique by the first item of Proposition 1. $\square$

The admission ceiling and weak-interaction condition are sufficient conditions for uniqueness. The first makes the problem on the fixed set equivalent to the original hospital problem; the second limits the combined response to rivals, accounting for both optimal loads and their conversion to admission rates. Smaller diversion bounds reduce the interaction coefficients when the other bounds are held fixed. However, the bounds use extrema over all rival profiles and all feasible bed-day allocations and can be conservative. Their failure does not imply multiple equilibria; it means that these conditions do not establish uniqueness. Under the stated conditions, simultaneous best-response iteration also converges to the unique equilibrium from any initial admission-rate profile in $\mathcal{A}$.

D Identification and Estimation Appendix

D.1 Derivations

We begin by deriving the likelihood in Equation (11). As in the main text, define $\bar{\sigma}_{idht}$ as the standard logit choice probability without the admissions constraint. The proof is for a single patient, diagnosis, and time, so we omit those subindices for clarity.

$$\begin{aligned}
\sigma_h &= \mathbb{E}\left[\frac{\tilde{a}_h e^{v_h}}{1 + \sum_{h' \in \mathcal{H}} \tilde{a}_{h'} e^{v_{h'}}}\right] \\
&= \mathbb{E}\left[\frac{\tilde{a}_h \bar{\sigma}_h}{\bar{\sigma}_0 + \sum_{h' \in \mathcal{H}} \tilde{a}_{h'} \bar{\sigma}_{h'}}\right] \\
&= a_h \bar{\sigma}_h \mathbb{E}\left[(1 + (\bar{\sigma}_0 + \bar{\sigma}_h - 1) + \sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{h'} \bar{\sigma}_{h'})^{-1}\right] \\
&= a_h \bar{\sigma}_h \mathbb{E}\left[(1 - \sum_{h' \in \mathcal{H} \setminus \{h\}} (1 - \tilde{a}_{h'}) \bar{\sigma}_{h'})^{-1}\right] \\
&= a_h \bar{\sigma}_h \mathbb{E}\left[\int_0^\infty \exp(-u(1 - \sum_{h' \in \mathcal{H} \setminus \{h\}} (1 - \tilde{a}_{h'}) \bar{\sigma}_{h'})) du\right] \\
&= a_h \bar{\sigma}_h \int_0^\infty e^{-u} \mathbb{E}\left[\prod_{h' \in \mathcal{H} \setminus \{h\}} \exp(u(1 - \tilde{a}_{h'}) \bar{\sigma}_{h'})\right] du \\
&= a_h \bar{\sigma}_h \int_0^\infty e^{-u} \prod_{h' \in \mathcal{H} \setminus \{h\}} (a_{h'} + (1 - a_{h'}) e^{u \bar{\sigma}_{h'}}) du
\end{aligned}$$

where the second equality follows from dividing the numerator and denominator by $(1 + \sum_{h \in \mathcal{H}} e^{v_h})$, the third follows from expanding the two values of $\tilde{a}_h$, the fourth from re-organizing shares, the fifth from the identity $y^{-1} = \int_0^\infty e^{-ty} dt$ for $y > 0$, the sixth from the linearity of expectations, and the last one from the independence of admission events.

D.2 Admission Fixed Point

Here we prove that the fixed-point equation presented in Main Text Section 6.1 is a contraction mapping, under some regularity conditions. We first establish these conditions, then prove the result.

Assumption A.3 (Acceptance Fixed Point Assumptions). *The following hold for every period t :*

1. **Regularity:** every hospital $h \in \mathcal{H}$ and line $j \in \mathcal{J}_h$ has $N_{hj}^{L\star} > 0$ and $\mathcal{I}_{hj}^* \neq \emptyset$. For every patient $i \in \mathcal{I}_{jt}$, $\ell_i^* > 0$ and $|v_{ih}| < \infty$ for every $h \in \mathcal{H}_i$.
2. **Interior Admission Rate:** $N_{hj}^{L\star} < \sum_{i \in \mathcal{I}_{hj}} \bar{\sigma}_{ih} \ell_i^*$ for every $h \in \mathcal{H}$ and $j \in \mathcal{J}_h$.
3. **Bounded Diversion:** $\sum_{i \in \mathcal{I}_{jt}^*} \frac{e^{v_{ih}}}{1 + e^{v_{ih}}} \ell_i^* < 2 \sum_{i \in \mathcal{I}_{jt}^*} \bar{\sigma}_{ih} \ell_i^*$.

The first assumption is minimal and states that every hospital we observe in the data has positive total length of stay, every admitted patient has positive length of stay, and preferences are finite. The second assumption states that the total admitted length of stay is strictly smaller than predicted under simple logit demand, which implies that $a_{hj} \in (0, 1)$. Finally, the last assumption states that a monopolist hospital obtains less than twice its present demand. Essentially, this implies that the market is spread out enough such that if all other hospitals were at capacity, more consumers would prefer the out-of-market option than the monopolist hospital.

Before proving the main result, we present an intermediary lemma.

Lemma A.2 (Elasticity Bound). *Under Assumption A.3.1, it holds that*

$$0 \leq \sum_{h' \in \mathcal{H} \setminus \{h\}} a_{h'j(i,h')} \left(-\frac{\partial R_{ih}}{\partial a_{h'j(i,h')}} \right) \leq \frac{e^{v_{ih}}}{1 + e^{v_{ih}}} - R_{ih}.$$

Proof. In what follows, we omit the line index for notational simplicity, and it is implicit that all arguments refer to the line in which a fixed patient i seeks care.

Note that

$$\begin{aligned} R_{ih} &= \mathbb{E} \left[\frac{e^{v_{ih}}}{1 + e^{v_{ih}} + \sum_{h' \neq h} e^{v_{ih'}} \tilde{a}_{h'}} \right] \\ &= a_{h'} \mathbb{E} \left[\frac{e^{v_{ih}}}{1 + e^{v_{ih}} + e^{v_{ih'}} + \sum_{h'' \notin \{h, h'\}} e^{v_{ih''}} \tilde{a}_{h''}} \right] \\ &\quad + (1 - a_{h'}) \mathbb{E} \left[\frac{e^{v_{ih}}}{1 + e^{v_{ih}} + \sum_{h'' \notin \{h, h'\}} e^{v_{ih''}} \tilde{a}_{h''}} \right]. \end{aligned}$$

Denote temporarily $D_{ih'} = 1 + e^{v_{ih}} + \sum_{h'' \notin \{h, h'\}} e^{v_{ih''}} \tilde{a}_{h''}$. We have that $\frac{\partial R_{ih}}{\partial a_{h'}} = -\mathbb{E} \left[\frac{e^{v_{ih} + v_{ih'}}}{D_{ih'}(D_{ih'} + e^{v_{ih'}})} \right] < 0$. Now note that $D_{ih'} \geq 1 + e^{v_{ih}}$; therefore,

$$\begin{aligned} \sum_{h' \in \mathcal{H} \setminus \{h\}} a_{h'} \left(-\frac{\partial R_{ih}}{\partial a_{h'}} \right) &= \sum_{h' \in \mathcal{H} \setminus \{h\}} a_{h'} \mathbb{E} \left[\frac{e^{v_{ih} + v_{ih'}}}{D_{ih'}(D_{ih'} + e^{v_{ih'}})} \right] = \sum_{h' \in \mathcal{H} \setminus \{h\}} \mathbb{E} \left[\frac{\tilde{a}_{h'} e^{v_{ih} + v_{ih'}}}{D_{ih'}(D_{ih'} + \tilde{a}_{h'} e^{v_{ih'}})} \right] \\ &= \mathbb{E} \left[\frac{e^{v_{ih}}}{\bar{D}_i} \sum_{h' \in \mathcal{H} \setminus \{h\}} \frac{\tilde{a}_{h'} e^{v_{ih'}}}{D_{ih'}} \right] \leq \mathbb{E} \left[\frac{e^{v_{ih}}}{\bar{D}_i} \frac{\sum_{h' \in \mathcal{H} \setminus \{h\}} \tilde{a}_{h'} e^{v_{ih'}}}{1 + e^{v_{ih}}} \right] \\ &= \mathbb{E} \left[\frac{e^{v_{ih}}}{\bar{D}_i} \frac{\bar{D}_i - (1 + e^{v_{ih}})}{1 + e^{v_{ih}}} \right] = \frac{e^{v_{ih}}}{1 + e^{v_{ih}}} - R_{ih} \end{aligned}$$

where $\bar{D}_i = 1 + e^{v_{ih}} + \sum_{h' \neq h} e^{v_{ih'}} \tilde{a}_{h'}$. □

Let $T_{hj}(\mathbf{a}) = \frac{N_{hj}^{L\star}}{\sum_{i \in \mathcal{I}_{jt}^*} R_{ih}(\mathbf{a}) \ell_i^*}$ denote the base fixed-point map. Note that $R_{ih}(\mathbf{a}) \geq \bar{o}_{ih} > 0$ and $R_{ih}(\mathbf{a}) < R_{ih}(\mathbf{0}) = \frac{e^{v_{ih}}}{1 + e^{v_{ih}}}$. Therefore, $\frac{N_{hj}^{L\star}}{\sum_{i \in \mathcal{I}_{jt}^*} \bar{o}_{ih} \ell_i^*} \geq T_{hj}(\mathbf{a}) \geq \frac{N_{hj}^{L\star}}{\sum_{i \in \mathcal{I}_{jt}^*} R_{ih}(\mathbf{0}) \ell_i^*}$. We denote the bounds on the map as $\underline{a}_{hj}$ and $\bar{a}_{hj}$. We now prove the main result.

Proposition A.2 (Admission Rate Contraction Mapping). *Suppose Assumption A.3 holds. Let $\mathcal{A} = \prod_{(h,j)} [\ln(\underline{a}_{hj}), \ln(\bar{a}_{hj})]$ and define the log-transformation of T , $\phi : \mathcal{A} \rightarrow \mathcal{A}$ as*

$$\phi_{hj}(\alpha) = \ln(N_{hj}^{L*}) - \ln\left(\sum_{i \in \mathcal{I}_{hj}^*} R_{ih}(e^\alpha) \ell_i^*\right).$$

Then, ϕ is a contraction on $(\mathcal{A}, \|\cdot\|_\infty)$.

Proof. Note that $\mathcal{A}$ is a non-empty, convex, and compact subset of $\mathbb{R}^{\sum_{h \in \mathcal{H}} J_h}$; endowed with the sup-metric, it is a complete metric space. It is straightforward to see that ϕ is C^∞ on the open set $(-\infty, 0)^{\sum_{h \in \mathcal{H}} J_h}$. Since $T(\prod_{h \in \mathcal{H}} [0, 1]^{J_h}) \subseteq \prod_{h \in \mathcal{H}} \prod_{j \in \mathcal{J}_h} [\underline{a}_{hj}, \bar{a}_{hj}]$, $\phi(\mathcal{A}) \subseteq \mathcal{A}$.

Next, we note that

$$\frac{\partial \phi_{hj}}{\partial \alpha_{h'j'}} = \frac{\sum_{i \in \mathcal{I}_j^*} a_{h'j'} \ell_i^* \left(-\frac{\partial R_{ih}}{\partial a_{h'j'}}\right)}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* R_{ih}} \geq 0.$$

We note that the diagonal of the implied Jacobian is zero, since R_{ih} does not depend on a_h , and the off-diagonal entries are nonnegative since higher acceptance at rivals can decrease admission at h . For each patient i , only the rival's line $j(i, h')$ contributes; derivatives with respect to the rival's other lines vanish. Now, summing the off-diagonal terms, we get

$$\begin{aligned} \|\nabla \phi_{hj}(\alpha)\|_1 &= \frac{\sum_{i \in \mathcal{I}_j^*} \sum_{h' \neq h} a_{h'j'} \ell_i^* \left(-\frac{\partial R_{ih}}{\partial a_{h'j'}}\right)}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* R_{ih}} \\ &\leq \frac{\sum_i \ell_i^* (R_{ih}(0) - R_{ih})}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* R_{ih}} = \frac{\sum_i \ell_i^* R_{ih}(0)}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* R_{ih}} - 1 \\ &\leq \frac{\sum_i \ell_i^* R_{ih}(0)}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* \bar{\sigma}_{ih}} - 1 \end{aligned}$$

where the first inequality follows from the lemma above and the second from $R_{ih} \geq \bar{\sigma}_{ih}$. By Assumption A.3 the last term $\beta \equiv \frac{\sum_i \ell_i^* R_{ih}(0)}{\sum_{i \in \mathcal{I}_j^*} \ell_i^* \bar{\sigma}_{ih}} - 1$ is positive and less than 1. Therefore, by the mean value theorem, we have that for every $\alpha, \alpha' \in \mathcal{A}$ there exists $\tilde{\alpha}$ on a segment between α and α' such that

$$|\phi_{hj}(\alpha) - \phi_{hj}(\alpha')| = |\nabla \phi_{hj}(\tilde{\alpha})(\alpha - \alpha')| \leq \|\nabla \phi_{hj}(\tilde{\alpha})\|_1 \|\alpha - \alpha'\|_\infty \leq \beta \|\alpha - \alpha'\|_\infty$$

where the first inequality is Hölder's inequality and the last one follows from the derivation above. Since β is positive and strictly smaller than unity, ϕ is a contraction map. $\square$

D.3 Demand Estimation Details

We estimate preferences in 24 service-line-market blocks, pooling quarters. We group observationally identical patient-quarters into choice events, retaining their hospital-choice counts and total length of stay. This preserves the likelihood and the contraction mapping described in Main Text Section 6.1. Choice sets contain hospitals serving the line that quarter. Emergency availability requires more than five emergency admissions in the hospital-year. Scheduled Medicare choices require accreditation; scheduled Medicaid and charitable-care choices require more than five scheduled admissions of the corresponding payer type in the year. Out-of-market admissions form the outside option, and we measure hospital distances relative to the nearest out-of-market provider.

We reserve randomly selected year-quarter blocks for estimation and use the remaining blocks for validation. Initial sampling fractions are one half for emergency care and four fifths for scheduled care, reduced in large markets to target 350,000 grouped events. Sampling entire quarters preserves hospital utilization totals. Within each split, we remove hospital-diagnosis cells with zero shares from choice sets; cells with positive estimation shares below 0.1% share a pooled fixed effect.

We initialize preferences with a standard logit model in which every hospital accepts every patient. We regress log share ratios on average patient characteristics to obtain starting values. We then alternate BFGS optimization of travel coefficients with an Anderson-accelerated adjustment of hospital-diagnosis effects to match observed shares. Fitted probabilities and observed bed-day totals initialize acceptance rates in the full model.

The full estimator jointly optimizes travel coefficients and hospital-diagnosis effects. At each proposed preference vector, it solves the length-of-stay contraction mapping and evaluates Equation (11), including outside-option choices. We iterate on log acceptance rates using a smooth approximation to the probability cap. Anderson acceleration combines recent iterations; ordinary iteration provides a fallback when acceleration is unstable. The outer optimizer is L-BFGS, with gradients that account for acceptance rates' dependence on preferences through implicit differentiation. A trust-region optimizer provides a fallback if L-BFGS fails to converge. The gradient stopping target is 10^{-4} for the scaled negative log-likelihood. The inner fixed-point target starts at 10^{-5} and can increase to 10^{-4} when numerical precision limits convergence.

Evaluating the likelihood requires integrating over hospitals' availability. We use 20-node Gauss-Laguerre quadrature for the one-dimensional integral in Equation (11). When preferences concentrate on a few hospitals, this rule can poorly represent states in which those hospitals reject a patient. We identify such choice events using the discrepancy between one and the sum of their admission probabilities. For these events, we enumerate the 16 availability configurations of the four hospitals with the largest unconstrained choice probabilities and integrate only over

the remaining hospitals. The selected hospitals remain fixed during each optimization pass. An initial discrepancy threshold of 10^{-3} selects events for correction; one additional pass can expand the corrected set using a 10^{-4} threshold. This concentrates the additional computation on choices for which rejection by a few preferred hospitals matters most.

After estimating preferences, we recover acceptance rates in both estimation and validation quarters using a more accurate integration rule. An exponential change of variables allows a trapezoid rule with spacing $1/2$ to resolve the long tails associated with small outside-option shares. We require a maximum log-acceptance fixed-point residual of 10^{-5} and admission probabilities that sum to one within 10^{-4} . Preferences remain fixed in validation quarters, with unestimated hospital-diagnosis effects set to zero, while observed utilization determines acceptance rates. Validation therefore assesses preference transfer conditional on utilization. Both sets of predictions supply request flows and acceptance rates for supply estimation. Standard errors use outer products of choice-event scores, with implicit acceptance-rate adjustments and a sandwich variance calculation clustered at the grouped choice event.

D.4 Supply Identification

In this appendix section, we prove the global identification of the hospital's cost parameters. The proof is hospital-by-hospital and omits a time index. In what follows, we fix a hospital h and denote by s the hospital's state, which indexes all the demand and revenue elements the hospital conditions on when solving its optimization problem. We note that we are abusing notation here, since s in the main text is briefly used for a population segment. We denote the distribution of s by $\mathcal{D}_h$, which is a marginal of the data-generating process underlying the data. The argument exploits the structure of the first-order condition in Equation (12): the net marginal revenue of an admission splits into a term that is polynomial in the admission rates and carries the linear parameters $\theta_{hj} = (c_{hj}^S, c_{hj}^F, c_{hj}^V, \tau_{hj})$, and the exponential marginal capacity cost term which carries the overflow rate c_h^O . Because polynomials and exponentials with distinct frequencies are linearly independent on open sets, sufficient variation in reimbursements—holding demand composition and capacity fixed—pins down the overflow rate; variation in case mix and in charitable-care demand then separates the remaining parameters.

We begin by establishing notation. The first-order condition for the admission rate problem of Equation (10) states that for any line with $a_{hj} \in (0, 1)$ it holds that:

$$\bar{p}_{hj} = \mathbf{x}'_{hj} \theta_{hj} + k_{hj}(c_h^O) \quad (\text{A.34})$$

where $\bar{p}_{hj} \equiv \sum_{k \in \mathcal{K}_j} \lambda_k R_{hk} p_{hk}$ is the derivative of expected revenue with respect to the acceptance rate, $\mathbf{x}'_{hj} \equiv (\sum_{k \in \mathcal{K}_j} \lambda_k R_{hk} \omega_k, \Lambda_{hj}, \Lambda_{hj}(1 + 2\Lambda_{hj}a_{hj}), -\Delta S_{hj}^u)$ collects all the linear terms, and $k_{hj}(c^O) \equiv \chi_{hj}(c_h^O) \exp(\chi_{hj}(c^O)a_{hj} - c_h^O b_{hj}) + v_{hj} \Lambda_{hj} \mu_{hj}^L$, where v_{hj} is the Lagrange multiplier on the line's ex-

ante feasibility constraint. Unlike in the main text, we work here with first-order conditions not expressed in terms of per-expected admission level. In addition, we establish three shorthands. First, the average resource intensity of the line's potential admissions,

$$\bar{\omega}_{hj} = \Lambda_{hj}^{-1} \sum_{k \in \mathcal{K}_j} \omega_k \lambda_k R_{hk},$$

so that the first entry of $\mathbf{x}_{hj}$ in Equation (A.34) is $\Lambda_{hj} \bar{\omega}_{hj}$. Second, we make the dependence of the main text's χ_{hj} on the overflow rate explicit, writing, for a rate t ,

$$\chi_{hj}(t) = \Lambda_{hj} (M_{hj}^L(t) - 1), \quad \text{so that } \chi_{hj} = \chi_{hj}(c_h^o).$$

Third, we define the slack margin $\delta_{hj}(t) = \chi_{hj}(t) - t \Lambda_{hj} \mu_{hj}^L$, which will separate slack from binding lines in the bed program below.

Combining the allocation stage (7) with the deterministic-equivalent line problem (10) gives the hospital's joint problem

$$\max_{\substack{\mathbf{a} \in [0,1]^h \\ \mathbf{b} \geq \mathbf{0}}} \sum_{j \in \mathcal{J}_h} \left[(\bar{p}_{hj} - \Lambda_{hj} (c_{hj}^S \bar{\omega}_{hj} + c_{hj}^F + c_{hj}^V) + \tau_{hj} \Delta S_{hj}^u) a_{hj} - c_{hj}^V \Lambda_{hj}^2 a_{hj}^2 - e^{\chi_{hj}(c_h^o) a_{hj} - c_h^o b_{hj}} \right] \quad (\text{A.35})$$

subject to $\Lambda_{hj} \mu_{hj}^L a_{hj} \leq b_{hj}$ for every j and $\sum_j b_{hj} \leq B_h$. In what follows, we omit the h index from c_h^o unless needed for clarity. When $c_{hj}^V > 0$ for all j , the objective is strictly jointly concave in $(\mathbf{a}, \mathbf{b})$ and the optimizer $(\mathbf{a}_h(s; \boldsymbol{\theta}, c^o), \mathbf{b}_h(s; \boldsymbol{\theta}, c^o))$ is unique by main text Proposition 1. We denote by $\mathbf{a}_h^*(s) := \mathbf{a}_h(s; \boldsymbol{\theta}^*, c^{o*})$ the vector of optimal admission rates at the true parameter vector.

Let $v_{hj} \geq 0$ denote the multiplier on line j 's ex-ante capacity constraint and $\eta_h \geq 0$ the multiplier on the hospital's total resource constraint. Because the overflow term is strictly decreasing in b_{hj} , the first-order condition in b_{hj} ,

$$c^o e^{\chi_{hj}(c^o) a_{hj} - c^o b_{hj}} = \eta_h - v_{hj}, \quad (\text{A.36})$$

forces $\eta_h > 0$, meaning that capacity always binds. Intuitively, capacity is always better allocated than not allocated, given each line's control over admission rates.

At any interior admission rate $a_{hj} \in (0, 1)$, stationarity in a_{hj} yields Equation (A.34), restated here for clarity

$$\bar{p}_{hj} = \mathbf{x}_{hj}(a_{hj})' \boldsymbol{\theta}_{hj} + k_{hj}, \quad \mathbf{x}_{hj}(a) := \begin{pmatrix} \Lambda_{hj} \bar{\omega}_{hj} \\ \Lambda_{hj} \\ \Lambda_{hj} (1 + 2\Lambda_{hj} a) \\ -\Delta S_{hj}^u \end{pmatrix}, \quad (\text{A.37})$$

where

$$k_{hj} := \chi_{hj}(c^o) e^{\chi_{hj}(c^o) a_{hj} - c^o b_{hj}} + v_{hj} \Lambda_{hj} \mu_{hj}^L \quad (\text{A.38})$$

is the marginal capacity cost: the marginal expected overflow cost of an admission plus the shadow value of the bed-days it consumes. Although k_{hj} is defined through the unobserved allocation (b_{hj}, v_{hj}) , Lemma A.4 below shows that it is a known closed-form functional of the admission rates, the total capacity, and the rate c^o alone; the internal allocation never needs to be observed.

Before stating our results, we define what a regular state is, what we mean by global identification, and our key set of assumptions.

Definition A.1 (Regular states; global identification). *The set of regular states is $\mathcal{R}_h := \{s \in \text{supp}(\mathcal{D}_h) : \mathbf{a}_h^*(s) \in (0, 1)^{J_h} \text{ and } \underline{B}(\mathbf{a}_h^*(s)) < B_h(s)\}$, where $\underline{B}(\mathbf{a}) := \sum_j \Lambda_{hj} \mu_{hj}^L a_{hj}$ is the expected admitted load. The parameter (θ^*, c^{o*}) is globally identified on Θ_h if every $(\theta', c^{o'}) \in \Theta_h$ satisfying $\mathbf{a}_h(s; \theta', c^{o'}) = \mathbf{a}_h(s; \theta^*, c^{o*})$ for all $s \in \text{supp}(\mathcal{D}_h)$ also satisfies $(\theta', c^{o'}) = (\theta^*, c^{o*})$.*

Assumption A.4 (Identification conditions).

1. (Affine surgical costs.) $c_{hk} = c_{hj(k)}^S \omega_k + c_{hj(k)}^F$ for every k , where $j(k)$ is the line serving type k .
2. (Common overflow rate.) c_h^o applies to every $j \in \mathcal{J}_h$.
3. (Nondegenerate primitives.) For every $s \in \text{supp}(\mathcal{D}_h)$ and $j \in \mathcal{J}_h$, $\Lambda_{hj}(s) > 0$; and each type's length of stay ℓ^k is strictly positive with a finite exponential moment, $\bar{c}_k := \sup\{t \geq 0 : \mathbb{E}[e^{t\ell^k}] < \infty\} > 0$, so that $\bar{c}_h^o := \min_k \bar{c}_k > 0$.
4. (Parameter space.) $(\theta, c^o) \in \Theta_h := \{(\theta, c^o) : c_{hj}^V > 0, \tau_{hj} \geq 0 \forall j \in \mathcal{J}_h, c^o \in (0, \bar{c}_h^o)\}$.
5. (Case-mix and charitable-demand relevance.) For each line j , the realized set

$$\left\{ \left(\bar{\omega}_{hj}(s), \Lambda_{hj}(s) a_{hj}^*(s), \Delta S_{hj}^u(s) / \Lambda_{hj}(s) \right) : s \in \mathcal{R}_h \right\} \subseteq \mathbb{R}^3$$

is not contained in a single affine plane (equivalently, it contains four affinely independent points).

6. (Conditional price variation in a full-slack region.) There exist values $(\Lambda_{hj}^0, \bar{\omega}_{hj}^0, M_{hj}^{L,0}(\cdot), \Delta S_{hj}^{u,0})_{j \in \mathcal{J}_h}$ and B_h^0 of the line-level composites, and a nonempty open set $U \subseteq \mathbb{R}^{J_h}$, such that the conditioned state set

$$O := \left\{ s \in \text{supp}(\mathcal{D}_h) : \left(\Lambda_{hj}, \bar{\omega}_{hj}, M_{hj}^L, \Delta S_{hj}^u, B_h \right)(s) = \left(\Lambda_{hj}^0, \bar{\omega}_{hj}^0, M_{hj}^{L,0}, \Delta S_{hj}^{u,0}, B_h^0 \right) \forall j, \bar{\mathbf{p}}_h(s) \in U \right\}$$

satisfies: (a) price richness, $\bar{\mathbf{p}}_h(O) = U$; and (b) interior, full-slack truth, for every $s \in O$, $\mathbf{a}^*(s) \in (0, 1)^{J_h}$ and

$$E_{\text{fs}}(\mathbf{a}^*(s), B_h^0; c^{o*}) < \delta_{hj}(c^{o*}) a_{hj}^*(s) \text{ for every } j, \quad E_{\text{fs}}(\mathbf{a}, B; c^o) := \frac{1}{J_h} \left[\sum_j \chi_{hj}(c^o) a_{hj} - c^o B \right].$$

Parts 1–4 are maintained model restrictions: part 1 restates the cost parametrization adopted in Section 6; part 2 restates the overflow-cost specification of Section 5, which makes bed-days

a fungible resource priced by a single internal shadow price, so that the clearing analysis below aggregates the hospital's capacity market into a one-dimensional index; part 3 restates the primitives of the model, matching the hypotheses of Proposition 1; and part 4 defines the space on which identification is claimed. Parts 5–6 are the substantive variation requirements. Part 6 is an “identification at a configuration” condition: reimbursement variation (e.g., from fee-schedule and payer-mix changes) that is rich in all J_h directions while market size, case mix, length-of-stay composition, charitable demand, and capacity are held fixed, on a region where the hospital operates with slack in every line; part 6(b) holds when capacity is large relative to expected load at the truth, for instance when arrival intensities are small relative to B_h^0 . The conditions in this assumption imply that lines are slack at states in O , which implies that these states are regular as defined above. Note also that part 5 cannot be satisfied by states in O alone: on O the triples of part 5 all lie on the vertical line through $(\bar{\omega}_{hj}^0, \cdot, \Delta S_{hj}^{u,0}/\Lambda_{hj}^0)$, so the case-mix and charitable-demand variation it requires must come from states outside O .

Having established our assumption, we move to prove two preliminary lemmas that will be used to establish identification.

Lemma A.3 (Curvature of the length-of-stay transform). *Under Assumption A.4.3, for every line j , every state, and every $t \in (0, \bar{c}_h^0)$: $\delta_{hj}(t) > 0$, and both $\chi_{hj}(t)$ and $\delta_{hj}(t)$ are strictly increasing on $(0, \bar{c}_h^0)$.*

Proof. The moment generating function for length of stay, $M_{hj}^L(t) = \sum_{k \in \mathcal{K}_j} \frac{\lambda_k R_{hk}}{\Lambda_{hj}} \mathbb{E}[e^{t\ell^k}]$, is finite on $[0, \bar{c}_h^0)$ with $M_{hj}^L(0) = 1$, $(M_{hj}^L)'(0) = \mu_{hj}^L$, and $(M_{hj}^L)''(t) = \mathbb{E}_{hj}[\ell^2 e^{t\ell}] > 0$ because lengths of stay are strictly positive. Strict convexity places M_{hj}^L strictly above its tangent at zero, $M_{hj}^L(t) > 1 + t\mu_{hj}^L$ for $t > 0$; multiplying by $\Lambda_{hj} > 0$ gives $\delta_{hj}(t) > 0$. Moreover $\chi'_{hj}(t) = \Lambda_{hj}(M_{hj}^L)'(t) > 0$ and $\delta'_{hj}(t) = \Lambda_{hj}((M_{hj}^L)'(t) - \mu_{hj}^L) > 0$, the latter because $(M_{hj}^L)'$ is strictly increasing with $(M_{hj}^L)'(0) = \mu_{hj}^L$. $\square$

Lemma A.4 (Bed allocation and capacity clearing). *Suppose Assumptions A.4.2 and A.4.3 hold. Fix a state, a rate $c^0 \in (0, \bar{c}_h^0)$, and $\mathbf{a} \in (0, 1)^{J_h}$ with slack $\underline{B}(\mathbf{a}) < B_h$. Given $\mathbf{a}$, the bed allocation solving (A.35) is the unique solution of the strictly concave program $\max_{\mathbf{b}} \{-\sum_j e^{\chi_{hj}(c^0)a_{hj} - c^0 b_{hj}}\}$ subject to the two constraint families, and:*

1. $\sum_j b_{hj} = B_h$ and $b_{hj} = \max\{(\chi_{hj}(c^0)a_{hj} - E)/c^0, \Lambda_{hj}\mu_{hj}^L a_{hj}\}$, where $E = E(\mathbf{a}, B_h; c^0)$ is the unique solution of the capacity-clearing equation

$$\sum_{j \in \mathcal{J}_h} \max\left\{\frac{\chi_{hj}(c^0)a_{hj} - E}{c^0}, \Lambda_{hj}\mu_{hj}^L a_{hj}\right\} = B_h. \quad (\text{A.39})$$

The multipliers are unique and given by $\eta_h = c^0 e^E$ and $v_{hj} = c^0 \max\{e^E - e^{\delta_{hj}(c^0)a_{hj}}, 0\}$; consequently

$$k_{hj}(\mathbf{a}, B_h; c^0) = \begin{cases} \chi_{hj}(c^0) e^E, & E \leq \delta_{hj}(c^0)a_{hj} \text{ (slack line),} \\ c^0 \Lambda_{hj}\mu_{hj}^L e^E + \delta_{hj}(c^0) e^{\delta_{hj}(c^0)a_{hj}}, & E \geq \delta_{hj}(c^0)a_{hj} \text{ (binding line),} \end{cases} \quad (\text{A.40})$$

the two branches agreeing when $E = \delta_{hj}(c^o)a_{hj}$.

2. At the clearing price E at least one line is strictly slack: $E < \delta_{hj}(c^o)a_{hj}$ for some j .
3. $(\mathbf{a}, B_h) \mapsto E(\mathbf{a}, B_h; c^o)$ is continuous on $\{\underline{B}(\mathbf{a}) < B_h\}$.
4. If $\tilde{\mathcal{J}}_h \subseteq \mathcal{J}_h$ is nonempty with $E \leq \delta_{hj}(c^o)a_{hj}$ for all $j \in \tilde{\mathcal{J}}_h$ and $E \geq \delta_{hj}(c^o)a_{hj}$ for all $j \notin \tilde{\mathcal{J}}_h$, then

$$E = E_{\tilde{\mathcal{J}}_h}(\mathbf{a}, B_h; c^o) := \frac{1}{\tilde{J}_h} \left[\sum_{j \in \tilde{\mathcal{J}}_h} \chi_{hj}(c^o)a_{hj} + c^o \left(\sum_{j \notin \tilde{\mathcal{J}}_h} \Lambda_{hj} \mu_{hj}^L a_{hj} - B_h \right) \right]. \quad (\text{A.41})$$

In particular, under full slack ($\tilde{\mathcal{J}}_h = \mathcal{J}_h$), $E = E_{\text{fs}}(\mathbf{a}, B_h; c^o)$ and $k_{hj} = \chi_{hj}(c^o)e^E$ for every j .

Proof. The bed program is strictly concave with linear constraints and a Slater point ($b_{hj} = \Lambda_{hj} \mu_{hj}^L a_{hj} + (B_h - \underline{B})/J_h$), so the Karush–Kuhn–Tucker conditions are necessary and sufficient, and the solution is unique. If $v_{hj} = 0$, (A.36) gives $b_{hj} = (\chi_{hj}(c^o)a_{hj} - E)/c^o$ with $E := \log(\eta_h/c^o)$, and feasibility $b_{hj} \geq \Lambda_{hj} \mu_{hj}^L a_{hj}$ reads $E \leq \delta_{hj}(c^o)a_{hj}$. If $v_{hj} > 0$, complementary slackness gives $b_{hj} = \Lambda_{hj} \mu_{hj}^L a_{hj}$ and (A.36) gives $v_{hj} = c^o(e^E - e^{\delta_{hj}(c^o)a_{hj}}) > 0$, i.e., $E > \delta_{hj}(c^o)a_{hj}$. Both cases are summarized by the stated max formula, whose sum over lines must exhaust the capacity that (A.36) forces to bind, giving (A.39); substituting the multiplier formulas into (A.38) yields (A.40) after collecting $\chi_{hj} - c^o \Lambda_{hj} \mu_{hj}^L = \delta_{hj}$.

Let $D(E)$ denote the left-hand side of (A.39), which is continuous and nonincreasing in E , diverges as $E \rightarrow -\infty$, equals $\underline{B} < B_h$ for $E \geq \max_j \delta_{hj}(c^o)a_{hj}$, and is strictly decreasing below that threshold (the line attaining the maximum is interior there); hence the clearing equation $D(E) = B_h$ has a unique solution, and, since $D(E) = B_h > \underline{B}$, some summand exceeds its floor, proving item 2 of the lemma.

For item 3, fix $(\mathbf{a}_0, B_0)$ with $\underline{B}(\mathbf{a}_0) < B_0$, write $E_0 := E(\mathbf{a}_0, B_0)$, and let $\varepsilon > 0$. By item 2, E_0 lies strictly below the threshold $\max_j \delta_{hj}(c^o)a_{0,hj}$, below which $D(\cdot; \mathbf{a}_0)$ is strictly decreasing; hence $D(E_0 - \varepsilon; \mathbf{a}_0) > B_0$, and $D(E_0 + \varepsilon; \mathbf{a}_0) < B_0$. Since $D(E; \mathbf{a})$ is jointly continuous, both strict inequalities are preserved on a neighborhood N of $(\mathbf{a}_0, B_0)$ within $\{\underline{B}(\mathbf{a}) < B\}$: $D(E_0 - \varepsilon; \mathbf{a}) > B > D(E_0 + \varepsilon; \mathbf{a})$ on N . For any $(\mathbf{a}, B) \in N$, monotonicity of $D(\cdot; \mathbf{a})$ then excludes roots outside $(E_0 - \varepsilon, E_0 + \varepsilon)$, so $|E(\mathbf{a}, B) - E_0| < \varepsilon$, proving continuity.

For item 4, under the stated weak inequalities, each summand of D equals the indicated branch (the branches coincide where an inequality binds with equality), so the clearing equation is linear in E and solves to (A.41); the expressions for k_{hj} follow from (A.40). $\square$

Lemma A.4 has a consequence worth emphasizing before proceeding: given the admission rates and the total capacity, the entire congestion side of the problem—the bed allocation, the clearing price E , the multipliers, and hence every k_{hj} —is a known functional of observed objects and of the overflow rate c^o alone. No component of $\boldsymbol{\theta}$ enters it, and no bed-level data are required

to evaluate it. Now consider two parameter vectors $(\theta^*, c^{o*}), (\theta', c^{o'}) \in \Theta_h$ that generate the same admission-rate response at a state where $\mathbf{a}^*(s)$ is interior and $\underline{B}(\mathbf{a}^*(s)) < B_h(s)$. Each parameter's bed allocation must solve the bed program of Lemma A.4 at the common $\mathbf{a}^*(s)$ under its own rate, so subtracting the two versions of (A.37) at the common observed $(\bar{p}_{hj}, \mathbf{a}^*)$ eliminates the price:

$$\mathbf{x}_{hj}(a_{hj}^*)' \Delta_j = k_{hj}(\mathbf{a}^*, B_h; c^{o*}) - k_{hj}(\mathbf{a}^*, B_h; c^{o'}), \quad \Delta_j := \theta'_{hj} - \theta^*_{hj}, \quad (\text{R})$$

where each side is evaluated at the same observed data.⁵

Finally, we need one additional technical result to show that, given the assumptions established so far, the observed set of acceptance rates forms an open set.

Proposition A.3 (Open acceptance set). *Suppose Assumptions A.4.1 to A.4.4 and A.4.6 hold. Then the response map $\bar{\mathbf{p}} \mapsto \mathbf{a}^*(\bar{\mathbf{p}})$ is C^1 on U with everywhere nonsingular derivative*

$$\partial \mathbf{a}^* / \partial \bar{\mathbf{p}} = \left(\text{diag}(2c_{hj}^{V*}(\Lambda_{hj}^0)^2) + \frac{1}{J_h} \chi^E \chi' \right)^{-1},$$

where $\chi = (\chi_{hi}(c^{o*}))_i$. Consequently $\mathcal{W} := \{\mathbf{a}^*(s) : s \in O\}$ is open in $\mathbb{R}^{J_h}$.

Proof. The feasible set of (A.35) is compact and does not depend on $\bar{\mathbf{p}}$, and the objective is jointly continuous, so by strict concavity and Berge's maximum theorem the optimizer is single-valued and upper hemicontinuous, hence a continuous function of $\bar{\mathbf{p}}$ on U (Aliprantis and Border, 2006, Theorems 17.31 and 17.6). By Assumption A.4.6(b), for every $\bar{\mathbf{p}} \in U$, the optimum is interior and full slack, so it solves the smooth system $F(\mathbf{a}; \bar{\mathbf{p}}) = \mathbf{0}$ with

$$F_j(\mathbf{a}; \bar{\mathbf{p}}) = \bar{p}_j - \mathbf{x}_{hj}(a_j)' \theta_{hj}^* - \chi_{hj}(c^{o*}) \exp(E_{fs}(\mathbf{a}, B_h^0; c^{o*})).$$

Its Jacobian in $\mathbf{a}$ is $-(\text{diag}(2c_{hj}^{V*}(\Lambda_{hj}^0)^2) + \frac{1}{J_h} \chi^E \chi')$, a negative of a positive diagonal plus a nonnegative rank-one matrix. Denote $\rho_j = 2c_{hj}^{V*}(\Lambda_{hj}^0)^2$; then the absolute value of the determinant of the Jacobian is $(\prod_j \rho_j) (1 + \frac{e^E}{J_h} \sum_i \chi_{hi}(c^{o*})^2 / \rho_i) > 0$ by the matrix determinant lemma (Horn and Johnson, 2013, §0.8.5). Fix $\bar{\mathbf{p}}_0 \in U$. The implicit function theorem yields neighborhoods $W_0 \ni \mathbf{a}^*(\bar{\mathbf{p}}_0)$ and $U_0 \ni \bar{\mathbf{p}}_0$ and a C^1 map $\phi : U_0 \rightarrow W_0$ that is the unique root of $F(\cdot; \bar{\mathbf{p}})$ in W_0 ; since $\mathbf{a}^*$ is continuous and solves $F = \mathbf{0}$ for every $\bar{\mathbf{p}} \in U$, it coincides with ϕ near $\bar{\mathbf{p}}_0$ and inherits the stated derivative. A C^1 map with nonsingular derivative is open by the inverse function theorem, so $\mathcal{W} = \mathbf{a}^*(U)$ is open. $\square$

We are now in a position to prove identification. The identification of the overflow rate rests

⁵Stationarity at an interior optimum requires only differentiability, so (R) does not presume anything about the rival beyond membership in Θ_h . The requirement $\underline{B}(\mathbf{a}^*) < B_h$ ensures, via items 1 and 2 of Lemma A.4, that the multipliers—and hence k_{hj} —are uniquely determined by the data and the rate; if every line constraint held with equality, the capacity multiplier would not be point-identified from the bed program and (R) could fail.

on the following classical independence property, which we state in the form we need. Its content is standard; we include a short proof for completeness. It invokes no model assumptions.

Lemma A.5 (Exponential–polynomial independence). *Let $r_1, \dots, r_m \in \mathbb{R}^d$ be pairwise distinct and $q_1, \dots, q_m$ real polynomials. If $\sum_{k=1}^m q_k(w) e^{\langle r_k, w \rangle} = 0$ for all w in a nonempty open set $V \subseteq \mathbb{R}^d$, then $q_k \equiv 0$ for every k .*

Proof. Step 1: reduction to a line segment with distinct projected frequencies. The hypothesis is an identity on an open subset of $\mathbb{R}^d$; to bring one-dimensional methods to bear, we restrict it to a segment, choosing the direction so that the frequencies remain distinct after projection. For $k \neq l$ the vector $r_k - r_l$ is nonzero, so $\{v \in \mathbb{R}^d : \langle r_k - r_l, v \rangle = 0\}$ is a proper hyperplane, and a finite union of proper hyperplanes cannot cover $\mathbb{R}^d$. Any v outside their union therefore satisfies $\langle r_k, v \rangle \neq \langle r_l, v \rangle$ for all $k \neq l$; write $\mu_k := \langle r_k, v \rangle$, pairwise distinct. Fix $w \in V$ and $\varepsilon > 0$ with $w + tv \in V$ for $|t| < \varepsilon$. Evaluating the hypothesis at $w + tv$ and using $e^{\langle r_k, w+tv \rangle} = e^{\langle r_k, w \rangle} e^{\mu_k t}$, we obtain, with $\tilde{q}_k(t) := q_k(w + tv) e^{\langle r_k, w \rangle}$ a univariate polynomial of degree $d_k := \deg \tilde{q}_k \leq \deg q_k$,

$$y(t) := \sum_{k=1}^m \tilde{q}_k(t) e^{\mu_k t} = 0 \quad \text{for all } |t| < \varepsilon. \quad (\text{A.42})$$

Step 2: y solves a linear constant-coefficient equation on all of $\mathbb{R}$. Consider the order- N operator $L := \prod_{k=1}^m (d/dt - \mu_k)^{1+d_k}$, $N := \sum_k (1 + d_k)$. For any smooth f and any μ , the product rule gives $(d/dt - \mu)(f(t)e^{\mu t}) = f'(t)e^{\mu t}$: the factor $(d/dt - \mu)$ passes through $e^{\mu t}$ and differentiates the prefactor. Iterating, $(d/dt - \mu_k)^{1+d_k}(t^j e^{\mu_k t}) = (t^j)^{(1+d_k)} e^{\mu_k t} = 0$ for $0 \leq j \leq d_k$, since the $(1 + d_k)$ -th derivative of a polynomial of degree at most d_k vanishes. Because the factors of L are polynomials in the single operator d/dt , they commute, so applying the k -th factor first shows $L(t^j e^{\mu_k t}) = 0$ for every k and $0 \leq j \leq d_k$. By linearity, $Ly = 0$ —and this holds on all of $\mathbb{R}$, since y is an entire function of t , even though (A.42) asserts $y = 0$ only on $(-\varepsilon, \varepsilon)$.

Step 3: vanishing on an interval upgrades to vanishing on $\mathbb{R}$. Because $Ly = 0$ is a linear homogeneous equation of order N with constant coefficients, its initial value problem admits a unique global solution (Coddington and Levinson, 1955, Ch. 3). By (A.42), y vanishes identically near $t = 0$, so $y(0) = y'(0) = \dots = y^{(N-1)}(0) = 0$; the zero function solves the same initial value problem, and uniqueness forces $y \equiv 0$ on $\mathbb{R}$.

Step 4: linear independence forces every coefficient to vanish. Since the μ_k are pairwise distinct, the characteristic polynomial of L is $\prod_k (z - \mu_k)^{1+d_k}$ with roots μ_k of multiplicity $1 + d_k$, and the N functions $\{t^j e^{\mu_k t} : 1 \leq k \leq m, 0 \leq j \leq d_k\}$ form a fundamental system for $Ly = 0$: a basis of its N -dimensional solution space, in particular linearly independent on $\mathbb{R}$ (Coddington and Levinson, 1955, Ch. 3). Expanding each polynomial as $\tilde{q}_k(t) = \sum_{j=0}^{d_k} c_{kj} t^j$, Step 3 states that $\sum_{k,j} c_{kj} t^j e^{\mu_k t}$ is the zero element of that solution space; expanding zero in a basis has only the trivial representation, so $c_{kj} = 0$ for all k, j , i.e., $\tilde{q}_k \equiv 0$ for every k .

Step 5: the base point sweeps out V . Evaluating $\tilde{q}_k \equiv 0$ at $t = 0$ gives $q_k(w) e^{\langle r_k, w \rangle} = 0$, hence $q_k(w) = 0$. Since $w \in V$ was arbitrary, each polynomial q_k vanishes on the nonempty open set V , and a polynomial vanishing on a nonempty open set is identically zero. $\square$

Proposition A.4 (The overflow rate is identified). *Suppose $J_h \geq 2$ and Assumptions A.4.1 to A.4.4 and A.4.6 hold. If $(\theta', c^{o'}) \in \Theta_h$ is observationally equivalent to (θ^*, c^{o*}) , then $c^{o'} = c^{o*}$.*

Proof. Work on the open set $\mathcal{W}$ of Proposition A.3 and write $w = \alpha^* \in \mathcal{W}$. On $\mathcal{O}$, the composites are fixed, so the left side of (R) is a degree-one polynomial in w with constant coefficients that depends only on its own coordinate,

$$q_j(w) := x_{hj}(w_j)' \Delta_j = \gamma_{0j} + \gamma_{1j} w_j, \quad \gamma_{0j} = \Lambda_{hj}^0 (\Delta_j^S \bar{\omega}_{hj}^0 + \Delta_j^F + \Delta_j^V) - \Delta S_{hj}^{u,0} \Delta_j^\tau, \quad \gamma_{1j} = 2\Delta_j^V (\Lambda_{hj}^0)^2,$$

while, by Assumption A.4.6(b) and Lemma A.4.4, the truth's side of (R) is

$$k_{hj}(w; c^{o*}) = \chi_{hj}(c^{o*}) e^{-c^{o*} B_h^0 / J_h} e^{\langle r^*, w \rangle}, \quad r^* := (\chi_{hj'}(c^{o*}) / J_h)_{j' \in \mathcal{J}_h} \in \mathbb{R}_{++}^{J_h}.$$

Since $\underline{B}(w) < B_h^0$ on $\mathcal{W}$, the rival's clearing price $E'(w) := E(w, B_h^0; c^{o'})$ is well defined and continuous there by Lemma A.4.1 and A.4.3.

Step 1: the rival's regime is constant on a nonempty open subset. For each nonempty $\tilde{\mathcal{J}}_h \subseteq \mathcal{J}_h$ define

$$G_{\tilde{\mathcal{J}}_h} := \{w \in \mathcal{W} : E'(w) \leq \delta_{hj}(c^{o'}) w_j \forall j \in \tilde{\mathcal{J}}_h, E'(w) \geq \delta_{hj}(c^{o'}) w_j \forall j \notin \tilde{\mathcal{J}}_h\},$$

a relatively closed subset of $\mathcal{W}$. Every $w \in \mathcal{W}$ lies in $G_{\tilde{\mathcal{J}}_h(w)}$, where $\tilde{\mathcal{J}}_h(w) := \{j : E'(w) < \delta_{hj}(c^{o'}) w_j\}$ is nonempty by Lemma A.4.2. An open subset of $\mathbb{R}^{J_h}$ is a Baire space, so the finite closed cover $\{G_{\tilde{\mathcal{J}}_h}\}_{\tilde{\mathcal{J}}_h}$ contains a set $G_{\tilde{\mathcal{J}}_h^0}$ with a nonempty open subset V (Munkres, 2000, §48). On V , Lemma A.4.4 applies to the rival with slack set $\tilde{\mathcal{J}}_h^0$, of cardinality $\tilde{J}_h^0$.

Step 2: $\tilde{\mathcal{J}}_h^0 = \mathcal{J}_h$. Suppose some line $j \notin \tilde{\mathcal{J}}_h^0$. By Lemma A.4.4, on V the line- j restriction (R) reads

$$\chi_{hj}(c^{o*}) e^{-c^{o*} B_h^0 / J_h} e^{\langle r^*, w \rangle} - c^{o'} \Lambda_{hj} \mu_{hj}^L e^{-c^{o'} B_h^0 / \tilde{J}_h^0} e^{\langle r^0, w \rangle} - \delta_{hj}(c^{o'}) e^{\langle d_j, w \rangle} - q_j(w) e^{\langle 0, w \rangle} = 0,$$

where $r^0 \in \mathbb{R}_{++}^{J_h}$ has entries $\chi_{hj'}(c^{o'}) / \tilde{J}_h^0$ for $j' \in \tilde{\mathcal{J}}_h^0$ and $c^{o'} \Lambda_{hj'} \mu_{hj'}^L / \tilde{J}_h^0$ for $j' \notin \tilde{\mathcal{J}}_h^0$, and $d_j := \delta_{hj}(c^{o'}) e_j$, with e_j the j -th standard basis vector. Since $J_h \geq 2$, there is a line $j' \neq j$ at whose coordinate d_j vanishes while r^* and r^0 are strictly positive, and $d_j \neq 0$ by Lemma A.3; hence $d_j \notin \{r^*, r^0, 0\}$. Grouping any coincident frequencies among $\{r^*, r^0, 0\}$ and applying Lemma A.5 on V , every grouped coefficient must vanish; in particular the coefficient of $e^{\langle d_j, w \rangle}$, the nonzero constant $\delta_{hj}(c^{o'})$, must vanish—a contradiction. Hence $\tilde{\mathcal{J}}_h^0 = \mathcal{J}_h$: the rival is full slack on V , and for every line j and $w \in V$, $k_{hj}(w; c^{o'}) = \chi_{hj}(c^{o'}) e^{-c^{o'} B_h^0 / J_h} e^{\langle r', w \rangle}$, with $r' := (\chi_{hj'}(c^{o'}) / J_h)_{j' \in \mathcal{J}_h}$.

Step 3: $c^{o'} = c^{o*}$. Suppose not. Each $\chi_{hj}(\cdot)$ is strictly increasing (Lemma A.3), so $r' \neq r^*$, and both differ from 0. Fixing any line j , restriction (R) on V is then a vanishing combination of exponentials with three pairwise-distinct frequencies and polynomial coefficients; by Lemma A.5 the coefficient of $e^{\langle r^*, w \rangle}$, namely $\chi_{hj}(c^{o*})e^{-c^{o*}B_h^0/J_h} > 0$, must vanish—a contradiction. Hence $c^{o'} = c^{o*}$. $\square$

Proposition A.5 (The linear parameters are identified). *Suppose Assumptions A.4.1 to A.4.5 hold and $(\theta', c^{o'}) \in \Theta_h$ is observationally equivalent to (θ^*, c^{o*}) with $c^{o'} = c^{o*}$. Then $\Delta_j = 0$ for every $j \in \mathcal{J}_h$, and hence $\theta' = \theta^*$.*

Proof. At any regular state $s \in \mathcal{R}_h$, Lemma A.4 determines the bed allocation, the multipliers, and hence every k_{hj} as functionals of the observed data and of the rate alone. With $c^{o'} = c^{o*}$ the two parameter vectors therefore share k_{hj} at every regular state, and (R) reduces to $x_{hj}(a_{hj}^*(s))' \Delta_j = 0$. Dividing by $\Lambda_{hj}(s) > 0$,

$$\Delta_j^S \bar{\omega}_{hj}(s) + (\Delta_j^F + \Delta_j^V) + 2\Delta_j^V \Lambda_{hj}(s) a_{hj}^*(s) - \Delta_j^\tau \frac{\Delta S_{hj}^u(s)}{\Lambda_{hj}(s)} = 0 \quad \text{for all } s \in \mathcal{R}_h,$$

an affine function of $(x, y, z) = (\bar{\omega}_{hj}, \Lambda_{hj} a_{hj}^*, \Delta S_{hj}^u / \Lambda_{hj}) \in \mathbb{R}^3$ vanishing on the realized set of triples. By Assumption A.4.5, that set contains four affinely independent points, so the affine function is identically zero: $\Delta_j^S = 0$, $\Delta_j^V = 0$, $\Delta_j^\tau = 0$, and then $\Delta_j^F = 0$. $\square$

Theorem A.1 (Global identification). *Suppose $J_h \geq 2$ and Assumption A.4 holds. Then (θ^*, c^{o*}) is globally identified on Θ_h in the sense of Definition A.1.*

Proof. Immediate from Propositions A.4 and A.5. $\square$

We conclude with a few remarks regarding the proof and its structure. First, we note that the identifying restrictions do not rely on the line-level allocations b_{hj} and the multipliers (η_h, ν_{hj}) , which appear only as internal optimality objects: at any regular state, Lemma A.4.1 shows they are the unique solution of the bed program given (a^*, B_h) and a candidate rate, so the marginal capacity cost $k_{hj}(a^*, B_h; c^0)$ is computable without observing b_{hj} . However, this approach has two caveats. First, it relies on the regularity condition $\underline{B}(a^*) < B_h$: at the knife edge $\underline{B}(a^*) = B_h$, the multipliers are not point-identified from the bed program, which is why Definition A.1 and Assumption A.4.5 restrict attention to regular states. Second, it relies on the common overflow rate of Assumption A.4.2, which collapses the internal capacity market into the single index $E(a, B_h; c^0)$; with line-specific rates, recovering the implied allocation from (a, B_h) alone would be substantially more challenging.

The second thing to note is that the full-slack requirement in Assumption A.4.6(b) is expositively convenient but not essential. What matters is that the truth's regime be constant on some open subset of $\mathcal{W}$. If the truth's constant regime contains a binding line j , identification of

c^o is immediate: the truth's own-coordinate exponential $\delta_{hj}(c^{o*})e^{\delta_{hj}(c^{o*})w_j}$ can only be matched, by Lemma A.5, by a rival term with the same frequency, which requires the rival to bind at j with $\delta_{hj}(c^{o'}) = \delta_{hj}(c^{o*})$; strict monotonicity of $\delta_{hj}(\cdot)$ (Lemma A.3) then gives $c^{o'} = c^{o*}$. If it contains none, the regime is full slack and Steps 2–3 of Proposition A.4 apply.

Finally, conditional reimbursement variation on the full-slack region does not, by itself, identify all supply-side primitives. Once $c^{o'} = c^{o*}$, restriction (R) on the open set V of Proposition A.4 reduces to $q_j \equiv 0$, i.e., $\Delta_j^V = 0$ and $\Lambda_{hj}^0(\Delta_j^S \bar{\omega}_{hj}^0 + \Delta_j^F) - \Delta S_{hj}^{u,0} \Delta_j^\tau = 0$. Therefore, price variation identifies the overflow rate c^o , the congestion cost c_{hj}^V , and the composite net per-admission cost $c_{hj}^S \bar{\omega}_{hj}^0 + c_{hj}^F - \tau_{hj} \Delta S_{hj}^{u,0} / \Lambda_{hj}^0$, but cannot separate the case-mix-scaled cost, the fixed cost, and the charitable preference. Variation in case mix and in charitable-care demand across regular states supplies the missing directions. This formalizes the observation in Section 6 that, absent variation in demand among those seeking charitable care, neither the cost of charitable care nor the hospital's preference for uninsured surplus is identified.

D.5 Supply Estimation

We estimate costs using general medical and surgical hospitals with at least 50 total beds from 2013 through 2019, when the required payment estimates are available. We require at least two retained quarters per hospital-year and two years per hospital, with emergency and scheduled insured care observed. Capacity equals staffed beds, excluding neonatal intensive-care and long-term-care beds, times days in the quarter, net of utilization by excluded out-of-state, urgent-care, and law-enforcement admissions. We exclude hospital-quarters whose predicted bed-day use exhausts or exceeds capacity, as the relaxed allocation requires positive aggregate slack. The sample contains 12,383 hospital-line-quarters from 159 hospitals.

Demand estimates determine the request flows $\lambda_{kt} R_{hkt}$ used to construct moments of expected revenue, complexity, and length of stay. A Gamma approximation matching each line's length-of-stay mean and variance yields $M_{hjt}^L(c^o)$, and thus makes expected overflow costs in Equation (10) available analytically. For uninsured surplus, we sum $-\lambda_{kt} \log(1 - R_{hkt}) / |\alpha_{k,j}|$ over uninsured types, where $\alpha_{k,j}$ is the travel coefficient. This approximation evaluates the logarithm at the mean request probability, capped at 0.99, and also enters hospitals' counterfactual objectives.

The specification lets c^S vary by hospital-year, $\bar{c}^F$ by service line, and c^V by hospital-line. Uninsured-surplus weights vary by system and spillover-treatment status, pooling standalone nonprofits and pooling standalone for-profit and public hospitals. Costs and the common shock standard deviation use positive transformations. For overflow, let $r_{hjt}^\Gamma = \mu_{hjt}^L / \text{Var}(\ell_{hjt})$ denote the baseline Gamma rate and q_{hy} the logistic transformation of an HRR-year index. We impose $c_{hjt}^o = \underline{c}^o + (r_{hjt}^\Gamma - \underline{c}^o) q_{hy}$, with $\underline{c}^o = 10^{-12}$. Thus the normalized index is common, while the effective rate varies across lines and quarters with length-of-stay composition. The upper bound ensures a finite moment-generating function.

For the proposed parameters and relaxation ρ , we profile out allocations and shadow prices using the main text’s smoothed complementarity conditions. Given η_{ht} , each line’s slack solves a strictly decreasing scalar equation; we adjust η_{ht} until slacks sum to $B_{ht} - \sum_j a_{hjt} \Lambda_{hjt} \mu_{hjt}^L$. The resulting cost-shock threshold is $e_{hjt}^\rho = p_{hjt} - \mathbf{x}_{hjt}' \tilde{\boldsymbol{\theta}}_{hjt} - \psi_{hjt}(c_{hjt}^o; \rho)$. With standard normal density ϕ , distribution Φ , and shock standard deviation σ_ε , each observation contributes

$$\ell_{hjt}^\rho = \begin{cases} \log \phi(e_{hjt}^\rho / \sigma_\varepsilon) - \log \sigma_\varepsilon, & 0 < a_{hjt} < 1, \\ \log \Phi(-e_{hjt}^\rho / \sigma_\varepsilon), & a_{hjt} = 0, \\ \log \Phi(e_{hjt}^\rho / \sigma_\varepsilon), & a_{hjt} = 1. \end{cases}$$

The corner contributions impose the appropriate one-sided cost restrictions. We maximize the average contribution weighted by Λ_{hjt} , giving greater weight to lines with larger expected request flows.

We solve the slack equations by safeguarded Newton steps and bisection in logs, and the shadow-price equation by bisection in logs. Computations use double precision and implicit derivatives of the roots. Inner solves target 10^{-10} absolute residuals, with scale-adjusted acceptance tolerances of 10^{-6} . The outer solver is L-BFGS-B, initialized from revenue-based costs. We reduce ρ/ρ_0 from 10^{-2} to 10^{-6} in five steps, starting each step from the preceding estimates; ρ_0 scales typical slack by its initial marginal value. Each step allows 5,000 iterations, targeting a gradient tolerance of 10^{-6} or relative objective improvement below 10^{-12} .

For reporting, we solve hospitals’ constrained choices over an eight-node Gauss–Hermite grid for each line’s normal cost shock. We weight the combinations by their binomial fit to observed admissions and implied requests. Reported costs and allocations average over these weighted nodes. Counterfactuals fix each line’s conditional mean cost shock and solve choices again under each policy.

E Counterfactual Appendix

This appendix contains the additional information and evidence on our counterfactual analyses.

E.1 Dollarizing Patient Surplus

To present dollar-value patient surplus estimates and contrast the gains from charitable care regulation with government spending, we need to convert mile-equivalent patient surplus to dollars. Our demand estimates allow us to compute the compensating variation in miles that would leave patients indifferent between the status quo and being allocated to different hospitals in counterfactual exercises. These estimates capture patient heterogeneity in patient travel and — by using travel distance as the numeraire — allow comparison across patients. However,

dollarizing these values requires estimating how patients trade off expenditure and travel.

Several articles estimate hospital-choice or referral models that allow estimating patients' dollar-valued disutility from travel. These estimates, however, share common empirical challenges relevant to the hospital setting. In particular, hospital patients rarely know which prices they will face when seeking care because hospital prices are opaque, mediated by coinsurance schedules, and tied to an uncertain bundle of services for each medical condition. For example, a patient choosing where to have a delivery must form expectations not only about the degree of complications she or the baby might have, but also about the price she will pay for each service, which requires knowledge of hospital prices and her position in her coinsurance schedule. In addition, referring physicians or insurers often mediate patient choices, mixing patients' preferences with those of the intermediaries.

Given these challenges, Prager (2020) stands out as the most appropriate for our setting. This article studies inpatient non-emergency admissions in Massachusetts in 2009 and 2012, where patients have fixed, hospital-tier-specific out-of-pocket copays. The copays are clearly stated, predictable, and observable before admission, allowing patients to forecast their out-of-pocket payment accurately. Using Prager's preferred specification and adjusting the estimates to our 2023-based dollars using CPI measures, we obtain a range of \$1,033 to \$1,823 per mile. While the article's estimates vary substantially across patient types, we use the lower estimate as a conservative measure of patients' disutility from travel. We caveat, however, that these estimates might contain the effects of physician referrals.

We benchmark our estimates to a range of alternative conversions obtained from Ho and Pakes (2014) and Gowrisankaran et al. (2015). The former studies privately insured maternity admissions from California during 2003. They use a revealed-preference moment-inequality estimator to recover travel disutility in dollars. Their estimator accommodates measurement error in expected hospital prices and flexible hospital-by-severity quality differences, but it relies on insurer prices rather than consumer prices. Converted from their 2003 dollars to 2023 dollars, the estimates imply an insurer-payment equivalent of approximately \$110 to \$571 per additional mile. However, the authors explicitly characterize the estimates as summarizing the preferences generated by this combined decision-making process rather than separately identifying patient preferences.

Gowrisankaran et al. (2015) use hospital-discharge and claims data from Northern Virginia covering 2003–2006. They construct patient-specific expected coinsurance rates from observed claims information and estimate a model in which utility depends on travel time and each patient's expected out-of-pocket payments. The paper reports that increasing travel time to all hospitals by one minute reduces estimated consumer surplus by approximately \$167.28 in the nominal dollars of the pooled sample period. With our average-sample-period CPI adjustment, this corresponds to approximately \$265 per additional travel minute in 2023 dollars. Converting this estimate to miles requires an external assumption about average driving speed: at 30 miles per hour, it

corresponds to approximately \$530 per mile, while assumed speeds of 20–40 miles per hour imply approximately \$397–\$795 per mile.

Given the clear observability of prices, the general inpatient setting, and the recency of the data, we consider the estimates from Prager (2020) the most appropriate for our setting. Nevertheless, we present the robustness of our findings to the full range of estimates in the main text.

E.2 Floor-and-Trade Proposition

Here we present a short proof of Proposition 2. First, define

$$Q_h^u(a) := \sum_{j \in \mathcal{J}_h} \sum_{k \in \mathcal{K}_j^u} a_{hj} \lambda_k R_{hk}(a) \omega_k, \quad Q^u(a) := \sum_{h \in \mathcal{H}^{np}} Q_h^u(a).$$

Market clearing imposes $\sum_{h \in \mathcal{H}^{np}} u_h = Q^u(a)$. Denote the left-hand side of the inequality (20) for hospital h by $\Delta \Pi_h(z^p)$. Note that given any positive market clearing price z^p , we have $\Delta \Pi_h(z^p) > 0$ since it is the difference between the hospital's optimal profit when per-unit uninsured revenue is higher but without any obligations, relative to the optimal profit under for-profit status after taxes and no revenue increase on the uninsured.

With the above notations, the inequality (20) implies that $\sum_{h \in \mathcal{H}^{np}} \Delta \Pi_h(z^p) \geq z^p Q^u(a)$. If letting $u_h = \frac{\Delta \Pi_h}{\sum_{h'} \Delta \Pi_{h'}} Q^u(a)$, then we have trivially that market clearing holds and that

$$\Delta \Pi_h \geq z^p u_h \iff \Delta \Pi_h \geq z^p \frac{\Delta \Pi_h}{\sum_{h'} \Delta \Pi_{h'}} Q^u(a) \iff \sum_{h'} \Delta \Pi_{h'} \geq z^p Q^u(a).$$

Hence, a market-clearing price and an obligations vector solve the floor-and-trade problem if and only if the aggregate condition holds.

E.3 Connection to Competition

The analysis in the main text highlights reallocations within a hospital following policy exposure, raising the question of competition's role in determining outcomes. Intuitively, competitive interaction might significantly affect policy effectiveness, but its direction is uncertain. For example, when a policy incentivizes one hospital to increase its emergency care resource allocation, it might displace profitable scheduled insured patients. This increases rival hospitals' benefit from increasing scheduled care resources to capitalize on the displaced demand. In turn, this raises the policy payment needed to incentivize those hospitals to increase their emergency capacity. On the other hand, when one hospital accepts more charitable care patients, it decreases the probability that others would be visited by these patients. This might increase the marginal profitability of an emergency bed, making it easier for the policy to incentivize additional emergency care resource allocation at other hospitals. Which force dominates depends on how hospitals' responsiveness to

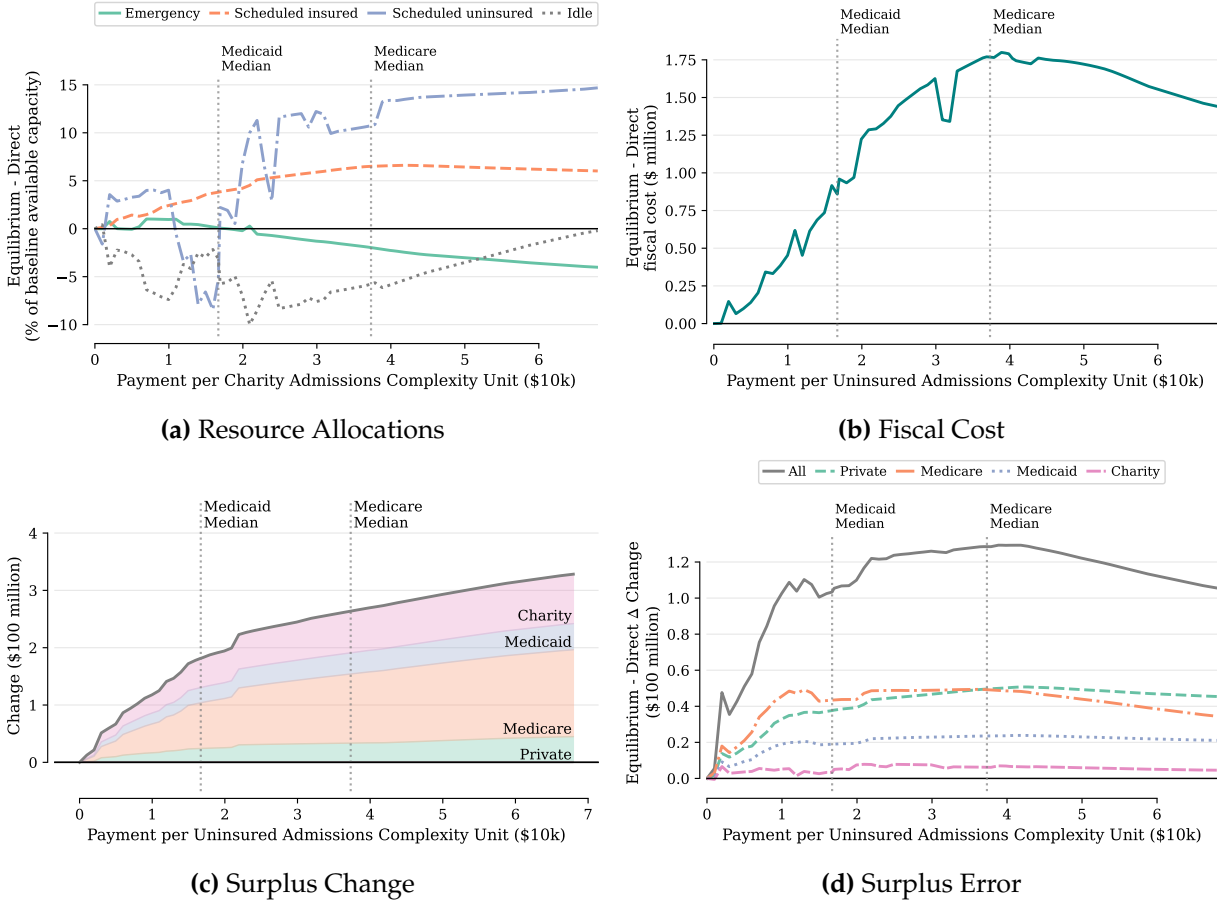


Figure A.15: Impact of Ignoring Competitive Effects of Pseudo-Insurance Policies

Notes: These figures show how ignoring policies' equilibrium effects affects the results.

the policy interacts with patient composition, risk, locations, and preferences.

To evaluate this, we solve for the direct effect of the pseudo-insurance policies and compare it to the equilibrium. The direct effect corresponds to the outcome when hospitals face the policy but assume patients' request likelihood remains at its status quo. As rivals' actions affect hospitals' profits only through their request levels and composition, this direct effect eliminates any competitive interaction among hospitals. Figure A.15 shows the resulting outcomes and compares them to the equilibrium estimates.

The results indicate that ignoring competitive effects significantly affects the policy evaluation. Figure A.15a shows that ignoring competitive effects significantly distorts the perspective on resource allocations, systematically underestimating the impact on scheduled insured and overestimating the impact on idle capacity. In particular, at Medicaid-equivalent rates, competitive effects increase access to insured scheduled care, decrease it for scheduled uninsured, and are largely neutral on emergency access. However, as policy payments increase, competitive effects shift fewer bed-days to emergency and more to scheduled care. Figure A.15b shows that these

competitive impacts translate into a modestly higher fiscal cost of the policy than would be expected without them. Crucially, Figures A.15c and A.15d show that competitive effects vastly increase surplus gains from pseudo-insurance policies; ignoring them underestimates the welfare gains of a Medicaid-equivalent policy by over \$100 million, missing almost a third of the policy's surplus gains.

E.4 Policies with Public Responses

This section briefly discusses the key results when public hospitals can respond to counterfactual policies. Table A.16 shows the main results, which can be compared with the main-text estimates in Table 3. This sample replicates the broad patterns of the main-text evidence. First, providing payment to hospitals that care for the uninsured benefits patients on average, with only a fraction of the value falling on patients receiving charitable care. Second, as in the main text, a policymaker who focuses solely on the surplus value of the pseudo-insurance policy for the uninsured would not implement any of the listed pseudo-insurance policies because the public cost is between two and five times the surplus it provides to the uninsured. Third, paying for otherwise charitable care increases available emergency-room capacity, decreases capacity for scheduled care for the insured, and increases it for scheduled care for the uninsured, overall decreasing idle capacity. Fourth, an unconstrained floor-and-trade policy reaches consumer surplus near Medicaid-equivalent rates at zero public cost, while a homogeneous policy achieves only a small fraction of those values.

The results also highlight some differences associated with public hospital participation. First, because public hospitals are more responsive to uninsured demand, patient-optimal pseudo-insurance pays slightly less than Medicaid rates. Second, because more hospitals participate in public payment programs, running Medicaid- or Medicare-equivalent pseudo-insurance systems is slightly more expensive. Third, because of how public participation affects private hospitals' incentives to provide care to the uninsured, patient surplus gains from pseudo-insurance policies are somewhat lower than in the main text result. Finally, because public hospitals in this scenario respond but do not participate in the floor-and-trade scheme, patient surplus gains are slightly larger, with more of the gains falling on the insured.

Table A.16: Counterfactual Policy Welfare Effects With Public Hospital Participation

	Regulatory	Δ Welfare (\$100M)				Δ Available Capacity (1k bed-days)			
	Price	Patients		Hospitals	Gov.	Emergency	Scheduled		Idle
	z^u, z^p	All	Charity				Insured	Uninsured	(p.p.)
Panel A: Pay-per-Complexity									
Medicaid-equivalent	1.676	2.302	0.495	1.392	1.543	99.059	-9.260	116.264	-5.070
Medicare-equivalent	3.759	3.071	0.745	3.205	3.475	157.479	-18.545	188.198	-8.048
Optimum at $\gamma^\pi = 0$	1.104	1.882	0.357	0.914	1.014	78.725	-5.792	60.048	-3.272
Optimum at $\gamma^\pi = 1.00$	4.165	3.300	0.782	3.543	3.851	164.859	-11.967	193.840	-8.530
Panel B: Floor-and-Trade									
Unconstrained	12.108	2.563	0.407	-0.199	0.000	82.915	-5.661	117.434	-4.790
Homogeneous	0.241	0.359	0.110	-0.014	0.000	13.158	-1.916	24.947	-0.890

Notes: Welfare is measured as the change relative to the baseline, in hundreds of millions of dollars. The Patients columns report the gains for all patients and charitable-care patients, respectively, with “Gov.” representing the change in fiscal cost. Available capacity is the sum across hospitals of the product of each line’s allocated bed-days and its acceptance rate, measured in thousands. The idle capacity ratio is defined as $(\sum_h B_h - \sum_{h,j} a_{hj} b_{hj}) / \sum_h B_h$, and the change is presented in percentage points. The second column reports the regulatory payment or credit price in \$10,000 units. All values are quarterly.

E.5 Implementation

We simulate policy equilibria for 2019 Q4 using estimated preferences, arrival rates, costs, and conditional mean cost shocks. The responding sample contains 133 hospitals with complete demand and supply inputs. Public hospitals and hospitals missing inputs for any line retain observed acceptance rates; their request flows can change as competitors respond. Across the main policies, preferences, illness rates, networks, physical capacity, and cost primitives remain fixed. We measure effects relative to a simulated equilibrium with no additional payment.

At a proposed acceptance vector, we recompute request probabilities, patient composition, revenues, and length-of-stay moments. The Gamma approximation changes with composition, holding the estimated effective overflow rate fixed. Conditional on these objects and shadow price η_h , Equation (10) has piecewise analytical solutions for interior, capacity-binding, and corner choices. The binding solution uses the principal branch of the Lambert W function. Bisection on $\log \eta_h$ allocates total bed-days across lines.

We solve the equilibrium fixed point using Anderson acceleration with eight past iterations and a mixing weight of one half. If this stalls, we use damped best responses, then Newton steps on the optimality conditions for persistently oscillating lines. The baseline starts from estimated acceptance rates; positive payments start from the nearest solved lower payment, inserting intermediate values when needed. This continuation provides an equilibrium selection convention when uniqueness is not assured. Retained equilibria satisfy $\max_{h,j} |BR_{hj}(\mathbf{a}) - a_{hj}| \leq 10^{-6}$. Capacity solves target 10^{-7} bed-days, with final feasibility checked against the larger of 10^{-5} bed-days and

$10^{-5}B_h$.

Request probabilities and expected patient surplus use an exponential change of variables and a unit-spaced trapezoid rule in double precision. Adaptive integration ranges yield 107 request and 112 surplus nodes; admission probabilities must sum to one within 10^{-3} . We convert expected log surplus into miles using each type's travel coefficient and apply the dollar conversion in Appendix E.1. Hospital preferences retain the uninsured-surplus approximation used in estimation, distinct from aggregate patient welfare.

For pseudo-insurance, payment z^u adds $z^u\omega_k$ per uninsured admission at responding hospitals. We start with 60 levels up to 1.575 times the median Medicare payment per complexity unit, insert exact Medicaid and Medicare benchmarks, and expand or refine around promising values. Price refinement allows 20 steps with a bracket-width target of 10^{-4} . Fiscal spending covers eligible admissions at the policy equilibrium. Alternative welfare weights rescore this common set of equilibria using Equation (17). Reported optima are conditional on the explored range and equilibrium selection.

Unconstrained floor-and-trade uses the same price-search strategy, with only nonprofits earning credits. At each price z^p , we solve for equilibrium and compute each nonprofit's best response to the same rivals after surrendering credit-market access. Equation (20) determines participation feasibility. Obligations are proportional to participation gains, as in Appendix E.2. We subtract credit transfers from welfare. The main specification puts 10% of profits at risk and excludes profit from activities outside modeled inpatient care.

The homogeneous policy assesses a common fraction of positive pre-tax profits, excluding credit income and uninsured-surplus benefits. This action-dependent assessment enters each best response. We search 30 obligation rates between zero and one half and refine near participation boundaries and welfare maxima. A nested scalar search selects the smallest detected positive market-clearing credit price, with a relative residual target of 10^{-8} . We verify individual participation and market clearing at the final actions. Credit transfers cancel, so both trading policies have zero fiscal spending.

The Medicaid exercise sequentially adds Medicaid payments, emergency-to-scheduled arrival ratios, network restrictions, and reassignment to insured scheduled care. Each stage re-solves the equilibrium; arrival adjustments preserve total illness within age-sex cells, and outcomes retain the original uninsured cohort's identity. The public-hospital exercise allows eligible public hospitals to respond while remaining outside credit markets.

Separate allocation benchmarks cover all four quarters of 2019, holding each line's fitted bed-day use $a_{hit}\Lambda_{hit}\mu_{hit}^L$ fixed, rounded upward. Patient groups are expanded into individual admissions with their mean length of stay. Deferred acceptance orders patient proposals by utility and hospital priorities by payer: private, Medicare, Medicaid, then charitable care, with random

tie-breaking. The dictatorial benchmark maximizes surplus by mixed-integer programming with SCIP and a five-minute limit per line-market-quarter, retaining a feasible assignment at least as good as deferred acceptance. The unconstrained benchmark assigns each patient to the preferred option.

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